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MEASUREMENT OF THE CHARM-YUKAWA COUPLING VIA CHARM QUARK-ASSOCIATED HIGGS BOSON PRODUCTION USING CMS RUN-2 DATA

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Abstract

A primary goal of the Large Hadron Collider, located in Geneva, Switzerland, is to measure the properties of the Higgs boson. Specifically, a measurement of the interaction strength between the Higgs boson and the charm quark, know as the Yukawa coupling of the charm quark, is a quantity of considerable interest in contemporary particle physics. The measurement of this coupling represents a key consistency check of the Standard Model of particle physics, where any deviation may provide exciting hints about the presence of new physics. In this thesis, such a measurement is presented using LHC Run-2 proton-proton collision data recorded by the CMS detector. Specifically, the charm quark-associated Higgs boson production process (cH), where the Higgs boson decays to four muons via a pair of Z bosons is targeted, which has previously not been explored at the LHC. Key challenges such as the low transverse momentum of the charm quark produced in the cH process are adressed using likelihood based jet selection methods. Additionally, an interpretation of the analysis in the context of effective field theory is presented, with a specific focus on the previously unconstrained chromomagnetic dipole operator of the charm quark, which may be uniquely constrained with the cH process.

Samenvatting

Een belangrijk doel van de Large Hadron Collider, gelegen in Genève, Zwitserland, is het meten van de eigenschappen van het Higgs boson. In het bijzonder is het bepalen van de interactiesterkte tussen het Higgs-boson en het charm quark, ook wel bekend als de charm-quark Yukawa-koppeling, een groot aandachtspunt in de hedendaagse deeltjesfysica. De meting van deze koppeling vertegenwoordigt een belangrijke test van de consistentie van het Standaardmodel van de deeltjesfysica, waarbij eventuele afwijkingen kunnen duiden op de aanwezigheid van nieuwe fysica. In dit proefschrift wordt een dergelijke meting voorgesteld met behulp van LHC Run-2 proton- protonbotsingsgegevens die zijn verzameld met de CMS-detector. Er wordt specifiek gekeken naar het proces van Higgs bosonproductie in associatie met een charm-quark (cH), een proces dat nog niet eerder onderzocht werd met de CMS data, waarbij het Higgs boson vervalt via $\mathrm{H} \to \mathrm{ZZ} \to 4$ muonen. Belangrijke uitdagingen, zoals het lage transversale momentum van het charm quark dat in het cH- proces wordt geproduceerd, worden aangepakt met behulp van op waarschijnlijkheid gebaseerde jetselectiemethoden. Daarnaast wordt een interpretatie van de analyse in de context van Effective Field Theory voorgesteld, met specifieke aandacht voor de chromomagnetische dipooloperator van het charm quark, waar we nu voor het eerst op een unieke wijze informatie over kunnen geven via het cH proces.

Acknowledgements

First and foremost, I would like to thank my (co-)supervisors, who have guided me through the past four years of my doctoral research at the IIHE. This includes Jorgen D'Hondt, Michael Tytgat and Gerrit Van Onsem, all of who have been extremely supportive with their continuous feedback and help, not only in the realm of particle physics but also beyond. This support and guidance has surely shaped me as a researcher who remains enthusiastic and excited about the future of particle physics.

I would also like to extend my gratitude to the countless other past and present IIHE members that have positively shaped this time, providing not only a supportive research environment but also plenty of memorable moments and interactions. I look forward to staying in contact with you, whether through a shared, continued path in the field or elsehow. This includes in particular, Nordin, with whom I have shared a parallel PhD path that has included many exciting endeavours, such as our shared trips to the US. Lastly, I want to thank my friends and family, who have continually supported me during the past four years and without whom this thesis certainly would not exist today.

The author's contributions

In the context of my doctoral studies, I have contributed to a variety of original research and work. While this has occured in the context of a large, international collaboration, the main contributions are here delineated explicitly:

- The analysis presented in Chapter 4, which is the primary focus of this thesis, was performed by me. Some aspects of the analysis were developed together with a Master's student I co-supervised, namely the reducible background estimation presented in Subsection 4.2.3.
- I am a primary contributor to [1]. This is a phenomenological paper exploring non-Standard Model interpretations of the Standard Model process that is analysed in this thesis. A demonstration of this work, fully translated into the context of the CMS detector for this thesis, is presented in Chapter 5.
- I developed a dedicated analysis framework, as detailed in Appendix Section A, with which the presented analyses are performed. A variant of this framework was also used for a signifcant portion of the analysis presented in [1].
- I was, for a number of years, the b-tagging and vertexing (BTV) group's offline validation contact and also more briefly, the data quality monitoring (DQM) contact. The work performed in these roles played an important part in ensuring that the data taken by the CMS experiment during Run 3 of the LHC was well-understood from the perspective of BTV algorithms. This work thus contributed to the reliable performance of jet flavour taggers in the Run-3 data taking era. A more detailed overview is given in Section 3.4.
- I presented a contribution at the 2025 edition of the bi-annual EPS-HEP conference. This consisted of an overview of recent results by the CMS collaboration in analyses that target the charm-Yukawa coupling. A corresponding conference proceeding was additionally published.

Papers & proceedings

- N. Breugelmans et. al. "Constraining new physics in charm quark associated Higgs boson production events using the Standard Model effective field theory approach". In prep.
- Heyen, F. (2026). Studying the Higgs-charm Yukawa coupling with the CMS detector. In Proceedings of The European Physical Society Conference on High Energy Physics — PoS(EPS-HEP2025) (p. 392). Sissa Medialab. The European Physical Society Conference on High Energy Physics. https://doi.org/10.22323/1.485.0392

Contents

Chapter 1

Introduction

The Standard Model (SM) of particle physics is the theory that best characterises our current understanding of fundamental particles and their interactions. It describes a broad range of phenomena and makes a plethora of predictions, many of which have been confirmed via measurement to great degrees of accuracy [2]. A notable feature of the SM is the Brout-Englert-Higgs (BEH) mechanism [3][4], which predicts the existence of a Higgs boson. The BEH mechanism is considered a central part of the SM as it provides a unique mechanism by which SM particles may acquire mass through their interaction with the Higgs field. As such, the experimental discovery of a Higgs-like scalar boson at the CERN Large Hadron Collider (LHC) in 2012 [5][6] was a major milestone in particle physics. Since this discovery, measuring the exact properties of this particle (henceforth simply referred to as the Higgs boson) has been a major goal of LHC experiments such as the CMS collaboration [7] as ultimately, physicists seek to determine whether the particle is indeed the Higgs boson described by the SM. A particular subset of these properties are the so-called Yukawa interactions between the Higgs boson and massive fermions, which are proportional to the mass of the fermions. As can be seen in Figure 1.1, a number of these have previously been measured and indeed align with the values predicted by the SM. However, the measurement of the Yukawa couplings of several of the lighter fermions still remain an open challenge, as these couplings decrease in strength with smaller fermion masses.

The next lightest fermion candidate for such a measurement is the charm quark. Consequently, the study of the Yukawa coupling between the Higgs boson and the charm quark is of significant interest [8]. This is as, experimentally speaking, relatively little is known about the interaction of the Higgs boson and lighter SM particles, such as the first and second-generation quarks. Studies of the Yukawa coupling of the charm quark thus provide an essential glimpse into previously unexplored territory, allowing us to better characterise the nature of the Higgs boson in this regime. Most importantly, potential deviations from the SM could be present in the couplings of the Higgs boson to lighter quarks. Since Higgs production rates at an experiment such as the LHC largely depend on the couplings of the Higgs boson to weak gauge bosons and third-generation quarks, such deviations may not be visible in our current experimental picture. As a result, the motivation to study the Yukawa coupling of the charm quark (henceforth referred to as the charm-Yukawa coupling) becomes clear.

Experimental approaches to measuring this coupling mostly fall into two categories. The first involves targeting processes in which the Higgs boson decays to a charm and anti-charm quark pair. With this strategy, the CMS collaboration recently reported that it is able to exclude

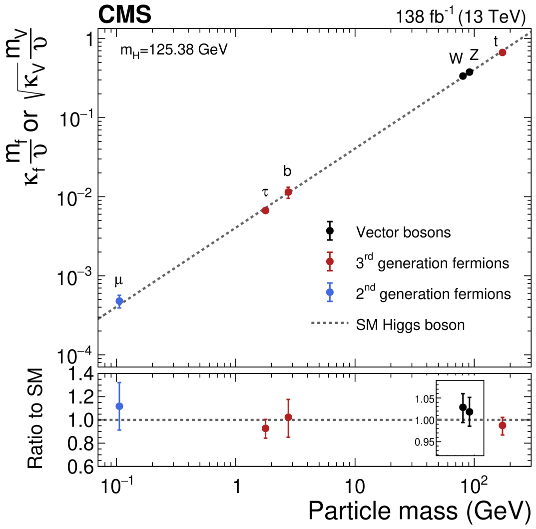


Figure 1.1: The measured coupling modifiers κ_f and κ_V of the coupling between the Higgs boson and fermions as well as heavy gauge bosons as functions of fermion or gauge boson mass m_f and m_V, where ν is the vacuum expectation value of the Higgs field. Figure taken from [7].

deviations of the charm-Yukawa coupling that are greater than 3.5 times of what is predicted by the SM, the most stringent constraint to date [9]. The second strategy is to target charm quark-associated Higgs production modes (henceforth referred to as cH production), where an interaction between a charm quark and Higgs boson occurs in the production of the Higgs boson. While analyses based on this strategy have thus far yielded comparatively weaker constraints on the charm-Yukawa coupling [10][11], they constitute important, complementary measurements. This is due to the strong statistical limitations that both types of strategies face, given that the cross section of processes involving the charm-Yukawa coupling are extremely small in comparison to the plethora of processes that occur at the LHC. As such, a final measurement of this coupling at the 5σ significance level may very well rely on a statistical combination that includes both strategies. While such a feat is still currently out of reach with the current datasets gathered at the LHC, it may come within reach with the significantly larger dataset expected at the High Luminosity LHC [12]. As a result, the preparation of well-understood analysis strategies to reach this important goal is clearly of importance.

Using data from the Run-2 dataset of the CMS detector, this work seeks to contribute to this endeavour by presenting such a complementary analysis of cH production in a previously unconsidered Higgs decay channel. In Chapter 2, an introduction to the SM and the charm-Yukawa coupling is given, together with a discussion of cH process. Additionally, an argument is presented indicating why the cH process may be uniquely suited to probing certain types of processes not described by the SM, which are here parametrised through so-called effective field theory. Subsequently in Chapter 3, the CMS detector is presented alongside the algorithms used to reconstruct the proton-proton collisions produced at the LHC. In Chapter 4, the analysis strategy that is employed in this work is outlined, along with the results that are derived by applying this strategy and a discussion thereof. Finally, a demonstration of the ability of this analysis strategy to constrain the presence of the aforementioned non-SM processes is demonstrated in Chapter 5.

Chapter 2

The Standard Model and the charm-Yukawa coupling

In this chapter, the theoretical underpinnings of the analysis presented in this work are discussed. This includes a brief discussion of the SM in Section 2.1, after which the charm-Yukawa coupling is introduced in Section 2.2. Following this, the cH process is discussed along with how it may be targeted by experiments such as the CMS detector Section 2.3. Finally, an effective field theory (EFT) interpretation of the cH process is discussed in Section 2.4.

2.1 The Standard Model of particle physics

The SM is formulated through the formalism of Quantum Field Theory (QFT). This is a formalism that combines concepts of classical field theory, quantum mechanics as well as special relativity into a single, coherent description of fundamental particles as excitations of underlying fields that pervade space-time. In this description, SM particles fall into two categories: fermions and bosons. The former are the massive particles which make up the matter of the universe while the latter are the force-carrying particles of the strong and electroweak forces. The distinction between these categories is made based on the spin of the particle, which may be of either half-integer or integer, respectively.

The fermion content of the SM consists of 12 unique particles. These include six leptons, namely the electron, muon and tau as well as their respective neutrinos alongside six different quarks that are distinguished by their so-called *flavour*. The different quark flavours include up, down, charm, strange, bottom and top, which specify a quark's mass eigenstate as well as electric charge. These fermions are typically arranged into three generations depicted as

$$\begin{pmatrix} e \\ \nu_e \end{pmatrix} \begin{pmatrix} \mu \\ \nu_\mu \end{pmatrix} \begin{pmatrix} \tau \\ \nu_\tau \end{pmatrix}, \begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}. \tag{2.1}$$

There are distinct differences between the leptons and quarks. Leptons carry integer (or no) electrical charge while quarks carry fractional electrical charges. More importantly, while both quarks and leptons may interact via the electroweak force, only the quarks interact via the strong force. A special feature of the strong force is that the potential energy associated with it grows as the distance between strongly interacting particles increases. This leads to a phenomenon referred to as *confinement*, where quarks almost exclusively form compositive states called *hadrons*. As a result, the emission of individual quarks or *gluons* (the force carrying particles of the strong force) in scattering processes typically produce complex cascades of particles referred to as *jets*, where the individual particles will ultimately form a variety of hadrons in a process referred to as *hadronisation*. This gives rise to a very complex phenomenology of proton-proton (pp) collisions such as those at the LHC. At very high energies, the processes that involve the strong force, typically referred to as quantum chromodynamics (QCD) processes, can be understood through perturbative descriptions. This refers to an expansion of the expressions that describe QCD processes in the strong coupling constant $\alpha_\mathrm{s}(Q^2)$, which has an inverse scaling relationship with the momentum transfer of a process Q. As a result, $\alpha_\mathrm{s}(Q^2)$ becomes larger at lower energies and phenomenological descriptions are relied on in this regime instead.

Lastly, the existence of anti-fermions must be mentioned. These carry the exact opposite quantum numbers (e.g. charge) as their fermion counterparts, though otherwise behave similarly (take the electron and positron for instance). For simplicity, references to a fermion in this work may be understood as referencing both the fermion and anti-fermion counterpart, unless otherwise indicated explicitly. Examples of the latter are e.g. referring explicitly to electrons e^- and positrons e^+ or charm quark c and anti-charm quark $\bar{\mathrm{c}}$ pairs.

There exist 13 unique bosons in the SM. These include the photon γ, $\mathrm{W}^\pm$ and Z which mediate the electroweak force as well as 8 gluons g that mediate the strong force. The final piece is the Higgs boson. Contrary to the force carriers, which all are spin 1, the Higgs boson is spin 0. The massive particles of the SM acquire their mass by interacting with the BEH field, excitations of which are identified as the Higgs boson, and is therefore a central element of the SM.

Considering the introduced particles and forces, the SM has a rich and detailed phenomenology. A great example of a mathematically rigorous delineation of this can be found for example in [13]. Given the focus of this work on the Yukawa coupling between the Higgs boson and charm quark, only this aspect of the SM is discussed in further detail.

2.2 The charm-Yukawa coupling

The quantity that defines the strength of the interaction between massive fermions and the Higgs boson is the so-called Yukawa coupling. To better understand this and associated concepts, some knowledge of the electroweak sector of the SM is required. These are discussed in this section while a comprehensive overview may be found in [14].

To understand the origin of the Yukawa couplings, a brief discussion of Lagrangian densities, gauge transformations and the role of symmetries in the SM is useful. The Lagrangian density $\mathcal{L}(\phi_i; a_i)$ is a quantity dependent on a set of fields ϕ_i and constants a_i from which the equations of motions for the particles associated with these fields may be derived. Commonly, theories of particles and their behaviour in a QFT are therefore defined through the formulation of a Lagrangian density. The form of this expression determines the nature of the particles that are included as well as their interactions.

The concept of gauge symmetries is a central component to the way in which particle interactions are introduced in the Lagrangian formulation of the SM. These originate from the fact that the quantum fields in a QFT carry phase information, which may depend on the space-time coordinate of the field. This phase information describes (local) degrees of freedom of the field and should have no effect on the physical observables of the system. Accordingly, the Lagrangian $\mathcal{L}$ that describes the system, should remain invariant under arbitrary phase transformations. These transformations are typically referred to as a choice of gauge and the corresponding invariance is accordingly referred to as a *local gauge symmetry*.

In the Lagrangian of the SM, invariance in the presence of local gauge symmetries is ensured through the addition of gauge fields. These gauge fields couple to the previously existing fields and effectively serve as mediators of phase information between space-time points of the original fields. It is exactly these gauge fields that we identify as the fields of the previously introduced force-mediating bosons, and which are required to maintain local gauge symmetry. A very interesting conclusion from this is that the dynamics of the bosons and the corresponding force are determined entirely by the structure of the local gauge symmetry that must be preserved. For the electroweak force, the corresponding symmetry is referred to as $SU(2)_L \,\mathrm{x}\, U(1)_Y$. Here, the L refers to the fact the weak force, described by the $SU(2)_L$ symmetry, only acts on left-handed particles. The handedness of particles, also referred to as the *chirality*, originates from the relativistic description of particles in space-time using the Poincaré symmetry group, which represents the underlying symmetries of special relativity. A physical interpretation of this property can be gained by considering a massless particle's *helicity*, which is the projection of the particle's spin onto its momentum vector. When the direction of the particle's spin coincides with the particle's momentum, it is considered to be right-handed. Accordingly, when the direction of the particle's spin is in the opposite direction of the momentum, it is considered to be left-handed. This, however, only holds true for massless particles, since for massive particles the helicity can depend on an observer's reference frame. Given that the weak force makes a distinction based upon this property, left and right-handed particles are represented in different ways in its formulation. These representations are referred to as doublets and singlets, respectively. In total, there are four bosons associated with the electroweak force. These are the photon γ, which mediates the electromagnetic force, as well as the electromagnetically charged $\mathrm{W}^{\pm}$ and electromagnetically neutral Z boson that mediate the weak force.

With these concepts in mind, the nature of the electroweak sector's Lagrangian in the SM may be discussed. Naively, the form of this would be given by

$$\mathcal{L}_{\mathrm{EW}} = i\overline{\psi}_L \gamma^\mu D^L_\mu \psi_L + i\overline{\psi}_R \gamma^\mu D^R_\mu \psi_R - \frac{1}{2}\mathrm{Tr}\left(W^a_{\mu\nu} W^{a\mu\nu}\right) - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}\,, \tag{2.2}$$

for a generic combination of a left-handed doublet ψ_L and right-handed singlet ψ_R. The individual elements of $\mathcal{L}_{\mathrm{EW}}$ are briefly summarised below:

g' :	coupling constant of $\mathrm{U}(1)_Y$
g :	coupling constant of $\mathrm{SU}(2)_L$
ψ_L,	left-handed doublet
ψ_R,	right-handed singlet
B_μ :	gauge field of $\mathrm{U}(1)_Y$
W^a_μ :	gauge fields of $\mathrm{SU}(2)_L$, $a = 1, 2, 3$
$W_{\mu\nu}$:	field strength tensor

$B_{\mu\nu}$: field strength tensor

$t^a = \frac{\sigma^a}{2}$, SU(2) generators

$Y_L = -1$, left chiral hypercharge

$Y_R = -2$, right chiral hypercharge

$D^L_\mu = \partial_\mu + ig'\frac{Y_L}{2}B_\mu + igt^a W^a_\mu$

$D^R_\mu = \partial_\mu + ig'\frac{Y_R}{2}B_\mu$

The terms $D^{L/R}_\mu$ are so-called covariant derivates that ensure the local $\mathrm{SU}(2)_L \,\mathrm{x}\, \mathrm{U}(1)_Y$ gauge symmetry is upheld for $\mathcal{L}_{EW}$. These contain the coupling constants g' and g, which determine the coupling strength of fermions to the B_μ and W^a_μ gauge fields, respectively. The generators of the $\mathrm{SU}(2)_L$ group are the Pauli matrices, with which arbitrary $\mathrm{SU}(2)_L$ gauge transformations may be described. Finally, the left and right chiral hypercharges are the charges associated with the $U(1)_Y$ symmetry of the electroweak force. Further details can be found in [14].

In this formulation, the observed charged gauge bosons $\mathrm{W}^\pm$ arise from linear combinations of the W_1 and W_2 gauge fields

$$W^\pm = \frac{1}{\sqrt{2}}(W_1 \mp iW_2), \tag{2.3}$$

while the Z boson and photon γ arise from linear combinations of the W_3 and B gauge fields achieved via a rotation

$$\begin{pmatrix} \gamma \\ Z \end{pmatrix} = \begin{pmatrix} \cos\theta_\mathrm{W} & \sin\theta_\mathrm{W} \\ -\sin\theta_\mathrm{W} & \cos\theta_\mathrm{W} \end{pmatrix} \begin{pmatrix} B \\ W_3 \end{pmatrix} \tag{2.4}$$

with the weak mixing angle θ_W.

The massive natures of the $\mathrm{W}^\pm$ and Z bosons, first reported in [15], are however incompatible with such a formulation. The reason is that naive mass term such as

$$m^2_W W^+_\mu W^{-,\mu} + \frac{1}{2}m^2_Z Z_\mu Z^\mu. \tag{2.5}$$

do not remain invariant under arbitrary $\mathrm{SU(2)_L}$ gauge transformations. This is as gauge fields A_μ generically transform as

$$A_\mu \to A'_\mu = A_\mu - \frac{1}{g}\partial_\mu \mathcal{V}(x), \tag{2.6}$$

where $\mathcal{V}(x)$ is some arbitrary phase depending on the spacetime coordinate x. Substituting Eq. (2.6) into Eq. (2.5) thus introduces additional terms that do not cancel. The same is true for fermion mass terms in the form of

$$m_f\overline{\psi}\psi. \tag{2.7}$$

Here, the invariance breaking terms arise from different transformation behaviours of ψ_L and ψ_R under $\mathrm{SU}(2)_L \,\mathrm{x}\, \mathrm{U}(1)_Y$ gauge transformations, since the ψ_L component participates in the weak force while ψ_R does not.

2.2.1 The Brout-Englert-Higgs mechanism

The BEH mechanism provides a way to circumvent the gauge symmetry breaking nature of the aforementioned generic mass terms. This is achieved through a process referred to as *spontaneous symmetry breaking*. A spontaneously broken symmetry refers to a symmetry that is upheld in a global view of the system (i.e. the overall Lagrangian density $\mathcal{L}_{\mathrm{EW}}$ remains invariant under a relevant gauge transformation) while the energetic ground state of the system explicitly breaks this symmetry. This is a process formally described by the Goldstone theorem [16], which states that each broken symmetry in a relativistic QFT generates an additional massless boson. These introduce additional degrees of freedom into the theory and are coined Goldstone bosons. The BEH mechanism exploits this by adding an additional term

$$\mathcal{L}_{\mathrm{Higgs}} = D_\mu \phi^\dagger D^\mu \phi - V(\phi) \tag{2.8}$$

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \lambda(\phi^\dagger \phi)^2 \tag{2.9}$$

to $\mathcal{L}_{\mathrm{EW}}$, that describes an additional complex field ϕ. This is a $\mathrm{SU}(2)_L$ doublet

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \tag{2.10}$$

with the scalar components ϕ^+ and ϕ^0. Here, $V(\phi)$ corresponds to the potential energy term of the field. Again, the covariant derivative

$$D_\mu = \partial_\mu + ig' \frac{Y_\phi}{2} B_\mu + igt^a W_\mu^a \tag{2.11}$$

ensures $\mathcal{L}_{\mathrm{Higgs}}$ remains locally gauge invariant under $\mathrm{SU}(2)_L \,\mathrm{x}\, \mathrm{U}(1)_Y$ transformations and therefore introduces couplings between the ϕ field and the B_μ and W_μ fields. The constants of the potential term Eq. (2.9) are chosen in such a way that the ground state of $V(\phi)$ is non-zero. This can be achieved by choosing them such that $\lambda > 0$ and $\mu^2 > 0$. The result is a ground state of V that is identified as the vacuum expectation value

$$v = \sqrt{\frac{\mu^2}{2\lambda}}\,. \tag{2.12}$$

The center of the potential is now an unstable local maximum and the only stable configuration can be found in the non-zero ground state. This is demonstrated visually in Figure 2.1. Through this, the symmetry of the potential is effectively broken. A popular choice of gauge for ϕ is

$$\phi = \begin{pmatrix} 0 \\ v + \frac{h}{\sqrt{2}} \end{pmatrix}, \tag{2.13}$$

where h is a new scalar field that is used to parametrise radial perturbations of the potential's ground state. This choice is referred to as the unitary gauge and h is identified as the field corresponding to the physical Higgs boson. By expanding Eq. (2.8) with this choice of ϕ, a range

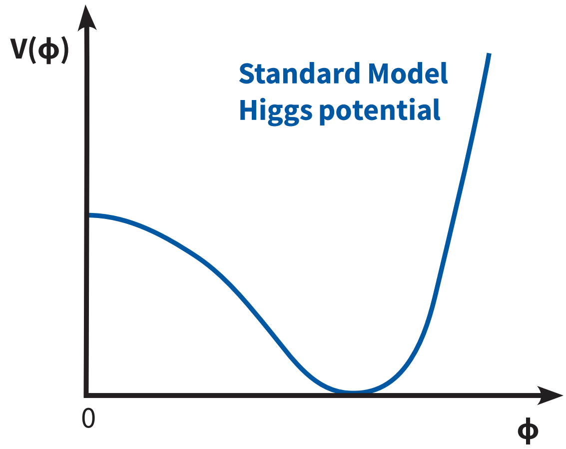


Figure 2.1: A sketch of the shape of the Higgs potential in the Standard Model, which is simplified by showing a cross section of the two-dimensional potential. A non-stable maximum at the center of a potential can be seen alongside a stable configuration in a non-zero ground state.

of terms are introduced to $\mathcal{L}_{\text{EW}}$. These contain a variety of interaction terms between the gauge fields and the Higgs field, as well as newly generated mass terms for the Z and W bosons

$$\left(\frac{g}{2}\right)^2 v^2 W^+_\mu W^{\mu-} = m^2_{\text{W}} W^+_\mu W^{\mu-} \tag{2.14}$$

$$\left(\frac{\sqrt{g^2+g'}}{2}\right)^2 v^2 Z_\mu Z^\mu = m^2_{\text{Z}} Z_\mu Z^\mu \,. \tag{2.15}$$

This can be understood to mean that the electroweak coupling constants g and g' along with v effectively determine the mass of the Z and $\text{W}^\pm$ bosons. A full description and compilation of all the terms of the electroweak Lagrangian density of the SM can be found in [14].

2.2.2 The Yukawa couplings

By including this additional contribution in our theory, mass terms for fermions may now be generated by including a term of the form

$$\mathcal{L}_{\text{Yukawa}} = -y_f \overline{\psi} \phi \psi \tag{2.16}$$

$$= -y_f v \overline{\psi} \psi (1 + \frac{1}{v}\frac{h}{\sqrt{2}}). \tag{2.17}$$

This expression remains invariant under $\text{SU}(2)_L \, \text{x} \, \text{U}(1)_Y$ gauge transformations due to the presence of ϕ. Similarly to the W and Z mass terms, the relation

$$m_f = y_f v \tag{2.18}$$

is obtained. A curious feature of the SM is that the Yukawa couplings y_f are free parameters of the theory. As a result these must be measured experimentally, with the measurement of the charm-Yukawa coupling y_c being the goal of this work. Since the charm quark mass has previously been determined from experiment to be $m_c = 1.27$ GeV [2], a measurement of y_c thus represents an important consistency test of the SM. To this end, one can exploit that an interaction between fermions and the Higgs field is introduced, as is evident from Eq. (2.17), with an interaction strength proportional to y_c. It is this feature that may be utilised by experiments at the LHC to measure y_c. The presence of such interactions is also how other Yukawa couplings, such as of the bottom and top quarks, are determined experimentally and have been found to be consistent with the SM expectation [17][18].

2.3 Measuring the charm-Yukawa coupling

By measuring the frequency of occurrence of physics processes in which the coupling between the Higgs boson and charm quark appears, y_c may be determined. As such, a suitable process must be found that can be detected by an experiment such as the CMS experiment, which records pp collisions at the LHC. These fall typically fall into two categories. The first consists of processes in which a Higgs boson decays into a charm and anti-charm quark pair ($H \to c\bar{c}$). This typically involves targeting processes in which the Higgs boson is produced in association with e.g. a top quark pair ($t\bar{t}H$) or weak gauge bosons (VH), to better identify the presence of the Higgs boson. The second category consists of processes in which a Higgs boson is produced in association with a charm quark. This latter category of processes is the focus of this work and is henceforth referred to as the cH process, which is described in Subsection 2.3.1. Subsequently, in Subsection 2.3.2, the κ-framework is introduced. This framework serves as a tool for analyses that target different charm-Yukawa sensitive processes, to interpret their results in terms of a single, common parameter denoted as κ_c.

2.3.1 The cH process

The cH process collectively encompasses processes in which a charm quark is produced alongside a Higgs boson in pp collisions. At leading order in the perturbative description of QCD, this consists of two processes sensitive to y_c, represented by the Feynman diagrams shown in Figure 2.2. The two diagrams, namely the s and t-channel diagrams, constitute the y_c sensitive contribution due to the presence of an interaction between the Higgs boson and the charm quark. There exist also additional cH processes, such as ones mediated through the effective coupling of the Higgs boson to gluons, which are not sensitive to y_c. A representative Feynman diagram can be seen in Figure 2.3. This accounts for approximately 80% of the inclusive cH cross section and thus represents a significant background to the cH process that is sensitive to the charm-Yukawa coupling. A contribution such as the one shown in Figure 2.3 can effectively be thought of as a higher order correction in perturbative QCD to the gluon fusion process, the role of which is discussed in Subsection 4.2.2.

Targeting the cH process to measure y_c is a relatively novel strategy in comparison to targeting $H \to c\bar{c}$ decays. A key advantage of the former approach is that contributions from the abundant QCD multijet background at the LHC are greatly reduced, since only a single jet resulting from a charm quark needs to be identified, as opposed to two. This is especially relevant given that the task of jet flavour identification (i.e. identifying the type of particle that initiated the jet) remains to be an experimentally challenging one. Additionally, since the sensitivity to y_c does not originate from the decay of the Higgs boson, the Higgs boson decay mode to target can be chosen freely. Especially signatures such as $H \to ZZ \to 4\mu$, which may be resolved cleanly by an experiment such as the CMS experiment, can be targeted. Motivated by this feature, this

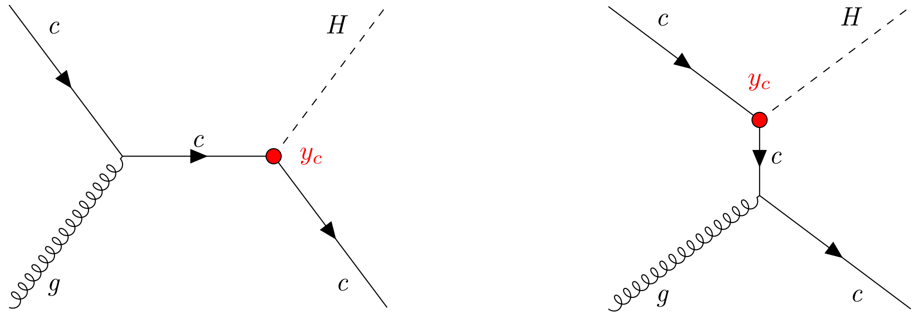


Figure 2.2: The leading-order cH processes through which y_{c} may be probed, since each diagram contains an interaction between a charm quark and Higgs boson, here denoted in red. The corresponding diagrams with an anti-charm quark $\bar{c}$ are implied.

work focusses on the H→ZZ→4μ decay mode in cH processes.

However, this type of analysis strategy also comes with notable drawbacks. A significant experimental difficulty results from the fact that the associated charm flavour jets are typically produced at lower transverse momenta p_T, as seen in Figure 2.4. These can be experimentally difficult to reconstruct well and thus a significant portion of the cH signal is expected to be lost due to resulting acceptance effects. Another drawback is that though Higgs boson decay channels such as H→ZZ→4μ may be very cleanly resolved in a detector, they come with very small branching ratios. The branching ratio indicates the probability with which a particle decays via a given decay mode and in this case is very small with $\mathcal{B}(\mathrm{H}\rightarrow\mathrm{ZZ}\rightarrow4\mu) = 0.3\%$ [2]. As a result, the already small cross section of the cH process is further reduced. Given these effects, a key challenge in targeting the cH process is anticipated to arise from the overall low signal process yields that can be expected in a statistical analysis of this process. Lastly, significant challenges are expected to also arise from that fact a majority of the cH cross section is not sensitive towards y_{c}. This means a key challenge of measuring y_{c} with the cH process consists not only of differentiating inclusive Higgs boson production from charm quark associated Higgs boson production, but also ultimately extracting the contributions that are sensitive to y_{c}.

2.3.2 The κ-framework

The κ-framework [20] is a commonly used framework to parametrise modifications to the couplings between the Higgs boson and other particles with respect to the expected SM values. It introduces the coupling modifier for the charm-Yukawa coupling as

$$\kappa_c = \frac{y_c}{y_c^{\mathrm{SM}}}, \tag{2.19}$$

where y_c is the measured Yukawa coupling and y_c^{SM} is the expected Yukawa coupling of the SM, calculated from the known charm quark mass. Accordingly, modifications to the Yukawa coupling of the charm quark are parametrised in this way as deviations from $\kappa_{\mathrm{c}} = 1$. However, y_{c} is not a quantity that can be measured directly. Instead, a signal strength measurement μ_{if} relative to the SM expectation, where i represents the production process and f represents the

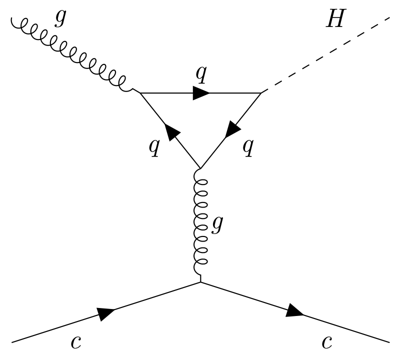


Figure 2.3: A Feynman diagram representing contributions to the cH process that are not sensitive to y_c, given the absence of an interaction between the Higgs boson and the charm quark. The quark loop seen in the middle is strongly dominated by top quarks, given their strong coupling to the Higgs boson in comparison to quarks of other flavours. Accordingly, the contribution from charm quarks in this loop is negligible. The corresponding diagrams with anti-particles are implied.

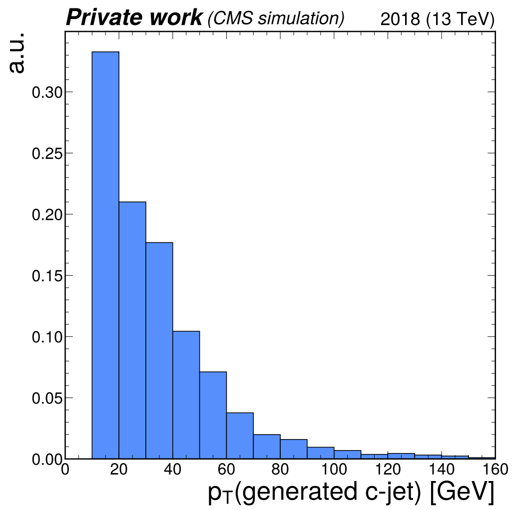


Figure 2.4: The transverse momentum of the generator level, charm quark-initiated jet, which is generated by a Madgraph5_aMC@NLO [19] simulation of the cH process within the CMS detector. This typically takes on small values, as the observed spectrum falls steeply with increasing transverse momentum of the jet. Note that the cut-off that can be seen at 10 GeV is not a physics effect. Rather, the reason for this is that at very low transverse momenta, jets cannot be well-reconstructed experimentally and as such, a minimum transverse momentum requirement is applied during simulation.

decay process, is made. As a result, a measurement of μ_{if} must be converted into an interpretation of κ_{c}. This is a step that contains some finer subtleties.

As described in [8], the rate of a Higgs production and decay process in relation to the expected SM signal (i.e. a signal strength) may be written as

$$\mu_{if} = \frac{\sigma_i \mathcal{B}_f}{(\sigma_i \mathcal{B}_f)^{\mathrm{SM}}}, \tag{2.20}$$

where σ_i is the production cross section in a given channel i and $\mathcal{B}_f$ is the decay branching ratio in a given channel f. This can alternatively be expressed as

$$\sigma_i \mathcal{B}_f = \kappa_{r,i}^2 \sigma_i^{\mathrm{SM}} \frac{\kappa_f^2 \Gamma_f^{\mathrm{SM}}}{\Gamma_{\mathrm{H}}}, \tag{2.21}$$

where modifications with respect to the SM expectation are introduced via the modifiers $\kappa_{r,i}$ and κ_f. The former modifies the SM production cross section σ_i^{SM}, while the latter modifies the partial SM decay width Γ_f^{SF} in the given decay mode. These decay widths can be understood as a quantity that is inversely proportional to a particle's mean lifetime, which is dependent on the decay mode. As a result, they can be used to characterise the branching ratios with which a particle decays. The denominator Γ_{H}, represents the total, summed decay width and can be written as

$$\begin{aligned}
\Gamma_{\mathrm{H}} =& \Gamma_{\mathrm{H}}^{\mathrm{SM}} (\kappa_{\mathrm{b}}^2 \mathcal{B}_{\mathrm{bb}}^{\mathrm{SM}} + \kappa_{\mathrm{W}}^2 \mathcal{B}_{\mathrm{WW}}^{\mathrm{SM}} + \kappa_{\mathrm{g}}^2 \mathcal{B}_{\mathrm{gg}}^{\mathrm{SM}} + \kappa_{\tau}^2 \mathcal{B}_{\tau\tau}^{\mathrm{SM}} + \kappa_{\mathrm{Z}}^2 \mathcal{B}_{\mathrm{ZZ}}^{\mathrm{SM}} + \kappa_{\mathrm{c}}^2 \mathcal{B}_{\mathrm{cc}}^{\mathrm{SM}} \\
&+ \kappa_{\gamma}^2 \mathcal{B}_{\gamma\gamma}^{\mathrm{SM}} + \kappa_{\mathrm{Z}\gamma}^2 \mathcal{B}_{\mathrm{Z}\gamma}^{\mathrm{SM}} + \kappa_{\mathrm{s}}^2 \mathcal{B}_{\mathrm{ss}}^{\mathrm{SM}} + \kappa_{\mu}^2 \mathcal{B}_{\mu\mu}^{\mathrm{SM}}) \qquad (2.22) \\
:=& \Gamma_{\mathrm{H}}^{\mathrm{SM}} \kappa_{\mathrm{H}}^2 \qquad (2.23)
\end{aligned}$$

Here, $\Gamma_{\mathrm{H}}^{\mathrm{SM}}$ is the total decay width of the Higgs boson in the SM and the $\mathcal{B}_f^{\mathrm{SM}}$ are the branching ratios of the possible decay modes. Note that the coupling of the Higgs boson to a pair of gluons or photons, which are induced via a loop similar to that shown in Figure 2.3, are included as independent quantities. Again, each partial width contribution to the total width comes with a modifier κ_i. Substituting these expressions into Eq. (2.20), the general rate modifier may be rewritten as

$$\mu_{if} = \frac{\kappa_{r,i}^2 \kappa_f^2}{\kappa_{\mathrm{H}}^2}. \tag{2.24}$$

Now, assuming that only modifications to the charm-Yukawa coupling play a role in Higgs boson production, and no modifications are made to the decay mode (e.g. $\mathrm{H} \to \mathrm{ZZ} \to 4\mu$), Eq. (2.24) becomes

$$\mu_{if} = \frac{\kappa_{\mathrm{c}}^2}{\kappa_{\mathrm{H}}^2} \tag{2.25}$$

for a signal strength measurement of the $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)$ process component that is sensitive to κ_{c}. This component is henceforth denoted as $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)_{\kappa_c}$ and is considered to be the process of interest.

Using the flat direction approach discussed in [8] and [21], a simplification of κ_{H} can be introduced. This approach is based on the finding that, when performing fits to existing Higgs boson production and decay rates, increases in the Yukawa couplings of light quarks (including the charm quark) can be compensated by increases in the couplings of the gauge bosons and heavy fermions. This is referred to as a "flat direction" in the fit, where observed Higgs boson production and decay rates can be modeled equally well for any value of κ_c by a respective scaling of all other processes. The individual modifiers in the sum of Eq. (2.22), with the exception of κ_c, as well as the modifiers in the nominator of Eq. (2.24), can thus be replaced with a single modifier κ. This allows Eq. (2.24) to be expressed as

$$\mu_{if} = \frac{\kappa^4}{\kappa^2(1-\mathcal{B}_\mathrm{cc}^\mathrm{SM}) + \kappa_\mathrm{c}^2\mathcal{B}_\mathrm{cc}^\mathrm{SM}}, \tag{2.26}$$

for the described scaling of Higgs production and decay with κ. This expression, for some measured μ, has a solution for κ given by

$$\kappa^2 = \frac{(1-\mathcal{B}_\mathrm{cc}^\mathrm{SM})\mu}{2} + \frac{\sqrt{(1-\mathcal{B}_\mathrm{cc}^\mathrm{SM})^2\mu^2 + 4\mu\mathcal{B}_\mathrm{cc}^\mathrm{SM}\kappa_\mathrm{c}^2}}{2}. \tag{2.27}$$

Here, the expected SM decay width $\mathcal{B}_\mathrm{cc}^\mathrm{SM} = 0.03$ can be substituted. Additionally, the fact that observed Higgs boson rates have been well measured to be close to their expected values (see e.g. [22]) can be reflected by setting $\mu \approx 1$. As a result, only a dependence on κ_c remains in the expression. Thus, by replacing κ_H in Eq. (2.25) with Eq. (2.27), a final expression relating a measured signal strength of the $\mathrm{cH(ZZ \to 4\mu)}_{\kappa_c}$ process to κ_c is obtained, given by

$$\mu_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H\to ZZ})_{\kappa_c}} = \frac{2\kappa_\mathrm{c}^2}{0.97 + \sqrt{(0.97)^2 + 4\cdot 0.03\kappa_\mathrm{c}^2}}. \tag{2.28}$$

Rearranging for κ_c gives

$$\kappa_\mathrm{c} = \pm\sqrt{0.97\mu_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H\to ZZ})_{\kappa_c}} + 0.03\mu^2_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H\to ZZ})_{\kappa_c}}}. \tag{2.29}$$

Effectively, this approach to interpreting κ_c from a signal strength measurement $\mu_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H\to ZZ})_{\kappa_c}}$ ensures compatibility with existing Higgs boson rate measurements. This is achieved by connecting the scaling behaviour of the κ_c-sensitive cH process with respect to κ_c , to the scaling of the total decay width of the Higgs boson. This is relevant as a non-unity value of κ_c leads to modifications of the Higgs boson partial decay widths, in turn leading to a slight modification of the quadratic relationship one might naively expect between κ_c and $\mu_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H\to ZZ})_{\kappa_c}}$.

2.4 The cH process in an EFT context

The cH process may also be interpreted in the context of Standard Model Effective Field Theory (SMEFT). In SMEFT, potential effects from physics processes not described by the SM (commonly referred to as beyond-the-SM or BSM physics) are parametrised in a mostly model-independent way. Specifically, the SMEFT framework can be used at colliders with a characteristic energy scale E to describe the effects of processes with a characteristic energy scale above E. This concept is illustrated in Figure 2.5.

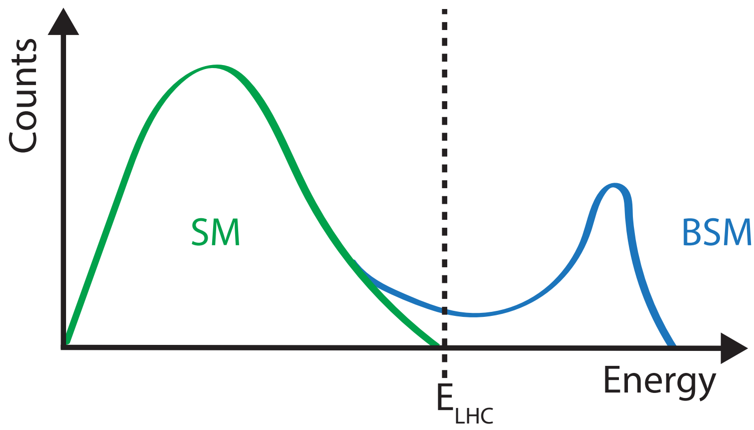


Figure 2.5: Illustration of how the presence of BSM physics, which is primarily visible beyond the reach of current collider energies (e.g. E_{LHC}), can lead to subtle modifications of SM observables. These effects can be parametrised by SMEFT.

Formally, SMEFT is a collection of all possible combinations of field interactions that obey the gauge invariance conditions of the SM. Generically, this can be expressed as an expansion in the energy scale of the new physics Λ

$$\mathcal{L}_{\mathrm{SMEFT}} = \mathcal{L}_{\mathrm{SM}} + \sum_{d>4} \sum_{i} \frac{C_i}{\Lambda^{d-4}} \hat{O}_i^d \tag{2.30}$$

where $\mathcal{L}_{\mathrm{SM}}$ is the SM Lagrangian density, O_i^d denotes a particular operator (i.e. a particular combination of fields) with a dimensionless coupling coefficient C_i (typically referred to as Wilson coefficients) and d denotes the dimension of the operator. The dimensionality is derived through a dimensional analysis of a Lagrangian and its fields, where energy dimensions of terms may be deduced from the requirement that the action

$$\mathrm{S} = \int \mathcal{L}\, d^4x \tag{2.31}$$

remains dimensionless. Accordingly, $\mathcal{L}_{\mathrm{SM}}$ is of energy dimension four. Since the SMEFT operators O_i^d all have energy dimensions higher than four and Λ comes with energy dimension one, the terms in the sum of Eq. (2.30) are scaled with $1/\Lambda^{d-4}$ to ensure the combination also has an energy dimension of four.

Typically, operators in SMEFT are grouped by their energy dimension. In $d = 5$, only one possible operator exists that violates lepton number [23] and is not relevant in this work. In $d = 6$ however, a plethora of valid operators exist. These are commonly described in the Warsaw basis [24], which is representation of $d = 5$ and $d = 6$ operators. In this basis, a total of 59 independent operators are found. Since $d = 7$ operators again violate lepton number and each additional dimension adds a suppressive factor of Λ^{-1}, a simplified SMEFT schema is commonly used in which only the contribution of $d = 6$ operators is considered in the expansion. Thus, Eq. (2.30) simplifies to

$$\mathcal{L}_{\mathrm{SMEFT}} = \mathcal{L}_{\mathrm{SM}} + \sum_i \frac{C_i}{\Lambda^2} \hat{O}_i^{(6)} \tag{2.32}$$

A good overview of SMEFT can be found in [25].

2.4.1 The chromomagnetic dipole operator

A particular operator relevant to this work is referred to as the chromomagnetic dipole (CMD) operator $\hat{O}_{qG}$, which introduces a four-particle interaction involving a Higgs boson, a gluon and a pair of quarks. For the charm quark, the CMD operator is written as

$$\hat{O}_{cG} = (\overline{q}_{2,L} \sigma^{\mu\nu} T^a c) \tilde{\phi} G^a_{\mu\nu}. \tag{2.33}$$

Here, $\overline{q}_{2,L}$ is the second generation, left-handed quark doublet, $\sigma^{\mu\nu} = i[\gamma_\mu, \gamma_\nu]/2$ with the Dirac matrices γ_μ , T^a are the generators of the SU(3) symmetry group, c is the right-handed charm quark field, $\tilde{\phi}$ is the adjoint Higgs doublet and $G^a_{\mu\nu}$ is the field strength tensor of the strong interaction. This operator may be uniquely bounded with the cH process due its unique chiral structure, which mixes left and right-handed spinors. Such a structure is otherwise only found in the Yukawa and quark-Higgs boson interaction terms of the SM.

To better understand this, it is worth considering other processes such as inclusive Higgs boson production, which have been successfully leveraged to set strong constraints on the top quark CMD operator $\hat{O}_{tG}$ [26]. Typically, the strategy that is used to probe even small Wilson coefficients, such as C_{tG}, is to exploit interference of the relevant (small) SMEFT contribution with a larger SM contribution. Though the pure SMEFT contribution itself may be small and experimentally negligible due to limited analysis sensitivity, the much larger contribution of the SM process it interferes with can result in a non-negligible interference effect with respect to the SM process. These interference effects occur when two processes are mediated in different ways, though they have the same initial and final states. However, the chiral structure of the CMD operator influences the effectiveness of this strategy. Since the $\hat{O}_{qG}$ operator effectively flips the chirality of the incoming and outgoing quarks, an additional *chirality flip* must be inserted for the SMEFT contribution to produce interference with the SM process. This is visualised in Figure 2.6. Such a chirality flip is proportional to the mass m_q of the respective quark. As a result, the interference contribution for a much lighter quark is significantly suppressed in comparison to the top quark, as also argued for the bottom quark in [27]. Effectively, this means that the processes that prove effective in targeting $\hat{O}_{tG}$ due to the large mass of the top quark, are thus much less sensitive to $\hat{O}_{cG}$. However, since the cH process itself contains the chirality flipping quark-Higgs boson vertex, interference terms between the EFT and SM contributions do not suffer from the above described effect. Furthermore, due to the very low expected cross section of the cH process, the contributions originating purely from $\hat{O}_{cG}$ may be comparatively large with respect to the SM cH process, even at small values of C_{cG}. Accordingly, the cH process may be an excellent target to constrain $\hat{O}_{cG}$.

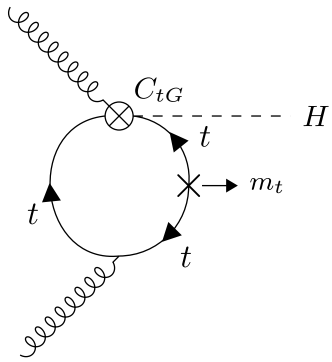


Figure 2.6: A modification of the gluon fusion process with a top quark loop by including the vertex introduced by the top quark CMD operator. Note that the arrows indicate chirality and not momentum flow. A chirality flip, denoted by the cross, proportional to the top quark mass m_t is required for the inclusion of the top quark CMD vertex.

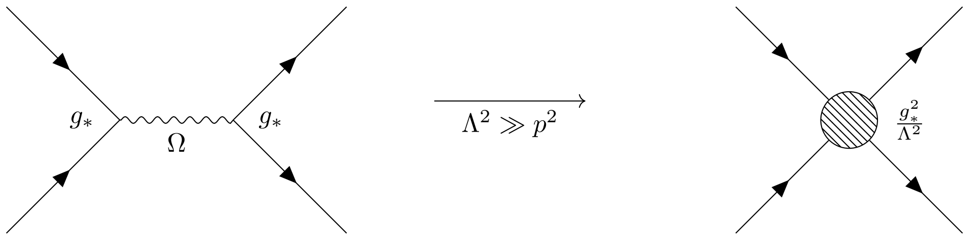


Figure 2.7: Feynman diagrams depicting a resonant process in which the new mediator particle Ω is created (left) and the approximate description of this in an EFT, where the diagram is reduced to a four-point interaction.

2.4.2 Validity of an EFT

In addition to EFT terms needing to satisfy the gauge invariance conditions of the SM, two additional key validity conditions are typically required of an EFT. The first is related to the fact that in an EFT, the particle nature of e.g. new, heavy mediator particles is simplified into the introduction of a new effective vertex. For example, a 2→2 particle resonant scattering via a new heavy mediator particle Ω associated with a newly introduced coupling constant g_*, is simplified via the introduction of a four-point interaction, as visualised in Figure 2.7. This corresponds to a first-order approximation of the new particle's mediator as

$$\frac{g_*}{p^2 - m_\Omega} \quad \xrightarrow[m_\Omega^2 \gg p^2]{} \quad -\frac{g_*}{m_\Omega}\left(1 + \frac{p^2}{m_\Omega^2} + \frac{p^4}{m_\Omega^4} + ...\right) \approx -\frac{g_*}{m_\Omega} \tag{2.34}$$

For the EFT description of this simplification to be valid, the energy involved in processes containing the effective vertex introduced by the relevant operator must thus lie well below m_Ω, which here represents the previously introduced new physics scale Λ. Practically, this can be achieved by placing an upper limit M_{cut} on the total energy that is considered in measurements of such processes. The requirement can be expressed as

$$M_{cut} < \Lambda. \tag{2.35}$$

A good estimator of M_{cut} is the invariant mass of the final state particles of a process. In case of the cH process, the invariant mass of the Higgs boson and jet system is a natural choice.

The second condition that must be met is related to the perturbativity of the theory. Concretely, this means that higher dimensional operators should constitute increasingly smaller corrections, so that the sum of operator contributions converges. In the case of this work where only d=6 operators are considered, this means ensuring contributions from d=8 operators and higher are sufficiently small. While this cannot be determined with certainty without explicit knowledge of the underlying theory the EFT is estimating, a popular choice is to require that at most $g_* \sim 4\pi$ [28].

These two conditions may be combined into a single, simultaneous requirement. In [28], an effective Lagrangian (ignoring relevant indices for simplicity) of the general form

$$\mathcal{L}_{\text{eff}} = \frac{\Lambda^4}{g_*^2}\mathcal{L}\left(\frac{D_\mu}{\Lambda}, \frac{g_h\phi}{\Lambda}, \frac{g_{\psi_{L,R}}\psi_{L,R}}{\Lambda^{3/2}}, \frac{gF_{\mu\nu}}{\Lambda^2}\right) \tag{2.36}$$

is obtained when a BSM coupling g_* is introduced. This provides a prescription for the powers of the couplings and Λ that are associated with the SM fields ϕ, ψ and $F_{\mu\nu}$, and the covariant derivate D_μ. Here, g represents the unaltered gauge field couplings of the SM, while $g_{\psi_{L,R}}$ and g_h represent the coupling of SM fermion and the Higgs doublet to the BSM theory. In a single BSM coupling scenario, this simplifies to $g_{\psi_{L,R}} = g_h = g_*$. Applying this prescription to the CMD operator gives

$$\hat{O}_{cG} \longrightarrow \frac{\Lambda^4}{g_\star^2}\left[\left(\frac{g_\star\psi_{L,R}}{\Lambda^{3/2}}\right)\cdot\left(\frac{g_\star\psi_{L,R}}{\Lambda^{3/2}}\right)\cdot\left(\frac{g_\star\phi}{\Lambda}\right)\cdot\left(\frac{g_sG}{\Lambda^2}\right)\right] \tag{2.37}$$

$$= \frac{g_\star g_s}{\Lambda^2}\left(\psi_{L,R}\cdot\psi_{L,R}\cdot\phi\cdot G\right) . \tag{2.38}$$

Reading off from Eq. (2.38), one can see that the coupling of the CMD operator is given by $g_\star g_s/\Lambda^2$. Comparing to Eq. (2.30) thus reveals that the CMD Wilson coefficient is given by $C_{cG} = g_* g_s$. By requiring the first discussed validity condition, the relation

$$\frac{C_{cG}}{\Lambda^2} < \frac{g_* g_s}{M_{cut}^2} \tag{2.39}$$

is obtained. Since both C_{cG} and Λ are a priori unknown, we can redefine $\tilde{C}_{cG} = \frac{C_{cG}}{\Lambda^2}$. With this redefinition, and by setting $g_* \sim 4\pi$, the expression

$$\frac{\left|\tilde{C}_{cG}\right| M_{cut^2}}{4\pi g_s} < 1 \tag{2.40}$$

can be used to define a plane in $\tilde{C}_{cG}$ and M_{cut} that satisfies both of the previously discussed conditions. This plane is visualised in Figure 2.8, showing regions in which the validity condition is (loosely) satisfied and in which it is not satisfied. Constraints placed on the CMD operator for a given M_{cut} value can be cross-checked against this to ensure the validity of the constraint.

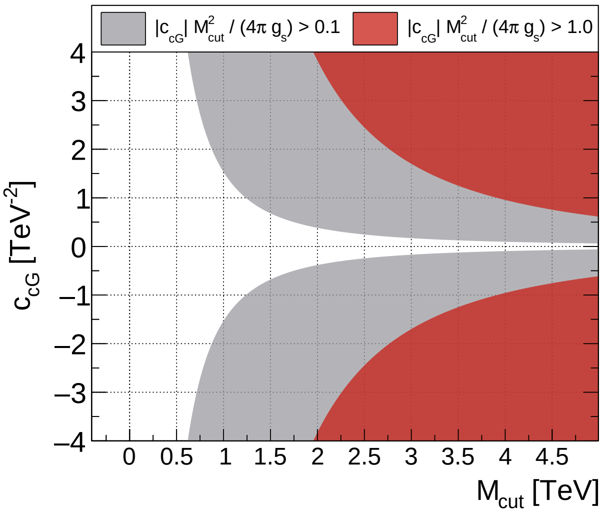


Figure 2.8: Diagram showing the plane spanned by $\tilde{C}_{cG}$ (here $\mathrm{c_{cG}}$) and M_{cut} for the CMD operator. The white region shows the region in which the perturbativity requirement is well satisfied. The grey region represents the transition from the white region to the red region, in which the perturbativity requirement is no longer valid. Figure taken from [1]

2.5 State of the art on κ_c

Given the previously described status of the charm-Yukawa coupling in our current, experimental picture of the Higgs boson, analyses targeting the charm-Yukawa coupling are becoming of increased interest. This pertains particularly to the cH process, since it is considered a relatively novel process. In reflection of this, recent, first-time analyses of the cH($\gamma\gamma$) and cH(WW) channels using Run-2 data recorded by the CMS experiment have been published. The corresponding results of both analyses are interpreted in terms of the charm-Yukawa coupling modifier κ_c and upper limits on κ_c at 95% CL are reported at $|\kappa_\mathrm{c}| = 38.1$ for cH($\gamma\gamma$) [10] and $|\kappa_\mathrm{c}| = 211$ for cH(WW) [11], respectively. The ATLAS collaboration has similarly recently published an analysis of the cH($\gamma\gamma$) channel using Run-2 data [29]. However, only inclusive cH production is considered in this analysis. A measured inclusive cH cross section of 5.3 pb is reported and no direct interpretation with respect to κ_c is made.

While the methods used to reconstruct Higgs boson decays in the cH($\gamma\gamma$), cH(WW) and also H(ZZ$\rightarrow 4\ell$) channels are well established [30][31][32], procedures with which to identify the associated charm quark-initiated jet of the cH process may yet be optimised. In both the existing cH($\gamma\gamma$) and cH(WW) analyses, this task is accomplished by performing a comparatively simple selection of jet candidates based on fixed discriminator thresholds that are obtained from jet flavour-tagging algorithms (these are introduced in Section 3.3). Given that the identification of charm quark-initiated jets via such methods remains to be relatively inefficient, the development of alternative methods is of significant interest and are explored in this work. A potential ansatz to this is performing a kinematics-based jet association that is agnostic towards jet flavour. Such an algorithm is devised in this work and is detailed in Subsection 4.1.2. This, in turn, also enables the use of the full spectrum of the jet flavour-tagging discriminator of the selected jet in a final statistical evaluation, as is described in Section 4.5. Such an alternative approach to identifying an associated charm-quark initiated jet could improve experimental sensitivity to the cH process. The reason for this is that the previous cH($\gamma\gamma$) and cH(WW) analysis efforts report large statistical uncertainties, as is expected when targeting processes with very low cross sections such as the cH process. As a result, approaches with a more inclusive jet selection method may improve analysis sensitivity, given that cH process contributions in which the associated charm quark-initiated jet cannot be well identified by jet-flavour identification methods are thus no longer discarded and may still contribute in a final statistical evaluation.

Also of note are the significant theoretical uncertainties associated with the current modelling of the cH process, which affects all analyses of the cH process independently of the decay mode of the Higgs boson. This includes an uncertainty originating from the modelling of the charm quark mass when the charm-quark originates from the proton, as well as more generally the modelling of the emission of massive quarks alongside a Higgs boson. The former is captured via a so-called *flavour scheme uncertainty*, as detailed in Subsection 4.2.1, while the latter is described in Subsection 4.2.2. These uncertainties are found to be leading systematic uncertainty sources in [10] and [11], and thus are similarly expected to significantly impact an analysis of the cH(ZZ$\rightarrow 4\mu$) channel. Improvements to these aspects of the process modelling thus remain to be of significant interest and potential approaches are discussed in Subsection 4.5.5.

It is also possible to constrain κ_c using more indirect methods. Significant deviations of the Yukawa couplings of the light flavour quarks (up, down, strange, charm) for example can lead to deviations of the inclusive Higgs production cross section. Accordingly, such potential deviations have previously been explored by the CMS collaboration in an analysis of inclusive Higgs boson production in the H(ZZ$\rightarrow 4\ell$) channel [33] and a constraint of $\kappa_c \in [-3.8, 3.2]$ is reported at a 95% CL. However, it should be noted that the interpretation with respect to κ_c that is made in such a way requires somewhat stronger signal modelling assumptions, since Higgs production can be modified in a variety of ways, including the modification of the Yukawa couplings of all other quark flavours. In practice, this means the Yukawa couplings of all other quarks must be set explicitly by hand in a fit, or simultaneously be allowed to take on best-fit values, and the inclusive production mechanisms must be assumed to be otherwise SM-like. Other, similar analyses have also been proposed, such as in [34], where the differential transverse momentum of the Higgs boson in inclusive Higgs boson production is also found to be sensitive to κ_c at a level similar to the sensitivity obtained in [33].

The most stringent, direct constraints to date on κ_c however currently originate from a recent combination of VH($c\bar{c}$) and $t\bar{t}$H($c\bar{c}$) channels by the CMS collaboration. Here, an observed 95% CL upper limit of $|\kappa_c| = 3.5$ is reported [9]. The ATLAS collaboration is likewise able to set a similar 95% CL upper limit of $|\kappa_c| = 4.2$, in an analysis of the VH($c\bar{c}$) channel [35]. The result of a first-time analysis of inclusive H($c\bar{c}$) decays by the LHCb collaboration also looks promising [36]. In this analysis, an upper limit of $|\kappa_c| = 43$ is derived, though using a much smaller dataset, indicating that such an approach could also yield significant sensitivity to κ_c. The comparatively strong constraints on κ_c that are derived in these analyses decays may originate from the fact that the final state particles, particularly the charm quark-initiated jets originating from H($c\bar{c}$) decays are associated with significantly higher transverse momenta than the associated charm quark-initiated jet of the cH process. As such, the reconstruction of these processes is less prone to detector acceptance effects associated with jets at low transverse momentum, as is the case for cH analyses. Though the respective VH($c\bar{c}$) and $t\bar{t}$H($c\bar{c}$) analyses must deal with significantly larger QCD backgrounds due to the presence of two charm quark-initiated jets in the signal signature, this trade-off may still be favourable due to the low signal yields associated with all charm-Yukawa sensitive processes. Furthermore, analyses of the $\mathrm{VH(c\bar{c})}$ and $\mathrm{t\bar{t}H(c\bar{c})}$ processes are not constrained by the significant flavour scheme uncertainty that affects the cH process.

Though the constraints derived in the aforementioned $\mathrm{cH}(\gamma\gamma)$ and cH(WW) analyses are significantly weaker in comparison to the channels involving direct $\mathrm{H(c\bar{c})}$ decays, or even the discussed indirect measurement methods, further analysis of the cH process nonetheless provide important complementary results. Moreover, analyses of the cH process may still contribute significantly in a statistical combination. This is especially important given that even at the High-Luminosity LHC (HL-LHC), the projected sensitivity to the charm-Yukawa coupling in individual channels is only starting to approach one [37]. Accordingly, the development and improvement of well-understood analysis methods, together with a combination of analysis channels, are key facets to exceeding the expectations set by such a projection. By performing a first-time analysis of the previously unexplored $\mathrm{cH(ZZ \to 4\mu)}$ channel, as well as exploring novel analysis methods that may also be applicable in other channels of the cH process, this work intends to contribute to making such a future measurement of the charm-Yukawa coupling possible. Furthermore, SMEFT interpretations of the cH process, which may be associated with modifications of κ_{c}, remain entirely unexplored in an experimental context. As such, this work seeks to make a first-time foray into such scenarios.

Chapter 3

The CMS experiment at the LHC

The Compact Muon Solenoid (CMS) detector [38] is a large, general purpose particle detector located at the Large Hadron Collider (LHC)[39] accelerator in Geneva, Switzerland. Run by the European Organisation for Nuclear Research (CERN), the LHC's ring spans a circumference of 27 km, making it the largest particle accelerator in the world. In their circular trajectory through the beam pipe, collimated bunches of $\sim 10^{11}$ protons are accelerated in both directions of the ring. At four different collision points, the trajectories of these proton bunches are crossed such that highly energetic pp collisions are produced. The CMS detector is located at one of these collision points. A sketch of the LHC accelerator complex can be seen in Figure 3.1. A detector such as the CMS detector effectively acts as a camera taking very complex snapshot of each collision. During Run-2 (2016 – 2018) of the LHC, approximately 30 protons collide on average per bunch crossing with a centre-of-mass energy of $\sqrt{s} = 13$ TeV. These collisions produce a plethora of particles, many of which decay to sets of particles of varying multiplicities themselves. As such, these collisions produce a complicated and varied phenomenology that requires a complex machine such at the CMS detector to fully capture. By recording the information from many millions of collisions, a multitude of different statistical analyses may be performed. This includes analyses of the Higgs boson and its properties, such as the charm-Yukawa coupling. To this end, this chapter gives an overview of the CMS detector and its subsystems, as well as the techniques used to reconstruct individual pp collisions.

Starting in Section 3.1, the individual components of the CMS detector are detailed, followed by a description of the reconstruction techniques that use the data obtained from these components in Section 3.2. Subsequently, the identification of charm quark-induced jets is specifically discussed in Section 3.3. Finally, an overview of techniques used to simulate pp collisions in the CMS detector is given in Section 3.5.

3.1 The CMS detector

The CMS detector is designed to be able to detect a wide range of signatures and is built from a set of complementary subdetectors. An overview of the detector may be seen in Figure 3.2. By combining data from these subdetectors, a comprehensive reconstruction of individual proton-proton collisions, commonly referred to as an *event*, may be made. The role and functioning of the individual subdetectors is covered in this section. While several of the detector components have undergone changes for the current Run-3 (2022 – 2026) of the LHC[42], the configuration relevant to this work is that of Run-2.

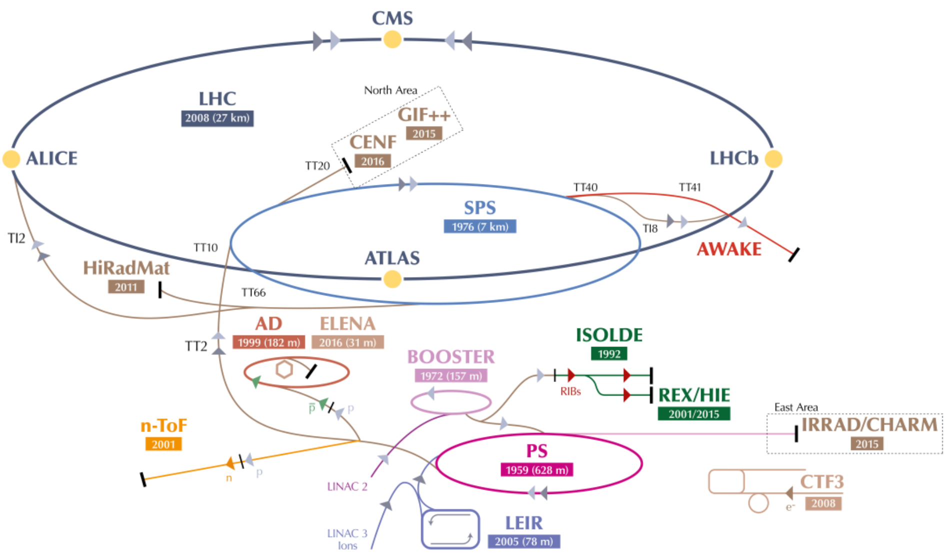


Figure 3.1: An overview of the LHC accelerator complex. Before entering the final LHC ring, particles must pass through a number of increasingly powerful set of accelerators. Figure taken from [40].

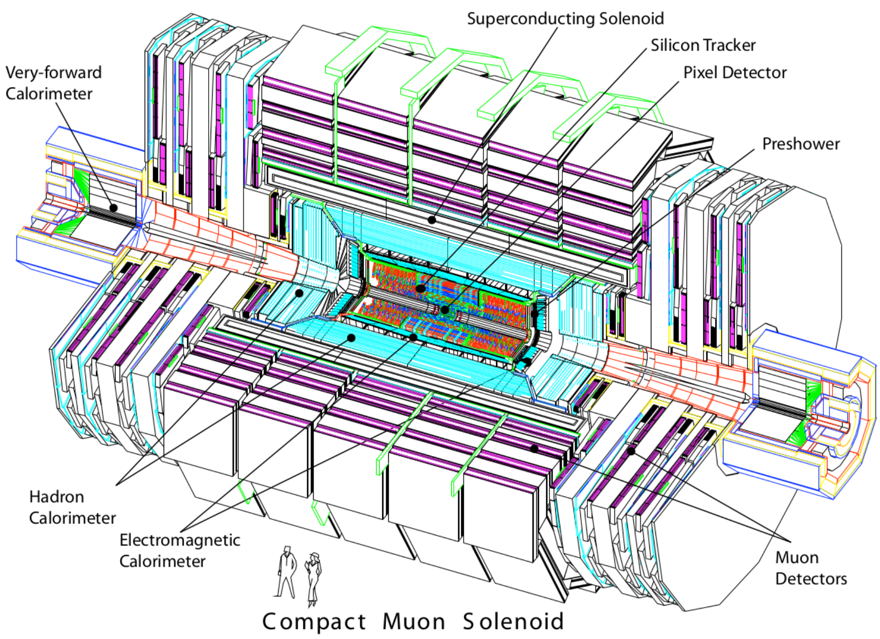


Figure 3.2: An overview of the CMS detector, showing the individual subsystems it is composed of. Figure taken from [41].

3.1.1 The CMS coordinate system

Due to the cylindrical nature of the CMS detector, using cylindrical coordinates to describe positions within the detector is a natural choice. Thus, the z coordinate describes the position along the beam pipe, r the radius, and ϕ the azimuthal angle, where the pp collision point is taken as the coordinate system's centre. Trajectories of particles with energy E that traverse into the plane perpendicular to z, may be described by the rapidity

$$y = \ln\sqrt{\frac{E + p_z c}{E - p_z c}}. \tag{3.1}$$

The behaviour of the rapidity is dictated by the momentum in the z-direction, p_z. When p_z is small, the rapidity becomes close to zero, while the rapidity tends to $\pm\infty$ for large p_z. However, this requires knowledge of both E and p_z, which can be difficult to measure due to the composite nature of protons. By assuming the particle is ultra-relativistic, as is typically the case at the LHC, it is possible to simplify this description and introduce the pseudorapidity

$$\eta = \ln\left(\tan\left(\frac{\theta}{2}\right)\right), \tag{3.2}$$

which is dependent solely on θ, the polar angle. A convenient feature of the (pseudo)rapidity is that differences of (pseudo)rapidity are Lorentz invariant and thus not dependent on the initial longitudinal boost of the proton-proton system, which is a priori not known due to the varying momenta fractions of its constituents. Together with the particle's transverse (to the beam axis) momentum p_T and mass m, a particles four-vector may be described by

$$p = \begin{pmatrix} m \\ p_T \\ \eta \\ \phi \end{pmatrix}. \tag{3.3}$$

Broadly speaking, the CMS detector may be split into two distinct regions for larger and smaller η, with a transition point that varies for each subdetector. The inner region or *barrel* consists of concentric layers around the beam pipe. The outer *endcap* region consists of two caps that close off the detector at either end. In this way, the CMS detector is designed for the best possible hermetic coverage around the collision point.

3.1.2 The silicon tracker

The silicon tracker [43] is the innermost system of the CMS detector and is situated closest to the beampipe. It is designed to track the trajectories of charged particles as they emerge from the collision point while producing minimal energy losses of the particles themselves. This subdetector is split into two main components, the pixel detector and silicon strip detector. A sketch of these components may be seen in Figure 3.3.

The pixel detector is situated directly around the beampipe and, as of 2017, consists of four circular layers of individual silicon pixels in the barrel region and three disk layers in the endcap

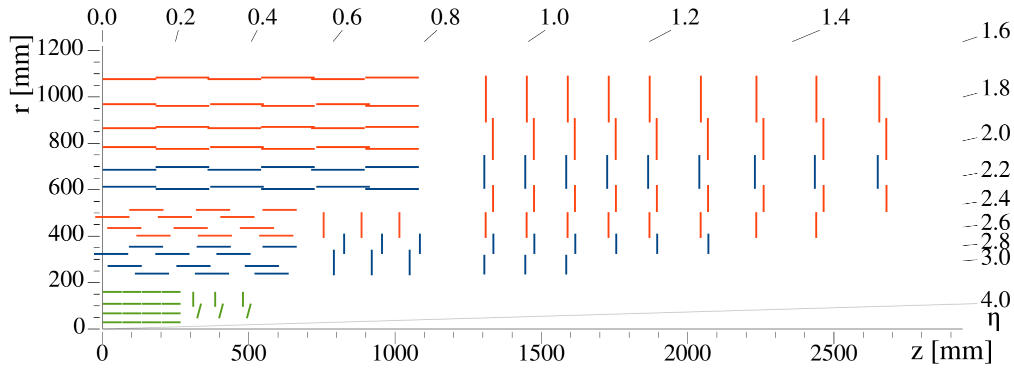


Figure 3.3: An overview of the CMS silicon tracker, shown in the r-z plane after its upgrade during Run-2. The pixel detector is denoted in green while the silicon strip detector is denoted in blue and orange. Figure taken from [43].

region. These consist of rectangular silicon sensors with a size of 100 x 150 μm^2. When a charged particle traverses through the active material of these sensors, an electrical signal is induced and recorded. This is typically referred to as a *hit*. The small pixel size allows for position measurements with a very high resolution at $\sim 10\mu$m in the $r\phi$ plane and $\sim 20\mu$m in the z direction [44]. An important feature of the pixel detector is its high radiation tolerance. This is necessary due to the close proximity of these modules to the beam pipe, where they must withstand being hit by the high particle flux produced by the multitude of pp collisions that the LHC delivers.

Following the pixel detector is the silicon strip detector. It is composed of silicon strips that increase in size at greater distances to the beam pipe. This is motivated by the fact that the particle flux that the strips are exposed to reduces with distance from the beam pipe. In the barrel region, this part of the detector consists of 10 layers of silicon strips, while in the endcap regions it consists of nine layers. The latter extendd the coverage of the tracking detector to $|\eta|= 2.5$.

The tracking system provides essential information for the reconstruction of events. As charged particles fly through the CMS detector, their trajectories are curved due to the magnetic field generated by the solenoid magnet (see Subsection 3.1.5). By measuring the curvature of these trajectories with this system, the transverse momentum p_T of particles can be reconstructed. Additionally, the tracker plays a key role in methods used to determine the nature of jets and the progenitor particles (i.e. quarks or gluons) from which these originate.

3.1.3 The electromagnetic calorimeter

The second innermost subsystem is the electromagnetic calorimeter (ECAL)[45][46], a sketch of which may be seen in Figure 3.4. It is designed to measure the energies of electromagnetic showers initiated by photons and electrons, with a coverage of up to $|\eta|$=3.0. The ECAL is a homogenous calorimeter, consisting of over 75,000 lead tungstate crystals. These crystals scintillate as charged particles pass through them. The produced photons are collected via photodiodes, thus generating an electrical signal. This signal may be evaluated to infer the energy that is deposited. Not only do the crystals scintillate but they are also extremely dense and, as a

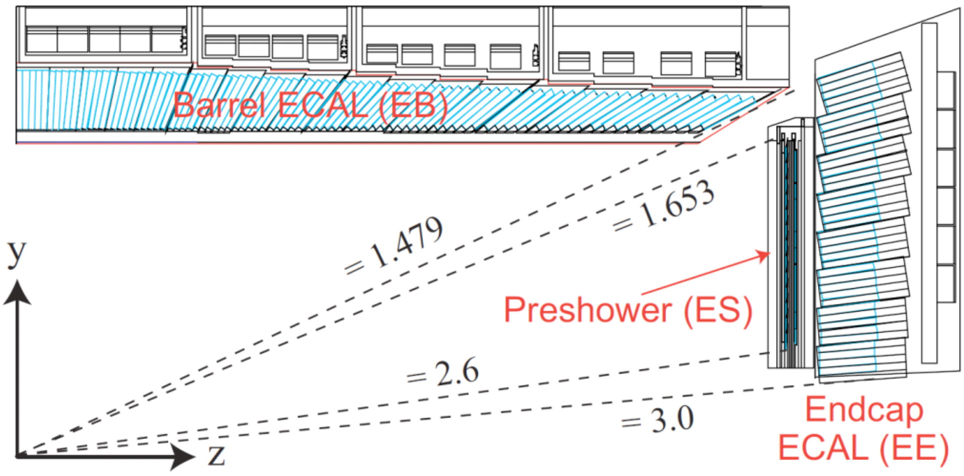


Figure 3.4: An overview of the CMS ECAL, shown in the $r(y)$-z plane. The dashed lines denote the coverage of the barrel and endcap ECAL region as well as the preshower detector. Figure taken from [47].

result, are very effective in absorbing the energy of incoming electrons and photons. This allows a very compact thickness of 23cm (22cm) in the barrel (endcap) region, which corresponds to ∼26 (∼25) radiation lengths. An additional component of the ECAL is the preshower detector. This consists of two layers of lead absorbers that are each followed by a plane of silicon strip sensors, and which helps to distinguish high energy photons from neutral pions. The latter decays into photon pairs and may mimic high energy photons in this part of the detector with an increased likelihood. The increased granularity of the preshower detector helps mitigate this effect. The energy resolution of the ECAL is $\sim$ 1–4%.

3.1.4 The hadronic calorimeter

Following the ECAL is the hadronic calorimeter (HCAL) [48]. It is designed to measure the presence and energy of hadrons, which typically traverse the ECAL with minor energy losses. It is the most hermetic part of the CMS detector with a coverage out to $|\eta|= 5.0$, in order to absorb almost all particle produced in the pp collision. The only exceptions to this are muons, which are particles that minimally deposit their energy, and neutrinos, which have an interaction probability so low that they cannot be measured with the CMS detector at all.

In contrast to the ECAL, the HCAL is a sampling calorimeter. This means layers of an absorber are interleaved with layers of a scintillator. Different materials are used in different parts of the calorimeter, which is split into the barrel ($|\eta| < 1.5$), endcap ($1.5 < |\eta| < 3.0$) and forward ($3.0 < |\eta| < 5.0$) regions. Since the HCAL component inside the magnet system does not sufficiently absorb all hadronic showers, the system also extends past the magnet. Due to the sampling nature of the calorimeter, a lower number of respective interaction lengths, and larger energy fluctuations in hadronic particle showers, the energy resolution of the HCAL is significantly worse than the ECAL. It lies in the order of 10–30%, with a dependence on the energy and pseudorapidity of the initiating particles.

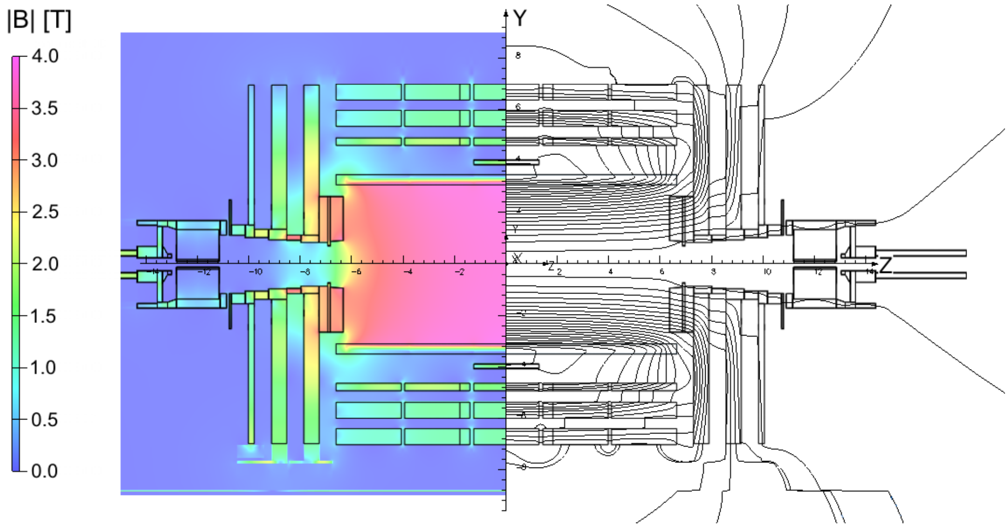


Figure 3.5: An overview of the magnetic flux (left) and magnetic field lines (right) inside the CMS detector, shown in the $r - z$ plane. Figure taken from [50].

3.1.5 The superconducting solenoid magnet

A key component of the CMS detector is the superconducting solenoid magnet [49]. It is responsible for maintaining a strong 3.8 T magnetic field that homogenously permeates the barrel of the detector. A measurement of the field strength can be seen in Figure 3.5. With its toroidal shape, the field is orientated along the z-axis and covers the 12.9 m long barrel region of the detector, curving the trajectories of charged particles emerging from the interaction point in the ϕ-direction. This allows for a measurement of the particles' transverse momentum, which together with the ϕ and η directions fully characterise a particle's momentum vector. The magnet itself is composed of niobium-titanum coils that are cooled to a temperature of 4.65 K, so that their superconducting properties take effect. The magnet is encased by a 12,000 t steel yoke that captures the magnetic field that is produced outside of the solenoid.

3.1.6 The muon chambers

The muon subdetector consists of a dedicated system of gaseous detectors [51][52], which are placed outside of the solenoid magnet. As suggested by the name of the CMS experiment, a strong focus is placed on the performance of this subdetector. The reason is that muons are often be produced in collisions that are of physics interest (such as in this work) and thus an emphasis is laid on detecting these with great efficiency. Due to muons being minimally ionising particles, they easily pass through the inner subdetector layers to reach the muon chambers, and information from the moun chambers as well as the tracker and calorimeters may be used to identify and reconstruct them.

Like the other subdetectors, the muon chambers are separated in a barrel ($|\eta| < 1.2$) and endcap ($1.2 < |\eta| < 2.4$) region, which are composed of drift tubes (DT) and cathode strip chambers (CSC), respectively. The drift tubes each consists of a gas volume containing a mixture of Argon and CO_2 in which a positively charged wire is stretched through the center. When charged particles such as muons traverse these tubes, the gas is ionised. Due to the positive charge of the wire,

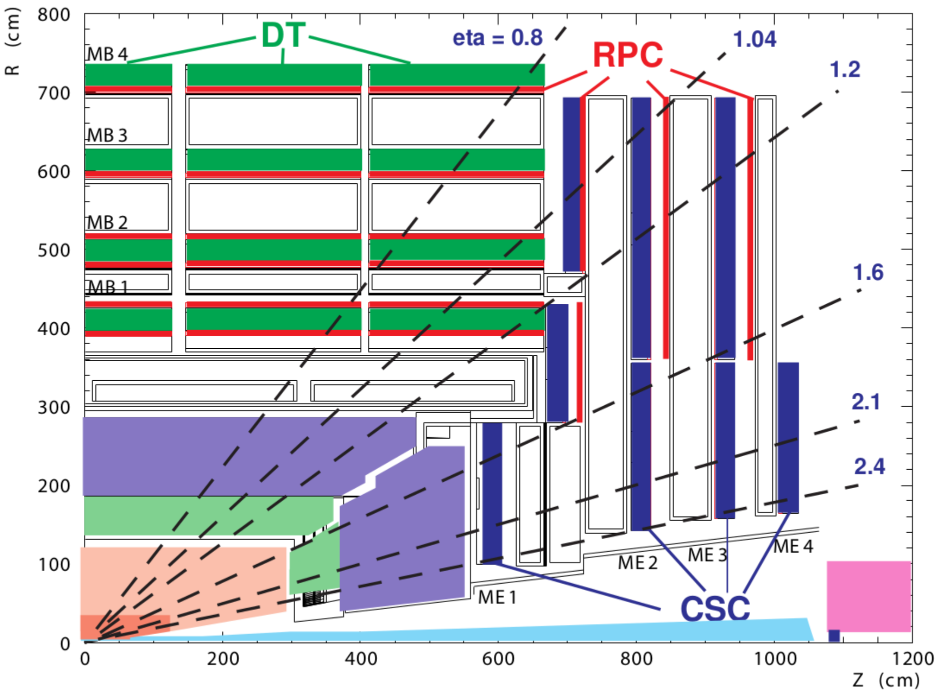


Figure 3.6: An overview of the CMS muon system, shown in the $r - z$ plane. Shown are the drift tube (DT), the cathode strip chambers (CSC) and resistive plate chambers (RPC). Figure taken from [41].

the resulting electrons drift towards the wire, producing an electrical signal. Thus the presence of muons may be determined by activation of the drift tubes. The cathode strip chambers on the other hand consist of layers of positively charged (anode) wires, which are arranged in a perpendicular fashion to a set of negatively charged (cathode) strips. Combining signals from both the wires and strips allows for a position measurement in both the r and ϕ direction. Both types of detector are supplemented by resistive plate chambers (RPC), which act as a trigger providing a precise timing resolution of ∼1ns. This makes it possible to unambiguously assign muons to individual bunch crossings. These consist of parallel, oppositely charged plastic plates that are coated with a conductive graphite layer and are contained in a gas volume. Ionisation of the gas due to the traversal of a charged particle thus leads to an electrical signal. An overview of the spatial arrangement of these systems can be seen in Figure 3.6. With this system, the bulk of muons may be measured with a precise momentum resolution of ∼1-2%.

3.1.7 The triggering system

The triggering system is an essential component in managing the data output of the CMS detector [53]. With a nominal collision rate of ∼40 MHz, the data rate the CMS detector provides is close to 40 TB/s. Not only is the storage of such a quantity of data unfeasible but a significant portion consists of low-energy scattering events that are not of interest to analysts. As such, the triggering system is implemented to extract a subset of events that are of physics interest.

The trigger systems is composed of two subsystems. The first the so-called level one (L1) trigger. This is a very fast hardware-based system that reduces the event rate to ~100 kHz by evaluating the presence of e.g. energetic muons or other interesting signatures, such as large energy deposits in the calorimeters in an event. A total time of 3.2μs is allocated to decide whether an event should be kept. Subtracting the time required for signal propagation in the detector, the L1 system must make a decision within 1μs. From the L1 trigger, the events are passed to a software based high-level trigger (HLT) system. This is composed of several thousand CPU cores that perform a simple reconstruction of the signatures of an event, to make a decision whether the event should be stored or not. Since different analyses are interested in different signatures, a set of trigger paths are defined where each path represents a different signature. As such, only one such path is required to be satisfied for an event to pass the HLT. Since the HLT is software-based, the trigger paths may be continuously updated depending on analysis needs. After the HLT, the event rate is reduced to ~100 Hz and the passing events are permanently stored on disks and ultimately, magnetic tape.

The HLT_IsoMu24 trigger path is utilised in this work. It requires the presence of at least one muon with a $p_T > 24$ GeV that also satisfies the relative isolation requirement $\mathcal{I}^{\mu}_{\mathrm{rel}} < 0.25$, as described in Subsection 3.2.3.

3.1.8 The 2018 dataset of the CMS detector

The dataset used in this work is comprised of the 2018 dataset, recorded by the CMS experiment during Run-2 of the LHC. This dataset consists of an integrated luminosity of 59.83 fb^{-1} worth of data at a center-of-mass energy of 13 TeV [54]. It is used in the estimation of background processes relevant in this work, as described in Subsection 4.2.3. It is also used in Section 4.3, where the model describing the estimation of contributions from processes that are backgrounds to the cH signal is compared to data.

3.2 Event reconstruction with the CMS detector

Events that pass the triggering system are stored and reconstructed using a more complicated set of reconstruction algorithms. An overview of the reconstruction techniques for the objects relevant to this work, namely muons and jets, is given in this section.

3.2.1 Track and vertex reconstruction

Particle tracks, describing the trajectories of particles through the detector, can be obtained by leveraging information from the pixel and strip detectors of the tracker [55]. By determining the track of a charged particle and thus the curvature of its trajectory in the detectors magnetic field, the particle's transverse momentum p_T may be implicitly determined using the Lorentz force

$$p_T = qBr. \tag{3.4}$$

Here, q is the charge of the particle, B is the magnetic field strength, and r is the radius determined from a particle's trajectory. Since track reconstruction is a computationally intensive procedure given the large number of permutations in which individual pixel or strip hits may be combined, this procedure is performed iteratively. Initially, tracks which are easily identifiable due to e.g. the relatively high p_T associated with them or their proximity to the interaction point, are identified by matching hits in the pixel and silicon strip subdetectors and performing a fitting procedure. The hits associated with these tracks are then removed from the collection of unassociated hits. This procedure is then repeated with looser fitting criteria so that hits that appear to originate from lower p_T tracks or those with an origin displaced from the collision point, may also be associated to tracks.

From the reconstructed tracks, common track origins or *vertices* may be identified. Since several pp collisions may occur in a single bunch crossing, this amounts to identifying the location of the individual collisions in an event. Tracks with a low perpendicular distance or low *impact parameter* to the center of the bunch crossing, and that satisfy requirements on the number of pixel and strip detector hits as well as the quality of the track fit, are chosen for this purpose. These tracks are clustered using a deterministic annealing algorithm [56] to produce a set of candidate vertices with some location along the z-axis. The vertex candidate which is associated with the highest $\sum p_T{}^2$ is assigned as the primary vertex of the bunch crossing. The remaining vertex candidates are referred to as *pileup vertices*, since they typically result from other, lower energy interactions produced in the bunch crossing. Particles that originate from these vertices are accordingly referred to as *pileup*.

3.2.2 The Particle Flow algorithm

The Particle Flow (PF) algorithm [57] is used to combine information from many of the different subsystems of the CMS detector to give an improved and holistic description of an event. This includes reconstructed tracks, the energy deposits in the ECAL and HCAL as well as hits in the muon chamber system. Since different types of particles will interact with the CMS detector's subsystems in unique ways, the properties of individual particles can be extrapolated from this information. These are briefly summarised in Table 3.1.

Table 3.1: Overview of particle signatures in the CMS detector

Particle	**Signature**
Muons	Muons produce tracks in the tracker as well as the muon system with minimal energy deposits in the calorimeters.
Electrons	Electrons produce tracks in the tracker as well as energy deposits in the ECAL with minimal deposits in the HCAL.
Photons	Photons do not produce tracks in the tracker due to being uncharged and primarily deposit their energy in the ECAL.
Charged hadrons	Charged hadrons produces tracks in the tracker, primarily depositing their energy in the HCAL.
Neutral hadrons	Neutral hadrons produce no tracks in the tracker, primarily depositing their energy in the HCAL.

A visual overview of these signatures and the particle type they correspond to can be found in Figure 3.7. The PF algorithm leverages exactly these properties. Initially, matched tracks in the tracker and muon systems are identified as muons and the corresponding components are removed from the event. Subsequently, matched tracks and energy deposit clusters in the ECAL

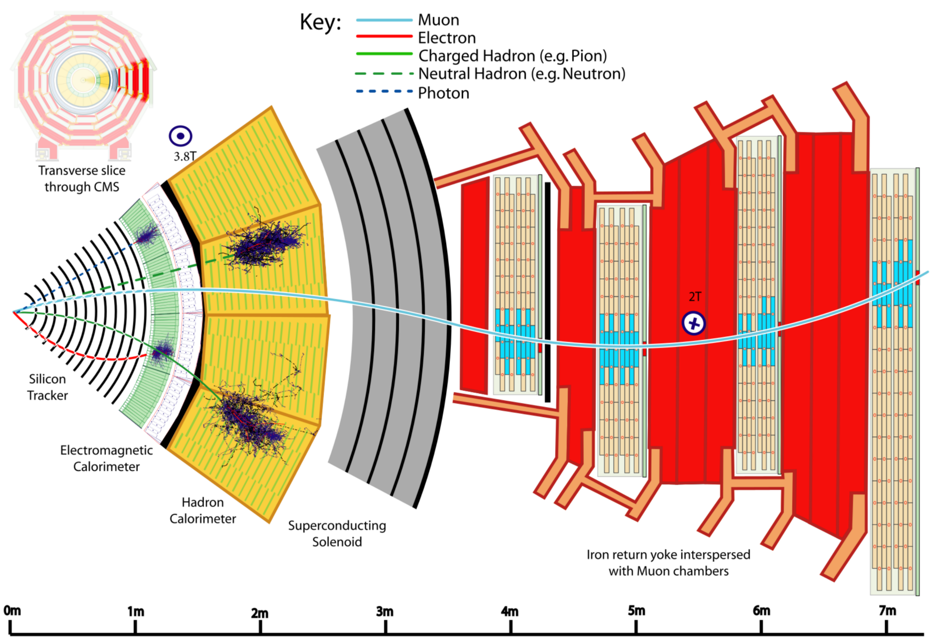


Figure 3.7: An transverse slice of the CMS detector, visualising the signatures that different particles produce in the different detector subsystems. Figure taken from [57].

are identified as electrons and the corresponding components are removed. An isolated cluster in the ECAL with no associated track is reconstructed as a photon candidate and the corresponding cluster is removed. This is expected to leave only charged and neutral hadrons. Clusters of energy deposits in the HCAL associated with a track are thus identified as charged hadrons. However, it frequently occurs that photons are produced in the decay of neutral hadrons. Thus, if the energy estimated from a track is considerably less than the associated cluster in the HCAL and there is a corresponding energy deposit in the ECAL, an additional photon candidate is reconstructed that is associated with the hadron. Finally, HCAL clusters with no associated track are reconstructed as neutral hadrons.

A more comprehensive description of the exact algorithms that are applied can be found in [57]. The following section describes in greater detail the reconstruction of objects relevant to this work. This includes muons, jets (which, due to their nature, typically consist of a collection of reconstructed objects), and missing transverse momentum.

3.2.3 Reconstruction and identification of muons

Since, in this work, muons are used to reconstruct the Higgs boson candidate produced in the cH process, they represent an important element of the presented analysis. Using the available information from the tracker and muon system, three different approaches may be used to initially reconstruct muon tracks.

- **Standalone muon tracks**: A standalone muon track refers to a fit of individual hits present in the muon detector.

- **Tracker muon tracks**: Tracker muon tracks are reconstructed by extrapolating tracks from the tracker to the muon detector, referred to as an *inside-out* approach. If a hit in the muon detector can be matched to the extrapolated track, then these matched tracks are identified as a tracker muon track. This approach seeks to reduce the impact from atmospheric muons traversing the detector, which may be falsely interpreted as standalone muon tracks.

- **Global muon tracks**: Global muon tracks are obtained through an *outside-in* approach, matching standalone muon tracks with tracker muon tracks through a comparison of the respective fitted track parameters. If the tracks are found to match, a combined fit of these tracks is performed. This approach reduces the impact from remnants of hadronic showers that reach the muon chambers, which may be incorrectly reconstructed as a tracker muon track.

Naturally, there is a large overlap between global and tracker muon tracks. Accordingly, if two muon tracks share the same track in the tracker then they are merged into a single object. The collection of standalone, tracker and global muons is passed to the previously introduced PF alogrithm which, by imposing additional quality requirements (further details in [57]), produces a set of reconstructed muon candidates.

A useful criterion in identifying muons that originate directly from the proton-proton interaction is the so-called relative isolation $\mathcal{I}^{\mu}_{\mathrm{rel}}$. This is defined as

$$\mathcal{I}^{\mu}_{\mathrm{rel}} = \Big(\sum p_{\mathrm{T}}^{\mathrm{charged}} + \max\big(0, \sum p_{\mathrm{T}}^{\mathrm{neutral}} + \sum p_{\mathrm{T}}^{\gamma} - p_{\mathrm{T}}^{\mathrm{PU}}\big)\Big)/p_{\mathrm{T}}^{\mu}. \tag{3.5}$$

Here, $\sum p_{\mathrm{T}}^{\mathrm{charged}}$ represents the scalar sum of the transverse momenta of charged hadron originating from the primary vertex of the event. The quantities $\sum p_{\mathrm{T}}^{\mathrm{neutral}}$ and $\sum p_{\mathrm{T}}^{\gamma}$ represent the respective transverse momenta sums for neutral hadrons and photons, respectively. These sums are calculated by accounting for transverse momentum contributions within a conical volume around the direction of the muon. The size of a cone between two positions i and j is defined as $\Delta R(i,j) = \sqrt{\Delta\eta(i,j)^2 + \Delta\phi(i,j)^2}$ and, in this case, the cone boundary around the muon direction is set at $\Delta R = 0.4$. The contribution to the relative isolation from pileup is estimated by subtracting $p_{\mathrm{T}}^{\mathrm{PU}} = 0.5 \sum_k p_{\mathrm{T}}^{k,\mathrm{charged}}$ in Eq. (3.5), where the sum over k represents charged hadron contributions not originating from the PV. The factor 0.5 corrects for different fractions of charged and neutral particles in the cone [58]. Lastly, p_{T}^{μ} represents the transverse momentum of the muon. The relative isolation is thus a variable that quantifies the presence of energy deposits in the ECAL and HCAL around the trajectory of the muon, relative to the p_T of the muon. Since muons are expected to produce such deposits only minimally, good muon candidates are typically associated with small values of $\mathcal{I}^{\mu}_{\mathrm{rel}}$.

Two sets of muon identification criteria are defined for this work:

- **Loose muons**: Loose muons are PF muons reconstructed from either a global or tracker muon track where the perpendicular distance of the extrapolated track to the event's primary vertex is less than 5mm in the z direction and less than 2 mm in the r direction.

- **Tight muons**: Tight muons are loose muons which are reconstructed exclusively from a global muon track. A number of additional criteria are also applied. This includes that the fit quality of the global muon track must be $\chi^2/\mathrm{ndf} < 10$, as well that the significance of the track's 3D impact parameter $\mathrm{SIP_{3D}} = \mathrm{IP}/\sigma_{\mathrm{IP}}$ satisfies $\mathrm{SIP_{3D}} < 4$. Here IP is the impact parameter or point of closest approach to the primary vertex and σ_{IP} is the associated uncertainty. By taking into account the associated uncertainty, the $\mathrm{SIP_{3D}}$ parameter is a more robust choice than solely the IP. Additionally, it is required the at least six layers with at least one pixel hit are registered in the tracker in the associated track as well as two segments hit in the muon detector. Lastly, a relative isolation requirement of $\mathcal{I}^{\mu}_{\mathrm{rel}} < 0.25$ is imposed.

The tight muon definition is used to select muons for reconstructing Higgs boson candidates in cH process candidate events. The loose definition on the other hand is used as part of the estimation of background processes that may mimic the cH process.

The efficiency with which the outlined muon identification criteria identifies muons can differ in simulation and data. As such, a calibration technique is applied to ensure that the efficiencies observed in simulation match those observed in data. This procedure is factorised into separate components and includes the previously introduced HLT_IsoMu24 trigger requirement [59], as well as the isolation and the other, additional criteria that are applied [60]. To perform the efficiency measurements, a so-called *tag-and-probe* method is applied. This involves identifying events in which a $Z \to \mu\mu$ decay is present and where at least one muon satisfies the selection criteria (the *tag* muon), while the second satisfies a loosened version of these criteria (the *probe* muon). The ratio of events in which the probe muon satisfies the selection criteria with respect to the loosened criteria is then interpreted as the corresponding efficiency.

3.2.4 Reconstruction and identification of jets

The quarks and gluons that are produced in pp collisions rapidly hadronise, typically producing collimated cones of particles that, as previously mentioned, are referred to as jets. These are reconstructed as composite objects containing individual hadron candidates, which are produced during the hadronisation process. Since the c quark of the cH process too will produce a jet, jet objects also represent an important aspect of the analysis presented in this work.

To produce jet objects, the hadrons reconstructed by the PF algorithm must be clustered. To ensure a minimal impact of pileup on this clustering, the contributions of pileup are mitigated through *charged hadron subtraction* [57]. This involves the removal of charged hadron contributions in the HCAL and ECAL, if these may be associated with any of the pileup vertices produced in the collision. This is described in greater detail in Subsection 3.2.1. Once this subtraction has been performed, the remaining PF hadrons are passed to the anti-k_{T} algorithm [61]. The anti-k_{T} algorithm is an iterative clustering algorithm that is based on a principle of minimal distances between particles. The distance d_{ij} between the particles i and j is defined as well as the distance d_{iB} between particle i and the beam. These are given by

$$d_{ij} = \min\left(\frac{1}{p^2_{\mathrm{T},i}}, \frac{1}{p^2_{\mathrm{T},j}}\right)\frac{\Delta^2_{ij}}{R^2} \tag{3.6}$$

$$d_{iB} = \frac{1}{p^2_{\mathrm{T},i}} \tag{3.7}$$

$$\Delta_{ij} = \sqrt{\Delta \mathrm{y}(i,j)^2 + \Delta\phi(i,j)^2}. \tag{3.8}$$

Here, y is the rapidity of a particle and R is a constant parameter that determines the cone size of the clustered jets. The default choice used with the CMS experiment is $R = 0.4$, which is also used in this work. Starting with the highest p_T object in the initial iteration, the distance d_{ij} with the closest PF candidate j is calculated. The two objects are clustered together and this process is repeated until a stopping condition $d_{ih} > d_{iB}$ is met. At this point, the jet is considered fully reconstructed and the PF candidates used in its clustering are removed for the reconstruction of subsequent jets.

Due to the presence of detector noise, unphysical low p_T jets can be erroneously reconstructed. This effect can be mitigated by applying additional criteria on reconstructed jets. This includes requiring that at least two PF candidates are clustered in the jet and that the jet's energy is not solely attributed to neutral hadrons or photons. These requirements remove almost all such unphysical jets while over 99% of physical jets fulfill them [62]. Additionally, a pileup discrimination algorithm is described in [62], of which the loose working point is applied to jets with $p_T < 50$ GeV in this work.

A calibration of jet energies is performed after reconstruction in both simulation and data [63]. This calibration accounts for pileup contributions in the clustering, the non-linearity of the detector response, and improper reconstruction of hadrons. A number of methods are used to derive sets of correction factors. An example is the use of events with a Z boson, the p_T of which may be precisely reconstructed via the Z→ $\mu\mu$ decay. When a single jet recoils against the Z boson candidate, the expected p_T balance of these two objects allows the p_T response of the jet reconstruction algorithm to be characterised and accordingly corrected. Additionally, significant discrepancies in the resolution of jets in simulation and data are observed, with the resolution being worse in the latter than the former. This is accounted for by a smearing method, in which the resolution of jets is artificially smeared in simulation so that a better comparison to data is achieved.

3.2.5 Missing transverse momentum

Due to the conservation of momentum, it is expected that the vectorial sum of momenta of all particles produced in a collision adds up to zero. However, this may not be the case when particles such as neutrinos are produced in the collision, as these cannot be measured by the detector. As a result, it can be useful to define the missing transverse momentum as

$$p_{\mathrm{T}}^{\mathrm{miss}} = \sum_{i}^{\mathrm{PF}} p_{\mathrm{T}}^{(i)}. \tag{3.9}$$

The presence of significant quantities of $p_{\mathrm{T}}^{\mathrm{miss}}$ may thus be used to identify the presence of neutrinos in an event.

3.3 Identification of charm quark-induced jets

To identify the charm quark-induced jet of the cH process, one must be able to discriminate against both bottom quark as well as both light quark and gluon-induced jets. This is a task colloquially referred to as *flavour tagging*, with a jet's *flavour* being determined by the type of particle that initiated it. Modern flavour tagging techniques typically use machine learning techniques to leverage key jet properties that may differentiate jets of different flavours, though this remains a challenging task. To discuss these properties, a definition of jet flavour is necessary. In the context of the CMS experiment, a ghost matching procedure [64] is applied to obtain such a definition for simulated events. This involves adding information from the event simulation to the reconstructed event. Specifically, hadrons containing bottom and charm quarks are identified in the simulation and added to the list of reconstructed PF candidates, albeit with negligible momenta. With this addition of so-called *ghost hadrons*, the jet clustering is once again performed. Due to the negligible momenta of the ghost hadrons, the clustering procedure itself is unaffected. However, the inclusion of the ghost hadrons can be used to construct the following definitions:

- **c jets**: If at least one charm (c) ghost hadron and no bottom (b) hadrons are clustered inside the jet, the jet is labelled as a c jet.
- **b jets**: If at least one b ghost hadron is clustered inside the jet, the jet is labelled as a b jet.
- **light jets**: If no ghost hadrons are clustered inside the jet, the jet is labelled as a light jet. Light jets may be initiated by quarks such as the up, down, or strange quark or by gluons. An additional, technical category of *pileup jets* exists, depending on whether so-called matching criteria between reconstructed and simulated jets are fulfilled, though they are subsumed into the light jets category for the purpose of this work.

As a result, the task of identifying c jets is twofold:

1. Discriminating *heavy-flavour* (HF) jets consisting of b jets and c jets against light jets.
2. Discriminating between b jets and c jets.

3.3.1 Properties of heavy-flavour jets

The term heavy-flavour originates from the mass of the bottom and charm quarks, which is an order of magnitude greater than the next heaviest quark, the strange quark. The c and b hadrons have relatively long lifetimes that allow them to travel an observable distance from the PV before decaying. The typical lifetime of a b hadron is 1.5 ps, while that of c hadrons ranges between 1 – 0.4 ps [2]. This oftentimes results in the presence of a secondary vertex (SV) that is measurably displaced from the collision point. The displacement associated with such a SV typically ranges between a few mm to $\sim$1 cm and is a key signature of HF jets. Accordingly, tracks originating from the decay of a HF induced jet typically originate from such a SV.

Another feature of HF jets is the presence of leptons in the jet. This results from the relatively large branching fractions of HF hadrons into states containing leptons. These are typically low-energy and are present in about 20% (10%) of b(c) jets. Accordingly, the identification of a low-energy electron or muon inside a jet serves as a good indicator that a jet originates from a HF hadron. Also of significance are the relatively high masses HF hadrons exhibit in comparison to their lighter counterparts. This results in HF induced jets often exhibiting a spatially broader energy flux compared to their lighter counterparts, due to stronger diffusion of momenta perpendicular to the flight direction. Additionally, the decay of HF hadrons is also associated with higher hadron multiplicities that carry significant transverse momenta. These features are illustrated in Figure 3.8. Additionally, distributions showing the significance of 2D impact parameters, which can be used to infer the presence of SVs, can be seen for b, c and light jets in Figure 3.9

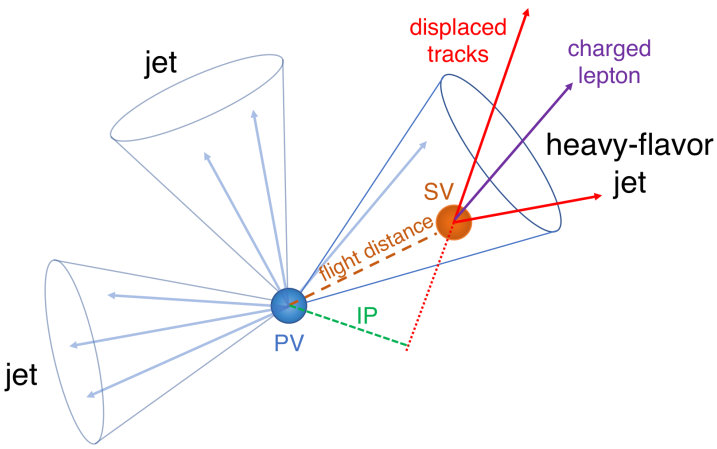


Figure 3.8: An illustration highlighting the properties of HF jets. The presence of a secondary vertex (SV), characterised by the impact parameter (IP) in green, as well as the presence of a lepton is highlighted. Figure taken from [65].

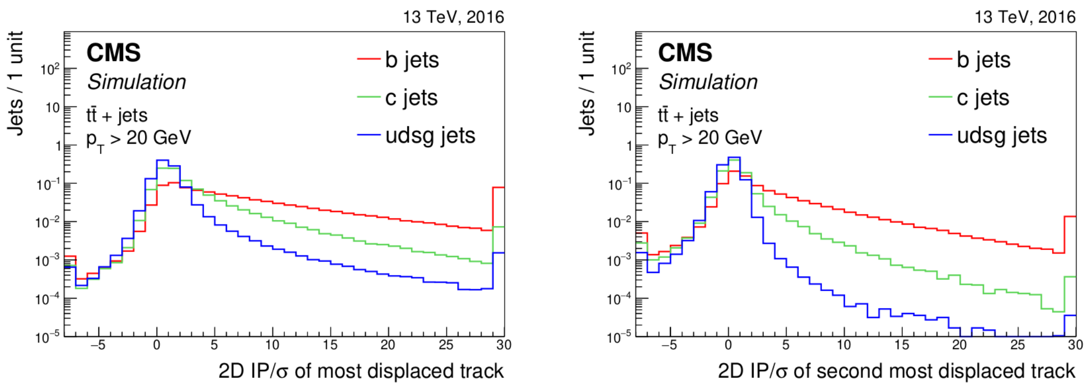


Figure 3.9: Plots showing the significance of the 2D impact parameter of the most and second most displaced tracks in a jet. As can be seen, these variables can differentiate b and c jets from light jets to a significant degree. Figure taken from [65].

3.3.2 The DeepJet algorithm

The DeepJet algorithm [66] is a machine learning algorithm used for jet-flavour identification in this work. It improves on previous neural network based algorithms [65] used by the CMS collaboration in the Run-2 period of the LHC. A notable feature compared to earlier algorithms is its use of lower level information such as use of track, PV and SV information, as well as PF candidate and event kinematics information. The network consists of three branches that individually process neutral and charged hadrons as well as secondary vertices before this information is combined with global variables. The network output consists of six output nodes, each representing an output class corresponding to a particular jet flavour category, which it is trained to identify. The output value of the nodes $\mathcal{P}(b/bb/lepb/c/l/g)$ for a given jet, are interpreted as the likelihood that a jet belongs to the respective class. These are defined as

- ***b/bb/lepb* (b jets):** These three classes represent subclasses of jets originating from a b hadron. The *b* class represents a jet originating from a single b hadron, the *bb* class originating from two b hadrons and *lepb* representing a jet originating from a b hadron with the presence of a soft lepton.
- ***c* (c jets)**: This class represents a jet originating from a c hadron.
- ***l, g* (light jets)**: These two classes represent light jets originating from a light quark or gluon, respectively.

From these output classes, two useful discriminators to identify c jets can be constructed. These are

$$\mathrm{CvsB} = \frac{\mathcal{P}(c)}{\mathcal{P}(c) + \mathcal{P}(b) + \mathcal{P}(bb) + \mathcal{P}(lepb)} \tag{3.10}$$

$$\mathrm{CvsL} = \frac{\mathcal{P}(c)}{\mathcal{P}(c) + \mathcal{P}(l) + \mathcal{P}(g)}, \tag{3.11}$$

each representing a discrimination of c jets against b jets and light jets, respectively. The performance of DeepJet, in terms of the CvsL and CvsB discriminators in simulated samples of top quark pair production, can be seen in Figure 3.10. These are characterised by analysing how the misidentification rate of c jets changes in dependence on the efficiency with which c jets are can be identified. A comparison to the DeepCSV algorithm (the previous iteration of a jet-flavour identification algorithm used by the CMS collaboration, presented in [65]) is included, highlighting the performance gain that the DeepJet algorithm achieves. The challenging, twofold nature of the c-tagging task is demonstrated in Figure 3.11, where the 2D distribution of the CvsL and CvsB discriminators for simulated b, c and light (udsg) jets are shown, respectively. All three categories exhibit significant overlap in terms of the discriminator distributions, indicating there is a significant trade-off between the efficiency and purity with which c jets may be identified.

Since neural network based algorithms are trained on simulated samples that do not perfectly describe their data counterpart, the neural network output must be calibrated with respect to data. To calibrate the entire shape of the algorithm's output distributions, the approach described in [67] is used. This involves targeting phase spaces enriched in b jets (top quark pair production), c jets (charm associated $\mathrm{W}^{\pm}$ production), and light jets (jet associated Drell-Yan production). Using simulation, the fractions of b, c, and light-flavour jets are determined in each phase space and an iterative fitting procedure, minimising differences between simulation and data, is performed. Through this method, correction factors that calibrate the algorithm's response are derived. These depend on the values of the two CvsL and CvsB discriminators

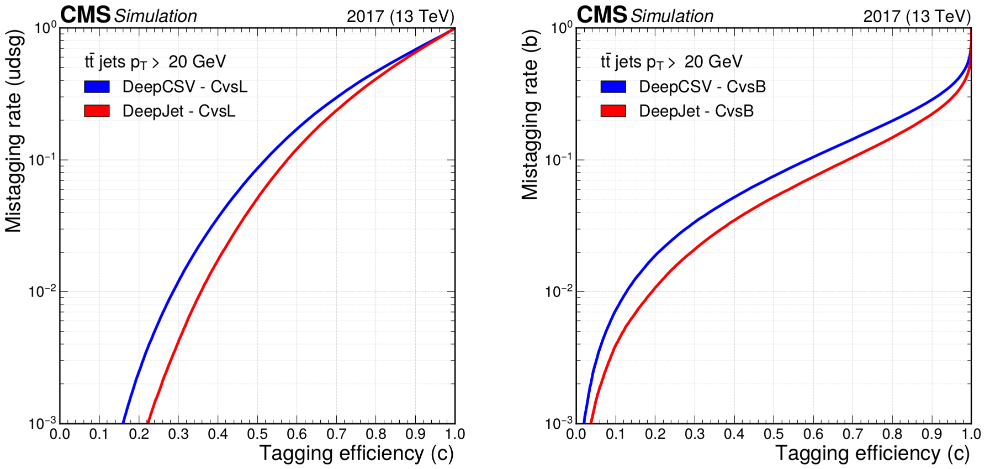


Figure 3.10: Performance of DeepJet algorithm in identifying c jets against b jets and light jets in simulated samples of top quark pair production ($t\bar{t}$), where both top quarks decay hadronically. The x-axis represents the efficiency with which c jets are identified, while the y-axis represents misidentification rate with respect to either b jets or light jets. Figure taken from [67].

as well as the true flavour of a simulated jet, as the correction factors can vary with all three parameters.

3.4 Validation of jet flavour tagging algorithms and data quality monitoring

The observable quantities that jet flavour tagging algorithms, such as the DeepJet, rely on are often themselves the products of other reconstruction algorithms. This includes, for example, particle tracks or PF candidate information. As these reconstruction algorithms are continually updated in the CMS software package [69], the potential effect of such changes must be regularly validated to ensure no unwanted or poorly understood behaviour is introduced. Concretely, this means the response of the jet flavour tagging algorithms is checked for each individual iteration of the CMS software by comparing it to a previous iteration and investigating any unexpected changes. Such an investigation typically consist of verifying changes to relevant observables, which are maintained by a variety of subgroups within the CMS collaboration, and gauging their impact on the behaviour of the respective jet flavour tagging algorithm. Accordingly, if changes are identified that significantly affect the performance of the flavour tagging algorithms, then they are flagged for further, strategic evaluation. The outcome of such an evaluation may be that the flavour tagging algorithms should be re-trained to try and recoup, for example, a performance loss. Since the discriminator distributions that the algorithm outputs may be altered by such a re-training, the algorithm is accordingly also re-calibrated. This is done using, for example, the procedure introduced in Subsection 3.3.2. As a result, the validation procedure is an important component in ensuring that the jet flavour tagging algorithm behaviour remains performant, well-understood and calibrated. Furthermore, the releases of the CMS software packages that are verified via such a procedure, and for which no significant issues identified, are then iteratively deployed for data taking. An example of a previous comparison between releases of the CMS software, which typically uses simulated samples, is shown in Figure 3.12 and Figure 3.13

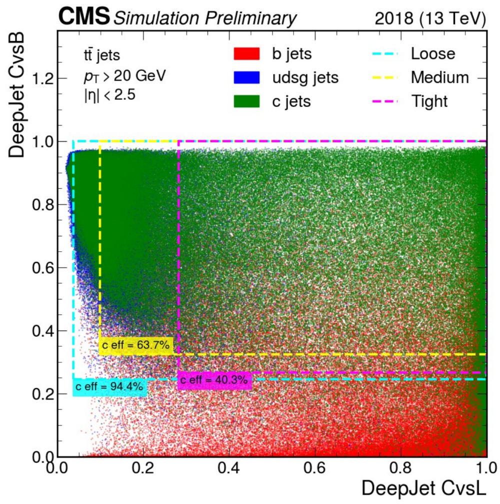


Figure 3.11: A 2D distribution of the CvsL and CvsB discriminators of the DeepJet algorithm, shown for simulated b jets (red), c jets (blue) and light jets (green) in simulated samples of top quark pair production ($t\bar{t}$). Additionally, three boxes representing different c jet identification efficiencies, each constituting fixed so-called working points, are shown. Figure taken from[68].

for the DeepJet algorithm.

While many different observables are typically compared per validated release, a good way to characterise the algorithm's response is via a so-called *receiver-operator characteristic* (ROC) curve for a given discriminator. These include, for example, the previously introduced CvsB and CvsL discriminators. Here, the BvsAll discriminator is additionally considered, which is analogously defined as:

$$\text{BvsAll} = \frac{\mathcal{P}(b) + \mathcal{P}(bb) + \mathcal{P}(blep)}{\mathcal{P}(b) + \mathcal{P}(bb) + \mathcal{P}(blep) + \mathcal{P}(c) + \mathcal{P}(g)}. \tag{3.12}$$

The ROC curve characterises the behaviour of the discriminator response by connecting a given efficiency with which an object can be identified, to a corresponding misidentification rate. This misidentification rate can be understood as the false positive rate of identifying said object. For the BvsAll discriminator, this corresponds to either correctly identifying a b jet or misidentifying a c jet or light jet, as a b jet. For the comparison shown in Figure 3.12 and Figure 3.13, some non-negligible differences in the ROC curve for all three discriminators were observed. This was, for example, identified to likely correlate to changes observed in quantities reconstructed with the silicon tracker, such as the number of pixel hits associated with jets. The outcome of such a validation is then reported centrally to the Physics Performance and Dataset group of the CMS collaboration, for further evaluation.

Another important aspect of these validation tasks is to investigate effects of changes in detector conditions on the jet flavour tagging algorithms. An example of this is the malfunctioning of a set tracker pixels of the CMS detector, which was identified mid Run-3. Such cases require thorough investigation of the potential performance impacts on jet flavour tagging algorithms, due to the reliance of these algorithms on the CMS pixel detector. In this case, for example, a

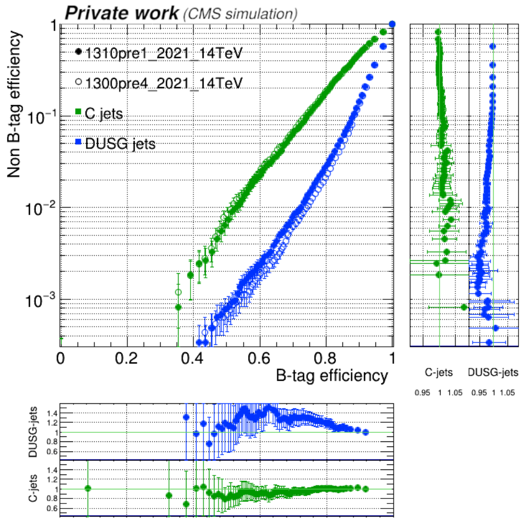
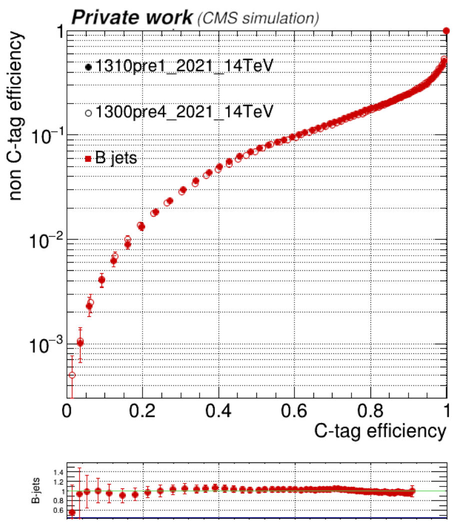
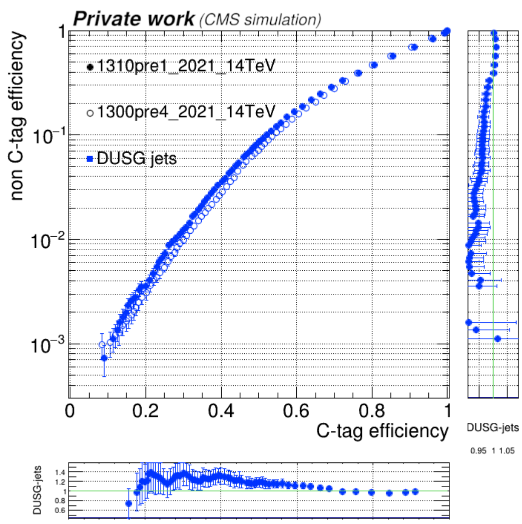


Figure 3.12: A comparison of two CMS software package releases, showing the ROC curve for the BvAll discriminator of the DeepJet algorithm (here referred to as DeepFlavour, an earlier iteration of the name). Separate curves are shown for c jets and light (DUSG) jets since the respective misidentification rates are significantly different. In the central panel, the filled markers represent the CMS software release that is being validated, while the unfilled markers represent the reference release. The additional panels below and on the right show the ratio between the two.



(a) The CvsB discriminator ROC curve.



(b) The CvsL discriminator ROC curve.

Figure 3.13: A comparison of two CMS software package releases, showing the ROC curve for the CvsB and CvsL discriminator of the DeepJet algorithm (here referred to as DeepFlavour, an earlier iteration of the name). In the central panel, the filled markers represent the CMS software release that is being validated, while the unfilled markers represent the reference release. The additional panels below and on the right show the ratio between the two.

significant loss in efficiency was observed in the performance of the flavour tagging algorithms in the regions of the malfunctioning pixels, due to an associated reduction in the efficiency of the tracker. This insight informed an attempt to recover this performance loss through a re-training of the flavour tagging algorithms, though this was not found to be successful. Instead it was observed that the SVs that indicate HF jets were simply no longer well-reconstructed in this region of the detector, leading to an unavoidable loss in flavour-tagging performance. Given also the more general loss in tracking performance related to this issue, jets in this region are instead vetoed for analysis.

While the described validation tasks are typically performed either only on simulation or with a set of previously recorded data, direct monitoring of data during the recording process is also possible. This allows for a more immediate picture of algorithm responses to data as it is recorded, and can also be performed for jet flavour tagging algorithms. Concretely, such monitoring is, for example, possible at the level of the Tier-0 processing system [70]. This is a system used for immediate and automatic data handling of the data generated by the CMS detector. It takes the output of the 'online' HLT system, which is written to a disk buffer, and performs initial reconstruction tasks before the data is finally passed to the fully 'offline' computing system for a comprehensive reconstruction. The Tier-0 system can hold data for approximately up to 48 hours and effectively acts an interface between the 'online' and 'offline' portions of the CMS detector's processing chain. Accordingly, it provides a suitable context for monitoring of BTV related data shortly after data-taking. The reason for this is that the steps required for such monitoring, such as the selection of b, c and light-jet enriched phases spaces on which the flavour tagging algorithm response is evaluated, are not necessarily viable at the HLT level where data can only be held for a very brief period of time. It is however viable in the context of the Tier-0 system and as such, there is an ongoing effort to implement such monitoring of BTV related quantities with the Tier-0 system.

In my role as validation and data-quality monitoring contact for the b-tagging and vertexing (BTV) subgroup of the CMS collaboration, I performed and contributed to the above described work over a number of years. This includes regularly performing the described validation procedures, providing software contributions to facilitate the monitoring of BTV algorithms during data taking at the Tier-0 level, and presenting the performance of BTV algorithms internally to the CMS collaboration. As a whole, this work has contributed to the ongoing credibility and well-understood nature of the jet flavour tagging algorithms provided by BTV. This is especially relevant given the strong need of jet flavour identification methods at the LHC, which are regularly used in analyses by the CMS collaboration. These analyses include this work as well as the other, previously discussed analyses that target the charm-Yukawa coupling.

3.5 Monte Carlo simulation of proton-proton collisions

To produce predictions of the CMS detector response as well as of the reconstruction algorithms to pp collisions at the LHC, a simulation of the collision and the CMS detector is required. Since particle interactions such as pp collisions are fundamentally stochastic, Monte Carlo methods [71] are used to simulate these. The concept of such a simulation relies on a phenomenological approach by sampling the known distributions of process and detector quantities and properties. In this way, detector and particle properties may be simulated with great detail. The simulation process occurs in discrete steps, each dealing with different aspects of the simulated process. These can be summarised as:

1. **The hard scattering process**: The hard scattering process refers to the immediate, high energy transfer scattering of two protons resulting in the production of new particles. To calculate this, two main ingredients are required. The first is a calculation of the matrix elements that describe the process in which proton constituents collide to produce additional particles. These matrix elements allow for the calculation of an expected cross section for the process and, in the simulation relevant to this work, are typically performed at a fixed order in perturbative QCD. Since this calculation depends on $\alpha_s(Q^2)$, a corresponding energy scale must be chosen at which it is evaluated. This is the renormalisation scale μ_R, which is typically chosen to reflect the energy scale of the process that is being described. The second ingredient arises from the fact that the proton itself is a complex object consisting not only of its valence quarks (two up-type quarks and one down-type quark), but also of a constantly changing ensemble of additional quarks and gluons that are regularly created and annihilated. This behaviour is parametrised via so-called *Parton Distribution Functions*. These are determined experimentally [72] and describe the likelihood with which a parton, that carries some fraction x of the protons total momentum, may be found in a proton at some energy scale Q^2. The evolution of the PDF with changing Q^2 can described by the DGLAP equations [73]. A cutoff, referred to as the factorisation scale μ_F, is defined that separates the description of the hard scattering process into a matrix element and PDF component. While the former description is perturbative in QCD, the latter is not. Software used to simulate the hard scattering are referred to as *event generators*. Commonly used event generators include Madgraph5_aMC@NLO [19] and POWHEG [74].

2. **Parton showering**: Particles such as quarks and gluons that are produced in the hard scattering carry the colour charge of the strong interaction. As a result, these may produce additional radiation or branch into other particles. While a most physically accurate description would be given by including these contributions in the calculation of the hard scattering process, this greatly increases the complexity of the calculation, which in complexer cases, quickly becomes unfeasible. As such, a *parton shower* model such as in the Pythia software package [75], is used instead to describe the splitting of a single mother particle into two daughter particles. In QCD, this describes gluon radiation ($q \to qg$) and gluon splitting ($g \to gg$ and $g \to q\bar{q}$) and in quantum electrodynamics (QED) describes Bremsstrahlung ($f \to f\gamma$) and pair creation ($\gamma \to f\bar{f}$). In case the parton showering originates from initial state partons, it is referred to as initial state radiation (ISR). Accordingly, parton showering originating from final state partons is referred to as final state radiation (FSR). In cases with final states containing multiple partons, there can be some ambiguity in the combination of matrix elements and parton showering, since both may describe the same underlying processes. For this, merging schemes are applied that resolve potential double counting of events. An implementation of this which is used in this work is the FxFx scheme [76]. Such a merging scheme, for a chosen merging scale μ_M, effectively ensures that a parton shower description is used for energies below μ_M and a matrix element description for energies above μ_M.

3. **Hadronisation**: At an energy around the QCD scale Λ_{QCD}, the perturbative parton shower prescription loses its validity as the running coupling of the strong force α_s becomes too strong. Here, the individual, colour-charged partons *hadronise* into colour-neutral states. Since this process currently cannot be described from first principles, a phenomenological description must be applied. In Pythia, the *Lund string* model is used [77]. It describes the interaction between two partons as a coloured field, the lines of which pass through a tube that is extended between the partons. The potential energy of the

tube (or string) is described by a term linear in the distance between the partons. Thus, if the partons are separated at a large enough distance and the potential energy is sufficiently large, the string may 'break' and new colourless quark-antiquark pairs are formed. This procedure may be repeated with these new parton pairs if they possess an invariant mass above some threshold.

4. **The underlying event**: A description of a variety of effects secondary to the hard scattering must be included in the simulation. These can have several origins such as secondary, lower-energy interactions of the pp collision, or remnants of the collided protons that hadronise also themselves. These effects are modeled using data [78].

5. **Pileup**: To estimate pileup contributions, a number of pileup interactions are added to every event, with the number being sampled from a Poisson distribution. The mean of this distribution itself is sampled from a distribution of the mean number of interactions per bunch crossing, which is determined from data. To ensure that the pileup profile that is simulated in this way indeed matches the profile observed in data, a reweighting procedure is performed to reproduce the measured profile shown in Figure 3.14.

6. **Detector simulation**: Finally, the detector response to the particles emerging from the previously described steps must be simulated. This is partially performed with the GEANT4 package [79], which is configured to model the CMS detector. This includes modelling the curving of particle trajectories due to the detector's magnet and the interaction of particles with the materials of the detector. Finally, the digitisation of the signals in the electronic modules of the subdetectors is modelled using the CMS software package [69].

A diagrammatic overview of how the hard scattering, parton showering, hadronsiation, and underlying event simulation steps look like can be found in Figure 3.15. Accordingly, the pileup contributions and the detector simulation are not represented in the diagram. The output of this simulation is passed to the reconstruction algorithms described in this section, thus allowing analysers to produce predictions of what the response of the detector to data looks like, along with the response of the reconstruction algorithms.

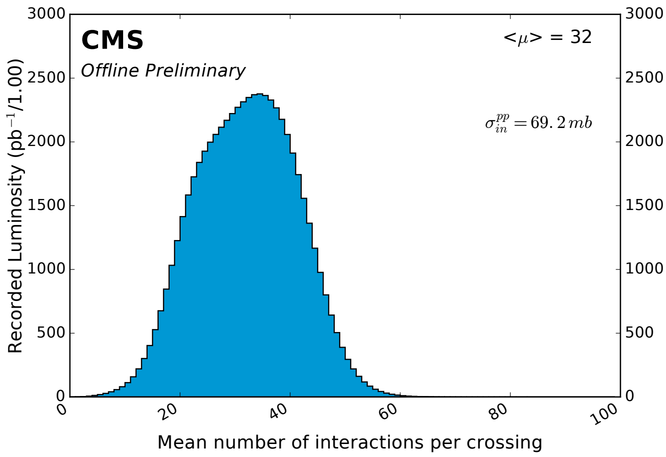


Figure 3.14: The distribution of the mean number of pileup interactions per bunch crossing, measured for the 2018 era of the Run-2 dataset of the CMS detector. Figure taken from [80].

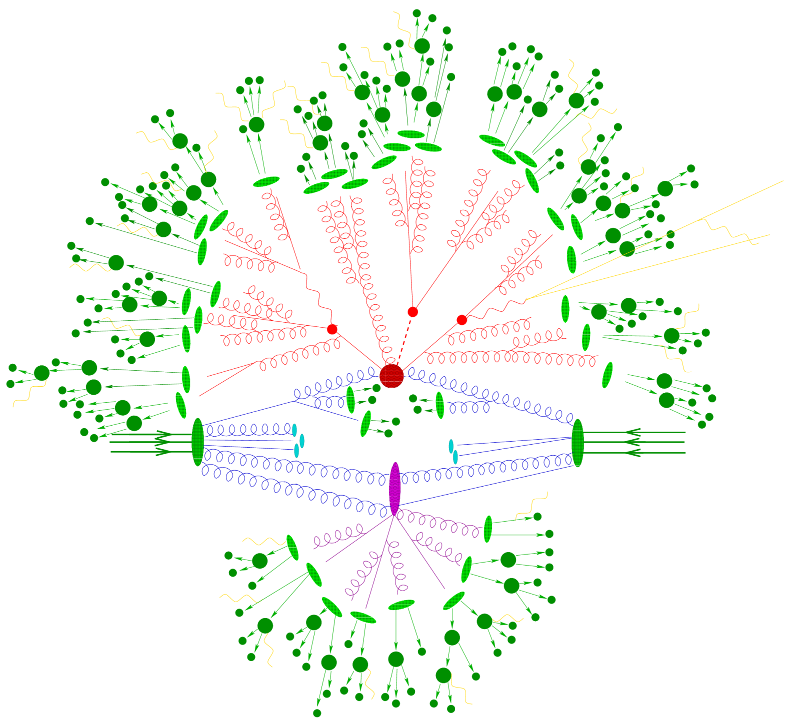


Figure 3.15: An overview of what an event simulation may look like. The initial state particles originating from the protons are shown in blue. The hard scattering process and parton showering is represented in red. Additional, soft scatterings are represented in purple. The hadronisation to form colourless hadrons (represented by dark green circles) is represented by the light green ellipses. Additional photon radiation is represented in yellow. Figure taken from [81].

Chapter 4

Search for the $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)_{\kappa_c}$ process

To probe the charm-Yukawa coupling with the $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)_{\kappa_c}$ process, a number of key analysis components are required. This starts with devising a method with which $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)$ candidate events may be selected. This is presented in Section 4.1, where both the Higgs boson candidate reconstruction as well as the selection of the jet originating from the associated charm quark are discussed. Subsequently, a model must be devised that predicts the event yields obtained from such a selection, as well as the associated distributions of observable quantities, when it is applied to a dataset. The components of this model are described in Section 4.2. This includes an estimation of the $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)_{\kappa_c}$ process, which is here referred to as the *signal process*, in addition to the contributions of a number of other non-signal processes. These latter processes are typically referred to as *background processes*, since they may mimic the signatures of the signal process and accordingly contribute to the selection of $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)$ candidate events that is made. This also includes the contribution of the $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)$ process not sensitive to κ_c. The model describing the background process contributions is then validated in Section 4.3 through a comparison to data in regions where no signal process contributions are expected.

Following this, a statistical evaluation method using discriminators derived from the presented event selection is discussed in Section 4.4. This statistical evaluation is based on the model of the signal and background processes and takes into account systematic uncertainties associated with this modelling, which are also examined. Finally in Section 4.5, assuming the absence of a measurable signal due to the very small cross section of the $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)_{\kappa_c}$ process, expected 95% CL upper limits on κ_c are derived by applying this method to an Asimov dataset. This is performed for a variety of scenarios, including extrapolations to full Run-2 and High-Luminosity LHC scenarios.

4.1 The $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)$ event selection

To reconstruct a $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)$ candidate event, a Higgs boson candidate needs to be reconstructed and a corresponding jet candidate needs to be identified. These two procedures are described in this section.

To reconstruct a Higgs boson (jet) candidate, an initial selection of muon (jet) objects must be made. The associated selection criteria that are applied to this end are summarised in Table 4.1, along with the HLT trigger path requirement used in this analysis. Additionally, the distributions of object quantities to which selection criteria are applied, such as the p_T and η of the muons and jets, as well as the ΔR of a jet and the closest muon, are shown in Figure 4.1. The objective of this selection is to identify events with well-reconstructed, isolated muons as well as at least one well-reconstructed jet. Following this initial selection, the corresponding objects are passed onto respective algorithms to select a final Higgs boson and jet candidate.

Table 4.1: Muon, jet object, and HLT path selection requirements.

Object	**Selection criteria**
Muons	$p_T > 5$ GeV $\lvert\eta\rvert < 2.4$ Tight muon identification criteria
Jets	$p_T > 25$ GeV $\lvert\eta\rvert < 2.5$ Jet ID Pile-up ID, loose working point ΔR(jet, selected muons) > 0.4 Jets in veto regions of detector are excluded
HLT	HLT_IsoMu24 trigger is passed

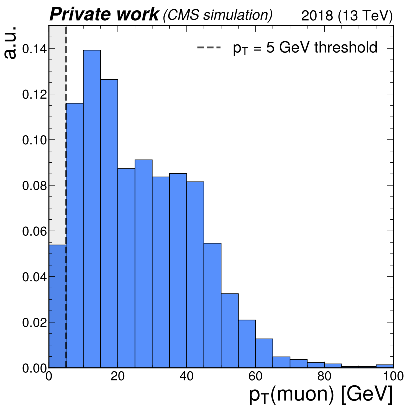
4.1.1 Higgs boson candidate selection

A Higgs boson reconstruction algorithm (and muon object selection) very similar to those presented and validated in [32] is implemented. This reconstruction is performed for events in which exactly four selected muons are present, to avoid introducing a potential bias in the mass of the Higgs boson candidate. This is relevant due to the potential contributions of background processes in which no Higgs boson is produced and that, accordingly, do not exhibit a resonant behaviour around the Higgs boson mass of $m(H) = 125$ GeV [2]. For events containing exactly four selected muons, the following reconstruction steps are applied:

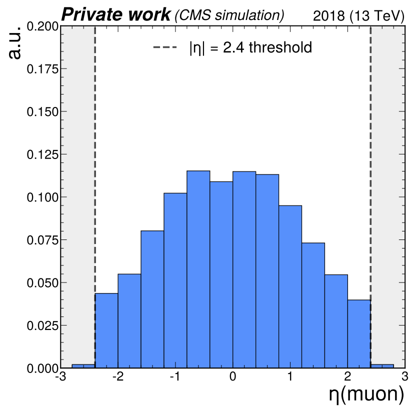
1. Of the four selected muons, the p_T-leading muon is required to satisfy $p_T > 20$ GeV and the sub-leading muon is required to satisfy $p_T > 10$ GeV. This choice is found to improve analysis sensitivity to Higgs boson decays in [32]. Additionally, the HLT_IsoMu24 trigger requirement must be met. This choice ensures that when analysing data, only events with at least one well-isolated muon, the p_T of which approximately matches the aforementioned p_T requirement, are considered. Lastly, to ensure two muons are not spuriously reconstructed from shared tracks, it is required that each muon candidate is separated from the others by ΔR >0.02.

2. Opposite-sign muon pairs are merged into Z boson candidates. At least two Z boson candidates must be reconstructed to proceed. Additionally, the invariant mass of any combination of opposite-sign muons must satisfy $m(\mu\mu) > 4$ GeV, to remove any contributions from low-mass resonances such as J/ψ [2].

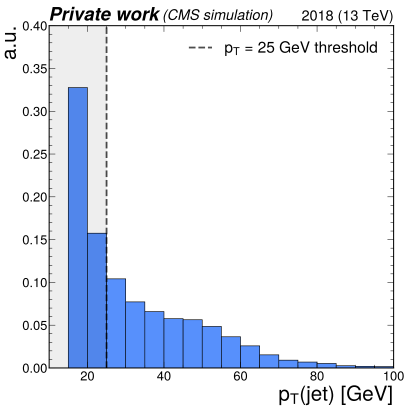
3. The Z boson candidate with a mass closest to the known Z boson mass of $m(Z) = 91.19$ GeV [2] is interpreted as the Z_1 candidate. The Z_1 candidate should satisfy 40 GeV <



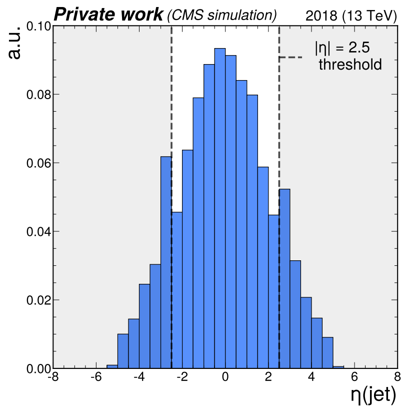
(a) p_T of reconstructed muons.



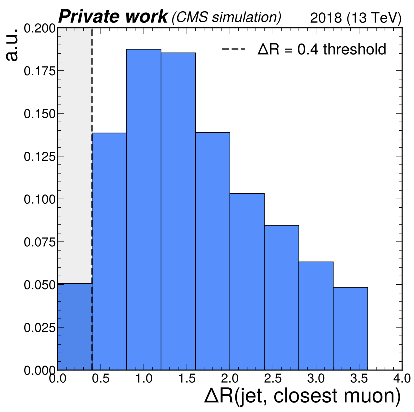
(b) η of reconstructed muons.



(c) p_T of reconstructed jets.

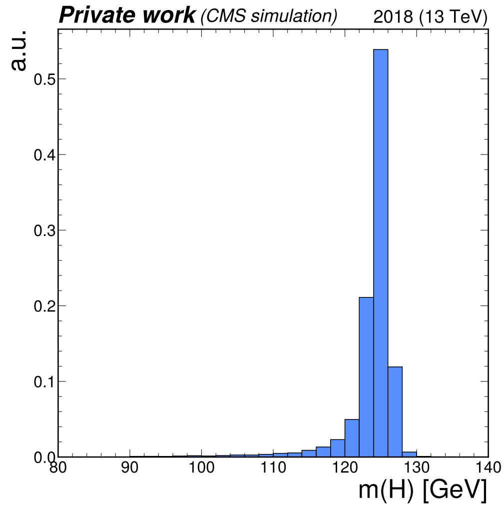


(d) η of reconstructed jets.

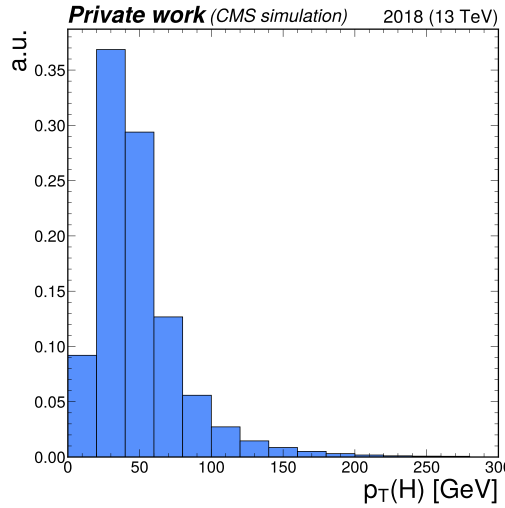


(e) ΔR between a reconstructed jet and the closest reconstructed muon.

Figure 4.1: Distributions of jet and muon quantities (inclusive in the number of jets and muons) to which selection criteria are applied during the initial muon and jet selection, shown for four-muon events in a simulation of the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ process. The thresholds associated with the selection criteria are marked by dashed lines and the region where contributions are removed are shaded.



(a) The $m(H)$ distribution.



(b) The $p_T(H)$ distribution.

Figure 4.2: The distribution of the reconstructed Higgs boson candidate mass $m(H)$ and transverse momentum $p_T(H)$ in simulated cH(ZZ→ $4\mu)_{\kappa_c}$ events. A clear peak around the nominal Higgs boson mass of 125 GeV is observed in the spectrum of $m(H)$.

$m(Z_1) <$ 120 GeV. The other candidate is taken as the Z_2 candidate, which is typically more off-shell (meaning its mass typically deviates more strongly from $m(Z)$) and thus the invariant di-muon mass requirement is relaxed to 12 GeV $< m(Z_1) <$ 120 GeV.

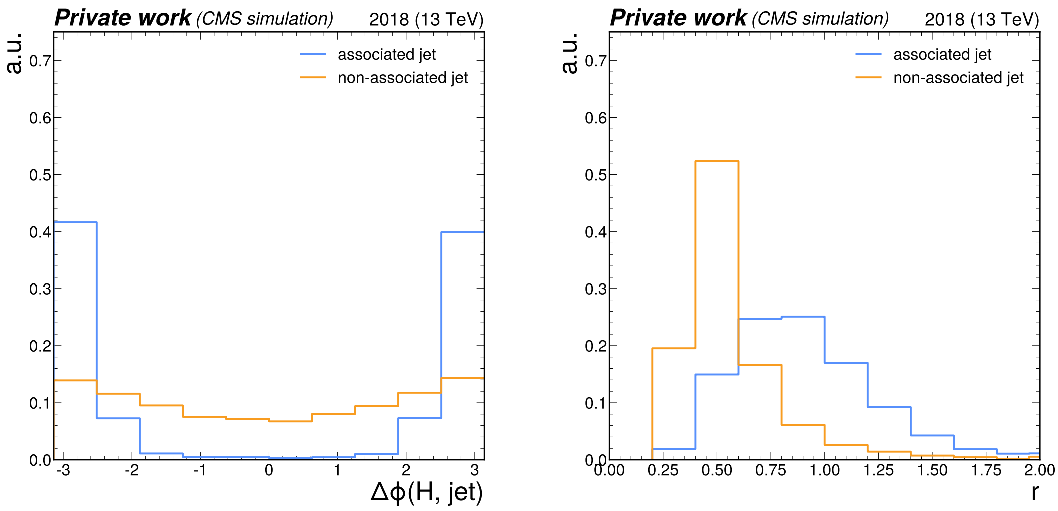
4. The Z_1 and Z_2 candidates are combined to form a Higgs boson candidate. The four-muon invariant mass of the Higgs boson candidate must satisfy $m(H) >$ 70 GeV.

The distributions of the reconstructed Higgs boson candidate mass and transverse momentum can be seen for simulated cH(ZZ→ $4\mu)_{\kappa_c}$ events in Figure 4.2. As expected, a peak around the known Higgs boson mass $m(H)$ = 125 GeV can be observed, with an elongated tail towards lower masses. This tail is expected to originate from energy losses that can be attributed to Bremsstrahlung of the individual muons as they traverse the detector.

4.1.2 Jet candidate selection

Once a Higgs boson candidate is reconstructed, a likelihood ratio algorithm is applied to best identify and select the jet that is associated with (i.e. recoils off) the reconstructed Higgs boson. This algorithm does not use jet-flavour identification methods and is based solely on kinematic properties of the jets, so as to minimise the introduction of any flavour bias in the selection. Specifically, two observables related to momentum conservation in the transverse plane are exploited:

1. The difference in azimuthal angle $\Delta\phi(H, jet)$ between the Higgs boson candidate H and the jet is used. As a result of momentum conservation, the Higgs boson and associated jet are expected to recoil off each other *back-to-back*, and thus $\Delta\phi(H, jet)$ is expected to be $\sim \pm\pi$.



(a) The $\Delta\phi(H, jet)$ template.

(b) The r template.

Figure 4.3: The $\Delta\phi(H, jet)$ and r templates in the 30–50 GeV bin of the Higgs boson candidate p_T.

2. Since the Higgs boson and associated jet recoil off each other, their p_T is expected to be approximately balanced. This information can be captured by the transverse momentum ratio $r = p_T(H)/p_T(jet)$.

To derive the relevant distributions to be used in a likelihood ratio, a parton-to-jet matching is performed in simulated $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)_{\kappa_c}$ events. This is achieved by, in each simulated event, taking the directional information of the simulated parton and matching it to a reconstructed jet with the matching requirement $\Delta R(jet, parton) < 0.3$. All jets which match the initial jet selection are considered for this process and a matching efficiency of ∼80% is achieved for events in which the parton is a charm quark, while an efficiency of ∼75% is achieved for events in which the parton is a gluon. A jet which is matched in this way is labelled as the *associated* jet, while the remaining non-matched jets are labelled as *non-associated* jets. Once this labelling is performed, the distributions of both kinematic variables are extracted as templates for both associated and non-associated jets and treated as probability density functions. To capture kinematic differences associated with higher and lower p_T Higgs boson candidates, this procedure is repeated in different bins of $p_T(H)$ listed in Table 4.2. An example of the templates that are extracted in this way can be seen in Figure 4.3 for the 30–50 GeV bin of the Higgs boson candidate p_T. The full set of templates is shown in Appendix Section B.

Using the extracted templates, a per-jet likelihood evaluation can be conducted in each event. For this, the likelihood ratio

$$\mathcal{L}(x) = \frac{\mathcal{L}_{associated}(x)}{\mathcal{L}_{non-associated}}(x), \text{ with } x \in \Big\{\Delta\phi(H, jet), r\Big\} \tag{4.1}$$

is defined for the two considered variables. From this follows the per-jet likelihood

$$\mathcal{L}(jet) = \mathcal{L}\Big(\Delta\phi\big(H, jet\big)\Big)\mathcal{L}\Big(r\Big). \tag{4.2}$$

Table 4.2: The $p_T(H)$ bins in which the jet selection procedure is performed.

Bin number	$p_T(H)$ range
1	0 – 15 GeV
2	15 – 30 GeV
3	30 – 50 GeV
4	50 – 100 GeV
5	100 – 200 GeV
6	>200 GeV

The jet with the highest associated likelihood in an event is selected as the jet candidate. The distribution of $\mathcal{L}(jet)$ for both associated jets and non-associated jets can be seen in Figure 4.4 for simulated cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ events in which an associated jet can be identified via a parton-to-jet matching. The distribution of the transverse momentum of the selected jet can be seen in Figure 4.5.

Of interest is the efficiency with which the jet associated with the Higgs boson is selected with the described jet selection method. For events in which more than one jet passes the jet selection criteria (which corresponds to ~25% of events), an efficiency of 68% is achieved for associated jet transverse momenta below 50 GeV, with the efficiency exceeding 90% for associated jet transverse momenta above 50 GeV, as seen in Figure 4.6. It should be noted that this calculation only accounts for events in which an associated parton is generated at the matrix element level (see Subsection 4.2.1 for more details) and in which the parton can be matched to a jet.

Cases of incorrect selection of the jet can be understood by considering the distributions of the $\Delta\phi(H, jet)$ and r templates. While for $\Delta\phi(H, jet)$ the distributions of the associated jets are consistently more peaked than for the non-associated jets, there is also some consistent overlap between the two. This is as the non-associated jets are approximately uniformly distributed in $\Delta\phi(H, jet)$, though a recoil effect for these can also be observed for higher $p_T(H)$ values. As a result, while $\Delta\phi(H, jet)$ separates the two categories reasonably well, the remaining overlap can contribute to an incorrect selection. A similar effect can be observed for r, though here the overlap in distributions at lower $p_T(H)$ is more pronounced. This can, together with the overlap observed for $\Delta\phi(H, jet)$, lead to an incorrect selection. However, at higher $p_T(H)$ the associated and non-associated distributions for r become noticeably more separated. This behaviour drives the improvement in jet selection efficiency for greater $p_T(associated\,jet)$, since $p_T(H)$ and $p_T(associated\,jet)$ are correlated.

With this, all components of cH(ZZ$\rightarrow 4\mu$) candidate events can be reconstructed. Events satisfying the described requirements are thus selected for further evaluation.

4.1.3 Reconstruction efficiency

A total selection efficiency of ~10% is achieved on the simulated cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ sample using the described methods. The losses in efficiency can be attributed to the following:

- Approximately 50% of generated events fall outside the geometrical detector acceptance due to the presence of less than four reconstructed muons.
- Of the 50% of generated events within the geometrical detector acceptance, approximately

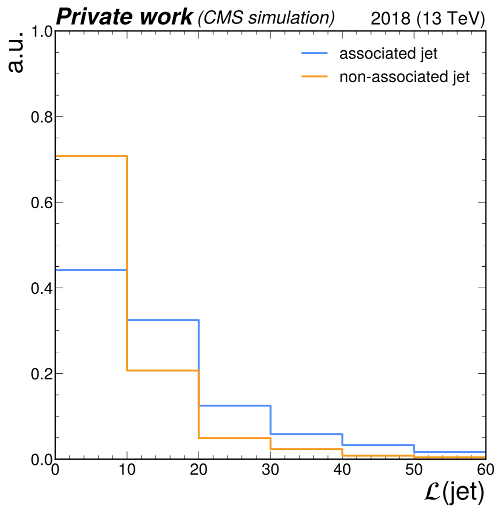


Figure 4.4: The distribution of the jet selection likelihood $\mathcal{L}$(jet) in simulated cH(ZZ → 4μ)$_{\kappa_c}$ events in which an associated jet can be identified via a parton-to-jet matching, shown separately for associated and non-associated jets. It can be seen that on average, applying the jet selection algorithm on the associated jet produces a higher likelihood value than on non-associated jets, as expected.

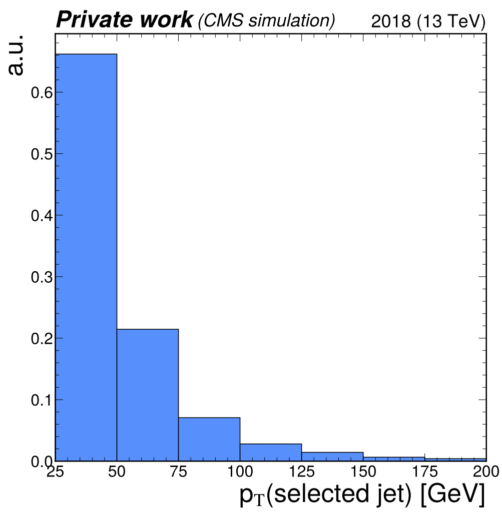


Figure 4.5: The distribution of the selected jet's transverse momentum in simulated cH(ZZ → 4μ)$_{\kappa_c}$ events. As expected, the majority of jets come at relatively low transverse momenta.

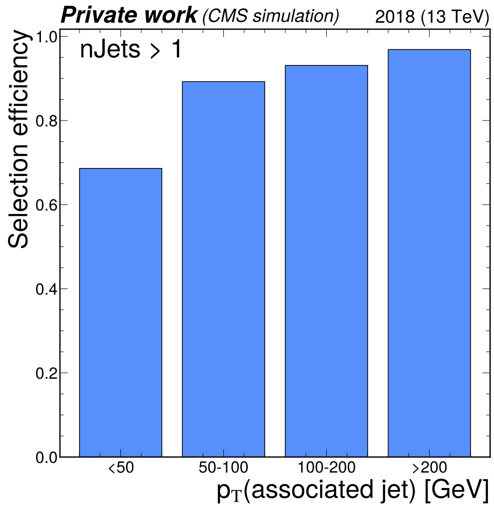


Figure 4.6: The efficiency with which the jet associated with the Higgs boson is selected via the described algorithm in simulated cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ events. This efficiency is shown in bins of the associated jet's transverse momentum for events in which more than one jet passes the jet selection requirements and an associated jet can be identified via a parton-to-jet matching.

50% do not fulfill the described muon selection requirements.

- Approximately 91% of these events pass the Higgs reconstruction, leading to an additional event yield loss of 9%.
- Of the remaining events, the event yield is reduced by close to 60% by applying the jet selection requirements that are imposed.

4.2 Signal and background estimation

The expected contributions of cH(ZZ$\rightarrow 4\mu$) candidate events that are obtained by applying the selection procedure described in Section 4.1 to a given dataset must be estimated. This includes estimating both the signal and background processes. First, the simulation of the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ process via Monte Carlo methods is discussed in Subsection 4.2.1. This is followed by a discussion of the estimation of background processes, which can be placed into two categories. The first consists of the *irreducible* background processes, which refer to processes in which four muons are produced in the hard scattering. These processes are also estimated via Monte Carlo methods and are discussed in Subsection 4.2.2. The second category consists of *reducible* background processes, which refers to processes in which less than four muons are produced in the hard scattering. The estimation of these, based on a data-driven approach, is discussed in Subsection 4.2.3.

4.2.1 Estimation of cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ process

The cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ process is estimated using a simulation generated by MadGraph5_aMC@NLO[19]. The following MadGraph5_aMC@NLO syntax is used, which illustrates some important concepts related to the simulation of the cH process:

```
import model loop_sm_MSbar_yb_yc-yc4FS
define p = g u u~ d d~ s s~ c c~
define j = g u u~ d d~ s s~ c c~
generate p p > h [QCD] @ 0
generate p p > h j [QCD] @ 1
```

In the first line, it can be read off that the loop_sm model is used, a model that implements perturbative calculations in QCD of SM processes at next-to-leading order (NLO). Only the Yukawa couplings of the bottom and charm quarks are included to ensure orthogonality of the cH simulation to simulations of other Higgs boson production processes such as the one initiated by gluon fusion. Additionally, a so-called *four flavour scheme* (4FS) version of the model is used. The flavour scheme denotes which quarks are included as constituents of the proton, in which they are approximated as massless. The 4FS includes the up, down, strange and charm quarks as proton constituents. In contrast to the 4FS, a three flavour scheme 3FS could also be used. Here, the charm quark is not included in the proton and initial state charm quarks must thus instead be produced via gluon splitting, i.e. $g \to c\bar{c}$. In the following two lines of the simulation configuration above, the proton and jet constituents are explicitly defined.

Finally in the last two lines, the processes included in the simulation are defined. These are, calculated at NLO in perturbative QCD, the $pp \to H$ and $pp \to H + j$ processes. Both are included to give the most accurate possible kinematic description of the cH process. The reasoning for this is related to the modelling of the c quark mass and can be better understood in terms of the previously introduced flavour schemes. If the mass of the c quark plays a significant role in the hard scattering, a description like in the 3FS is most apt. Here the charm quark must originate from gluon splitting and is thus described as a massive particle. However, in such a description the charm quark can never be generated as an initial state particle and processes such as $c\bar{c} \to H$ cannot be generated at NLO in perturbative QCD. Conversely, if the mass of the c quark does not play a significant role in the hard scattering, the 4FS description is the most apt. In this case the mass can be approximated to zero and the charm quark can be properly described as an initial state particle originating from the proton.

A best overall description of the cH process is obtained by including both these possibilities in the simulation. This is achieved by including both the $pp \to H$ and $pp \to H + j$ processes at NLO in perturbative QCD in the 4FS. In this way, the 3FS-like description can be generated via NLO contributions to $pp \to H$, while 4FS-like contributions are described by both the $pp \to H$ and $pp \to H + j$ contributions. However, this approach also introduces double counting of processes. These are accounted for using a reweighting procedure, where double counting from parton shower contributions are also accounted for with the FxFx merging scheme.

The types of event topologies generated by these commands can be best understood by categorising them according to the partons that initiate them. In this way, four categories can be identified:

- $\boldsymbol{c\bar{c}}$: At LO, this topology arises from an initial state c and anti-c quark that interact to produce a Higgs boson. Final state partons must thus be produced via NLO contributions to the matrix element (i.e. gluon radiation) or via parton showering. Notably, the charm quarks associated with the Higgs boson are not present in the final state of this type of topology. This means that for the described event selection strategy, this analysis is only be sensitive to these topologies via a jet association that is 'incorrect' by construction. A visualisation of this type of event topology can be seen in Figure 4.8a. It is the most prominently represented topology in the generated sample.

- ***cg***: At LO, this topology arises from an initial state (anti-)c quark that emits a Higgs boson and absorbs a gluon. This topology corresponds to the LO cH process that is discussed in Subsection 2.3.1. Additional final state partons may be produced via the NLO contributions to the matrix element or via parton showering. A visualisation of this type of event topology can be seen in Figure 4.8b. It is the second most prominently represented topology in the generated sample.
- ***gg***: This topology arises from two initial state gluons which split to form $c\bar{c}$ pairs that interact to produce a Higgs boson. Since this is already considered a NLO contribution to the matrix element by the generator, additional final state partons can only be generated via parton showering. A visualisation of this type of event topology can be seen in Figure 4.8c. It contributes to only a small fraction of events in the generated sample.
- ***cl***: This topology arises from an initial state (anti-)c quark that interacts with an additional initial state light quark via gluon exchange, which proces a Higgs boson. Since this is already considered a NLO contribution to the matrix element by the generator, additional final state partons can only be generated via parton showering. A visualisation of this type of event topology can be seen in Figure 4.8d. Like the *gg* topology, it contributes to only a small fraction of events in the generated sample.

An overview of the relative occurrence of these topologies in the generated sample can be seen in Figure 4.9a. Clearly, the $c\bar{c}$ topology dominates with a relative proportion $> 60\%$, while the *cg* topology only comprises $\sim 35\%$ of events. Since the jet identification effectively plays no role in the former topology, it is worth considering how the selection described in Section 4.1 affects the relative occurrence of each topology. This is shown in Figure 4.9b. It is observed that the selection algorithm (specifically the jet selection) introduces a significant bias towards *cg* topologies, as expected and desired. This effect, specifically of the jet selection, is demonstrated in Figure 4.7. Here, the p_T spectrum of the p_T-leading jet in each event, split by topology type, is shown with no event selection applied. The only exception to this is the $\Delta R(jet,\, selected\; muons) > 0.4$ requirement, which is applied since energy deposits in the calorimeters around the muon trajectory (from, for example, Bremsstrahlung of the muon) may be misidentified as a jet, thus falsifying this spectrum. As can be observed from Figure 4.7, the $p_T >$ 25 GeV requirement on jets is primarily responsible for the shift in the relative occurrence of $c\bar{c}$ and *cg* topologies before and after applying the event selection criteria. However, clearly a relative proportion of close to 40% of $c\bar{c}$ topologies still remains after the selection is applied. These may still contribute towards the sensitivity of the analysis to the charm-Yukawa coupling accordingly, though they are typically associated with light flavour jets. This motivates the use of an inclusive statistical evaluation strategy, exemplified by the use of the full CvsB and CvsL DeepJet discriminator distributions of the selected jet.

To capture the effects of uncertainties associated with the previously described choice of a FS in the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ simulation, the approach taken in the previous analyses of the cH(WW) and cH($\gamma\gamma$) channels is also applied here. This method uses two additional signal samples that specifically simulate the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ process exclusively in either the 3FS or 4FS. This corresponds to configurations in which the charm *must* originate from gluon splitting or originate directly from the proton, respectively. These effectively represent the respective extremes of the

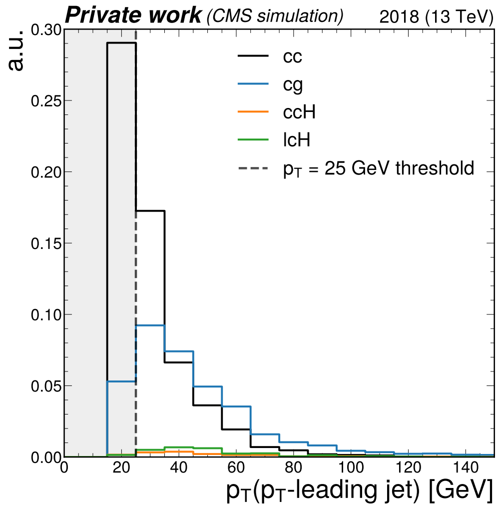
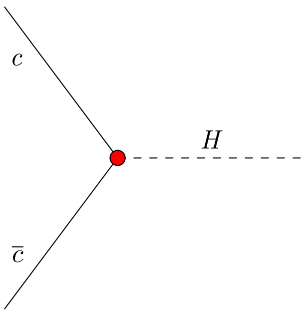


Figure 4.7: The p_T spectrum of the p_T-leading jet, split by topology type (normalised to the sum of all types), in a simulation of the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ process before any selection criteria are applied, with the exception of the $\Delta R(jet, \, selected\ muons) > 0.4$ requirement. As can be seen, applying the $p_T > 25$ GeV criterion on these jets, indicated by the dashed line and shaded area, leads to a significant change in the relative occurrence of the $c\bar{c}$ and cg topologies.

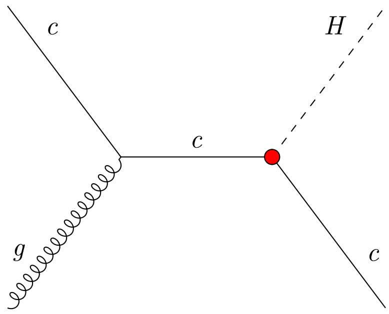
description used in the nominal sample, which combines both descriptions. From these additional samples, an uncertainty envelope is constructed for the discriminator distributions used in the statistical evaluation presented in Section 4.4. This envelope is determined by taking the relative discrepancy between the 3FS and 4FS samples and interpreting it as an uncertainty band around the nominal 4FS FxFx sample distributions. This leads to very large yield uncertainties in the signal modelling, with an almost 60% total uncertainty on the signal yield being observed with respect to the nominal signal yield. Note that this approach may overestimate the FS uncertainty, given the heuristic way in which it is derived. The reason that this approach is taken is that currently, no better motivated approaches have been proposed. Potential analysis improvements to alleviate the effects of this FS uncertainty are further discussed in Subsection 4.5.5. An overview of all used signal samples is given in Table 4.3, in which the inclusive decay of the Higgs boson decay to leptons ℓ is simulated.

Table 4.3: cH(ZZ$\rightarrow 4\ell)_{\kappa_c}$ samples used in this work

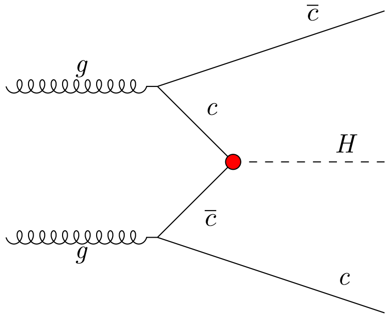
Process	$\sigma\cdot$BR (fb)	**Number of simulated events**
cH(ZZ$\rightarrow 4\ell)_{\kappa_c}$ 4FS FxFx	0.025	4,854,537
cH(ZZ$\rightarrow 4\ell)_{\kappa_c}$ 3FS	0.014	1,941,000
cH(ZZ$\rightarrow 4\ell)_{\kappa_c}$ 4FS	0.022	1,628,000



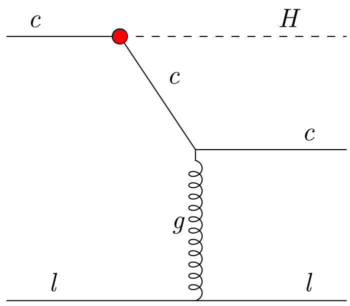
(a) The $c\overline{c}$ topology at lowest order.



(b) The cg topology at lowest order.

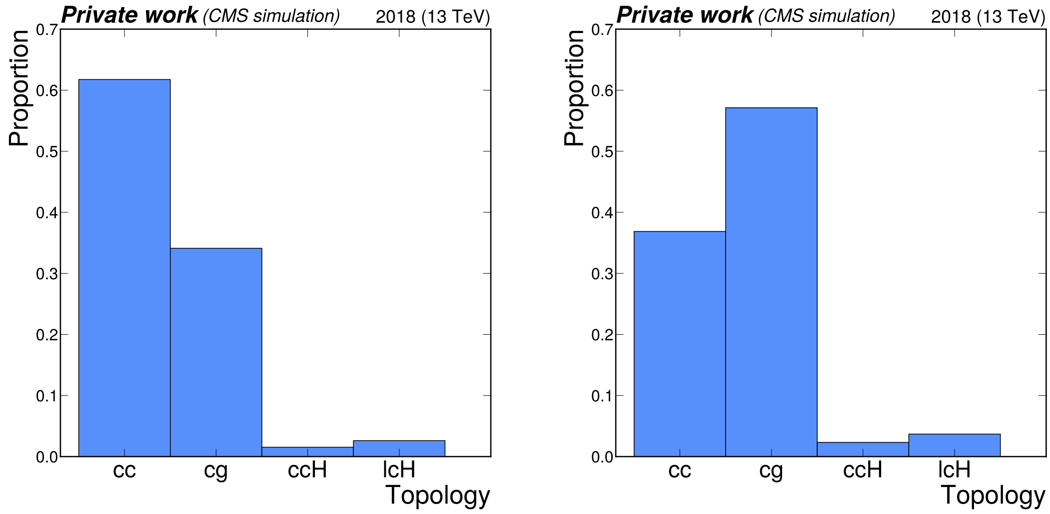


(c) The gg topology at lowest order.



(d) The cl topology at lowest order.

Figure 4.8: Examples of Feynman diagrams that illustrate the four different event topologies generated in the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ sample at the lowest order in perturbative QCD. The red dots represent the occurrence of the y_c coupling.



(a) Before applying event selection.

(b) After applying event selection.

Figure 4.9: The relative occurrence of the $c\bar{c}$, cg, gg and cl event topologies in the simulated $\mathrm{cH(ZZ \to 4\mu)}_{\kappa_c}$ sample before and after applying the event selection.

4.2.2 Estimation of irreducible backgrounds

Irreducible background processes are background processes that produce the same prompt final state particles as the signal process in question. Thus, processes which produce four final state muons along with the presence of one or several jet(s) constitute the irreducible background. These again fall into two categories, namely those where the four muons originate from a Higgs boson and those in which they do not. When analysing for example the mass spectrum of Higgs boson candidates, the processes of the former category are thus clearly resonant around the Higgs mass of 125 GeV, while those of the latter category may take on a more continuous, broader shape. The irreducible backgrounds considered in this work can be summarised as:

- **Gluon fusion (ggH)**: In these processes, a pair of gluons produce an intermediary (top) quark loop that produces a Higgs boson. While technically the charm quark can also appear in this loop, this contribution is not simulated. The reason for this is that the relative contribution of each quark flavour to the loop depends on the strength its respective Yukawa coupling. Given the very small mass of the charm quark in comparison to the mass of the top quark of ~173 GeV [2], the former's contribution to the production cross section is considered negligible. As a result, gluon fusion processes are considered to be insensitive to the charm-Yukawa coupling.

  The Higgs boson that is produced in gluon fusion processes may decay to four muons, while parton radiation may produce additional jets. An example Feynman diagram of the ggH process can be seen in Figure 4.10a.

- **Top quark pair associated Higgs production (ttH)**: In ttH processes, a pair of top quarks is produced alongside a Higgs boson. The Higgs boson decay may produce four muons while the decay of the top quark pairs produces additional jets, such as b jets. An example Feynman diagram of the ttH process can be seen in Figure 4.10b.

- **Vector boson associated Higgs production (VH)**: The vector boson produced alongside a Higgs boson may refer to W$^{\pm}$ and Z bosons. Thus, three different process types contribute to this background, namely W^{+}H, W^{+}H and ZH. The Higgs boson decay may produce four muons, while a W$^{\pm}$ boson decay as well as a Z boson decay may produce additional jets. An example Feynman diagram of these processes can be seen in Figure 4.10c.

- **Vector boson fusion (qqH)**: In vector boson fusion processes, two vector bosons emitted by initial state quarks fuse to produce a Higgs boson. The Higgs boson decay may produce four muons while additional final state quarks that produce jets are also present. An example Feynman diagram of the qqH process can be seen in Figure 4.10d.

- **Top quark plus quark associated Higgs production (tqH)**: In tqH processes, a top quark as well as a quark of a different flavour are produced alongside a Higgs boson. The Higgs boson decay may produce four muons while the additional quarks will produce jets. An example Feynman diagram of the tqH process can be seen in Figure 4.10e.

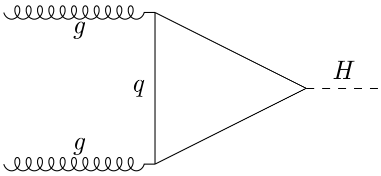
- **ZZ production from gluons (gg→ZZ)**: A pair of Z bosons can be produced from a pair of initial state gluons via an intermediate quark loop. The Z boson decays may produce four muons while additional parton radiation may produce jets. An example Feynman diagram of the gg→ ZZ process can be seen in Figure 4.10f.

- **ZZ production from quarks (qq→ZZ)**: A pair of Z bosons can be produced from two initial state quarks. The Z boson decays may produce four muons while additional parton radiation may produce jets. An example Feynman diagram of the qq→ZZ process can be seen in Figure 4.10g.

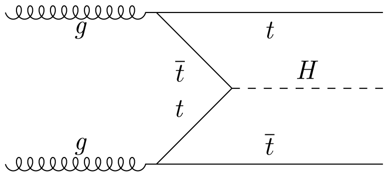
- **The bH process**: The bH process is analogous to the cH process, with the difference being that the associated quark is a bottom quark, instead of a charm quark. As a result, this process similarly produces a jet, while the Higgs boson may similarly decay to four muons. Note that, like for the cH(ZZ → 4μ)$_{\kappa_c}$ estimation, this estimation of the bH process only accounts for contributions sensitive to the Yukawa coupling of the bottom quark. The corresponding insensitive contributions are modelled like the κ_c-insensitive cH(ZZ → 4μ) contributions, as discussed below.

The irreducible backgrounds are estimated using Monte Carlo simulation at NLO in perturbative QCD. The exception to this is the gg→ZZ process, for which only a sample with LO accuracy is available. To correct for this, a K-factor of 1.7 is applied to the cross section, to extrapolate the cross section to NLO [82]. Similarly, a K-factor of 1.1 is applied to the NLO cross section of the gg→ZZ process, extrapolating the cross section to next-to-next-to-leading (NNLO) order in perturbative QCD [83]. This is done to account for the significant, non-negligible contributions to the qq→ZZ process at NNLO. An overview of the samples that are used can be found in Table 4.4. Like with the cH(ZZ → 4μ)$_{\kappa_c}$ process, additional samples are used for the bH($ZZ \to 4\mu$) background, to account for an uncertainty related to the choice of flavour scheme. However, here the five flavour scheme (5FS) is now the nominal flavour scheme, so that the bottom quark is included in the proton, and the 4FS is used to estimate the bH contribution where the initial state bottom quark originates from gluon splitting.

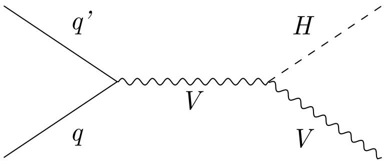
Though it may not immediately be apparent, the κ_c-insensitive cH(ZZ → 4μ) contributions are effectively also modelled by this set of estimations. The most clear and prominent example of



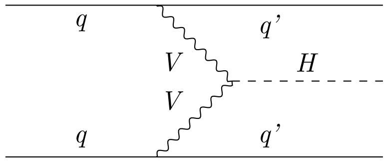
(a) The ggH process.



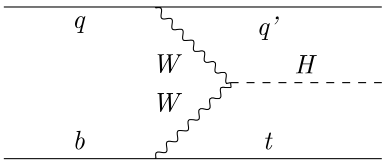
(b) The $t\bar{t}$H process.



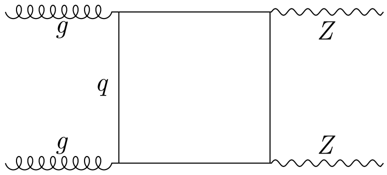
(c) The VH process.



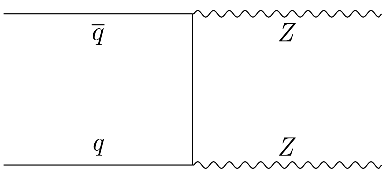
(d) The qqH process.



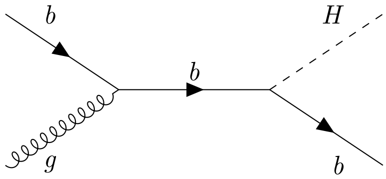
(e) The tqH process.



(f) The gg→ZZ process.



(g) The qq→ZZ process.

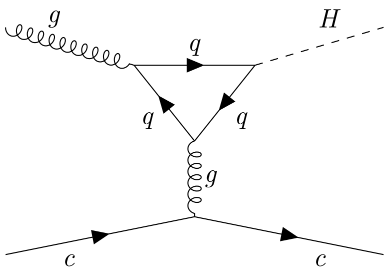


(h) The bH process.

Figure 4.10: Representative Feynman diagrams of the irreducible background processes relevant to this analysis. This includes the ggH, $t\bar{t}$H, VH, qqH, tqH, gg→ZZ, qq→ZZ and bH processes. When no final state parton is explicitly present in the diagram it is implied via, for example, parton radiation.

this is the gluon fusion process estimation. Here, additional charm quarks may be produced alongside the Higgs boson, either from higher order QCD corrections or via parton powering. An example of the former is an NLO gluon fusion contribution where the Higgs boson that is produced via a quark loop indirectly recoils off an initial state charm quark via an intermediary gluon that couples to the quark loop and charm quark. This corresponds to the left example Feynman diagram shown in Figure 4.11 and will clearly produce a cH(ZZ$\rightarrow 4\mu$) final state, though no direct interaction between the Higgs boson and the charm quark takes place. An example of the parton showering case on the other hand, is where an additional, radiated gluon splits to a charm quark pair, again producing a final state with a Higgs boson and at least one additional charm quark. This is visualised in the right diagram of Figure 4.11. Note that such a case is of course not exclusive to the gluon fusion process, but may take place for any Higgs boson production mode. Finally, processes such as vector boson fusion or tqH production may also produce cH(ZZ$\rightarrow 4\mu$) final states, since the additional quarks involved in these production processes may also be charm quarks.

The fact that κ_c-insensitive cH(ZZ$\rightarrow 4\mu$) contributions can evidently originate from a range of different processes that produce the same signature as the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ process makes this background a challenging aspect of this analysis. Accordingly, an extraction of the signal that is indeed sensitive to κ_c thus has a significant dependent on the correct modelling of the processes in question, given that the κ_c-insensitive and κ_c-sensitive cH(ZZ$\rightarrow 4\mu$) contributions cannot be experimentally well-distinguished. However, there is a general difficulty associated with modelling Higgs boson production in association with the emission of light quarks such as the charm quark. Specifically, this difficulty arises from the strong scale dependence of the relevant calculations at a fixed order in perturbative QCD, which is a product of the charm quark and the Higgs boson being associated with very different energy scales, given their large mass difference. As a result, the process modelling becomes strongly dependent on the choice of renormalisation and factorisation scales. This strong dependence, in turn, presents as a significant source of systematic uncertainty. This uncertainty is most extreme in the case of the gluon fusion process estimation, for which a yield changing effect of approximately 40% is observed for the uncertainty associated with the renormalisation scale. Since the gluon fusion process is expected to be the primary background process in the region sensitive to the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ process, this thus represents a key systematic uncertainty of the analysis presented here.



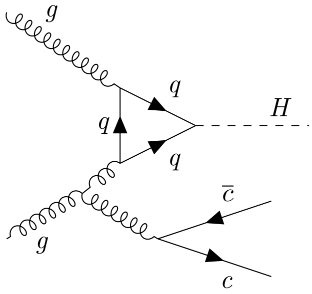


Figure 4.11: Feynman diagrams that represent how contributions to the gluon fusion process may produce cH final states that are insensitive towards κ_c. The left diagram shows an example of an explicit higher order perturbative QCD correction to the gluon fusion process that produces a cH final state. The right diagram shows an example where a cH final state is produced due to gluon radiation and subsequent gluon splitting $g \to c\bar{c}$, which may be described via parton showering.

Table 4.4: Simulated processes used for background estimation in this work. The listed processes represent the irreducible backgrounds to the cH(ZZ$\to$ 4μ) process, with the exception of the WZ$\to 3\ell\nu$ process, which is used solely in the estimation of the irreducible backgrounds.

Process	σ·BR	**Number of simulated events**
ggH(ZZ→4L)	12.18 fb	940,000
ttH(ZZ→4L)	0.393 fb	490,043
W^-H(ZZ→4L)	0.147 fb	193,893
W^+H(ZZ→4L)	0.232 fb	298,647
ZH(ZZ→4L)	0.668 fb	544,550
qqH(ZZ→4L)	1.044 fb	477,000
tqH(ZZ→4L)	0.0213 fb	969,000
gg→ZZ(4μ)	1.59 fb	786,136
gg→ZZ(4τ)	1.59 fb	493,998
gg→ZZ(2μ2τ)	3.19 fb	434,000
qq→ZZ/Zγ^* →4L	1.256 pb	84,366,000
bH 5FS FxFx	0.192 fb	3,869,701
bH 4FS	0.099 fb	2,000,000
bH 5FS	0.153 fb	1,944,000
WZ→ $3\ell\nu$	4.43 pb	9,821,283

4.2.3 Estimation of reducible backgrounds

Reducible background processes are background processes that do not produce the same final state particles as the signal process but where misidentification of physics objects can still falsify the sought-after signature. Since jets are typically abundant in most collisions, this amounts to the misidentification of additional muons for this analysis. Major contributions to the reducible background are expected to come from the Drell-Yan process, as well as other processes which may produce at least two leptons such as $t\bar{t}$ production [84]. Example Feynman diagrams of these two processes can be seen in Figure 4.12. However, the simulation of misidentified muons

(a) The Drell-Yan process. (b) The $\mathrm{t\bar{t}}$ process.

Figure 4.12: Representative Feynman diagrams of the Drell-Yan and $\mathrm{t\bar{t}}$ processes, which are expected to contribute significantly to the reducible background of the $\mathrm{cH(ZZ \to 4\mu)}$ analysis. Jets are produced via, for example, parton radiation for the Drell-Yan process, while the decay of the top quarks in the $\mathrm{t\bar{t}}$ process automatically produces (b) jets from the hadronisation of the b quarks that top quarks predominantly decay to.

is subject to significant modelling uncertainties. This is as muon misidentification can occur from, for example, hadronic activity that enters the muon chambers. The low-energy behaviour of the resulting hadronic interactions is not well understood and thus difficult to model.

As such, data-driven methods can be employed instead. This involves determining the misidentification rate of muons in data and applying it to a sideband region, from which the contributions of reducible backgrounds are extrapolated into the signal region. Here, a sideband region refers to a region in which no signal process contributions are expected, while the signal region refers to the region in which the signal process is expected to be found. The methodology used to this end is presented in this section and follows that of [32].

Determination of muon misidentification rate

To determine the misidentification of muons with respect to the tight muon requirement outlined in Subsection 3.2.3, a three-muon selection is applied to data. Specifically, events with a $Z \to \mu^+\mu^-$ decay that also contain a third reconstructed muon are targeted. Since hard-scattering processes that produce a Z boson are not expected to produce any additional muons, the third reconstructed muon (referred to as the *probe muon*) is assumed to be one that is misidentified as such. To determine a misidentification rate, the ratio of probe muons that pass the tight muon requirement with respect to those that pass the loose muon requirement (also per the definition in Subsection 3.2.3) is calculated. This procedure is performed in bins of the probe muon p_T for the barrel ($|\eta| \leq 1.2$) and endcap ($|\eta| > 1.2$) regions, respectively. The exact reconstruction algorithm that is applied is the following:

1. Events that contain at least two muons that pass the tight identification requirement, and where the third passes at least the loose identification requirement, are chosen. The p_T-leading muon is required to satisfy $p_T > 20$ GeV and the sub-leading muon is required to satisfy $p_T > 10$ GeV. Additionally, the HLT_IsoMu24 trigger requirement must also be met. Lastly, to ensure two muons are not spuriously reconstructed from shared tracks, it is required that each muon candidate is separated from the others by $\Delta\mathrm{R} > 0.02$.

2. Opposite-sign muon pairs are merged into Z boson candidates and the candidate closest to the nominal Z boson mass is taken as the final boson Z candidate. Additionally, the

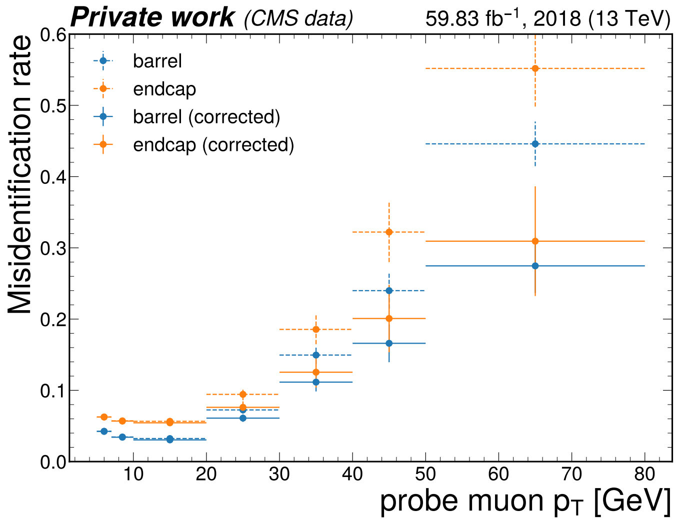


Figure 4.13: The muon misidentification rate in bins of the probe muon transverse momentum, in both the barrel and endcap regions. The doted lines represent the uncorrected misidentification rate while the solid lines represent the misidentification rate where contributions from WZ$\rightarrow 3\ell\nu$ processes are subtracted.

invariant mass of any combination of opposite-sign muons must satisfy $m(\mu\mu) > 4$ GeV, to remove any contributions from low mass resonances such as J/ψ.

3. The remaining, third muon not selected as part of the Z boson candidate is taken as the probe muon.

The misidentification rate of the probe muon that is determined in this way in bins of the probe muon p_T, can be seen in Figure 4.13. However, contributions from processes that indeed produce three muons in the hard-scattering process must be subtracted. This consists primarily of WZ$\rightarrow 3\ell\nu$ processes that artificially inflate the calculated misidentification rate at higher probe muon p_T[84]. This contribution is subtracted using simulation. It is this corrected version of the misidentification rate that is used in the following section.

Application of muon misidentification rate

The muon misidentification rate is applied to a sideband region to estimate the contribution of reducible backgrounds to the previously described cH(ZZ$\rightarrow 4\mu$) selection. It is useful to introduce some of the related terminology at this point. The four-pass (4P) region henceforth refers to the inclusive region that is defined via the cH(ZZ$\rightarrow 4\mu$) selection. The three-pass-one-fail (3P1F) and two-pass-two-fail (2P2F) regions respectively refer to regions in which the cH(ZZ$\rightarrow 4\mu$) reconstruction is performed as previously described but where only three (two) of the muons satisfy the tight identification criteria and the remaining one (two) muon(s) satisfy only the loose identification criteria. The 3P1F and 2P2F are collectively referred to as the application region (AR).

The extrapolation of the AR to the 4P region is performed using the previously determined misidentification rate. The prescription for this application can be obtained from the misidentification rate f_i that is defined as

$$f = \frac{N_{\text{tight}}}{N_{\text{loose}}}. \tag{4.3}$$

Here N_{loose} and N_{tight} are the number of probe muons in a given bin that pass the loose and tight identification criteria, respectively. From this, the relation

$$N_{\text{tight}} = N_{\text{loose}} f \tag{4.4}$$

follows. Since one is interested in the contributions of muons which pass the loose but not the tight identification requirement in the AR, Eq. (4.4) can be reinterpreted as

$$N_{\text{loose}-\text{not}-\text{tight}} = N_{\text{loose}}(1-f). \tag{4.5}$$

By substituting this back into Eq. (4.4), the desired prescription is found:

$$N_{\text{tight}} = N_{\text{loose}-\text{not}-\text{tight}} \frac{f}{(1-f)}. \tag{4.6}$$

Thus, for each muon that fails the tight identification requirements but passes the loose ones in the 3P1F and 2P2F regions, the weight $f/(1-f)$ is applied, where f is the p_T and η dependent muon misidentification rate. This leads to the following expressions for the individual contributions of the AR to the 4P region:

1. **2P2F**: Since this region contains two muons that pass the loose identification criteria but not the tight, the weight $f/(1-f)$ must be applied twice. The total contribution of this region in the 4P region can thus be written as

$$N_{4\text{P}}^{(2\text{P2F})} = \sum_{k}^{N_{2\text{P2F}}} \frac{f_k^{(3)}}{(1-f_k^{(3)})} \frac{f_k^{(4)}}{(1-f_k^{(4)})}, \tag{4.7}$$

   where the $f_k^{(3/4)}$ is the misidentification rate associated with each of the non-passing muon for the k-th event. Major contributors to the 2P2F region are the Drell-Yan and $t\bar{t}$ processes, which produce only two prompt muons.

2. **3P1F**: Since this region contains only one muon that passes the loose identification criteria but not the tight, the weight $f/(1-f)$ is only applied once. The total contribution of this region in the 4P region can thus be written as

$$N_{4\text{P}}^{(3\text{P1F})} = \sum_{k}^{N_{3\text{P1F}}} \frac{f_k^{(4)}}{(1-f_k^{(4)})}, \tag{4.8}$$

   where the $f_k^{(4)}$ is the misidentification rate associated with the non-passing muon for the k-th event. The major contributor to the 3P1F region consists of WZ→ $3\ell\nu$ due to the presence of three prompt leptons.

To obtain the total contribution of the AR to the 4P region, potential contaminations of the 2P2F and 3P1F regions must be accounted for. The first source of contamination is the potential overlap of the 3P1F region with contributions from the 2P2F region, where an additional muon has been misidentified in the former. This may lead to an overestimation of the 3P1F region. Such contributions can be estimated via the term

$$N_{\text{exp.}}^{(\text{3P1F})} = \sum_{k}^{N_{\text{2P2F}}} \left(\frac{f_k^{(3)}}{(1-f_k^{(3)})} + \frac{f_k^{(4)}}{(1-f_k^{(4)})}\right). \tag{4.9}$$

Effectively, here contributions from the 2P2F region are weighed with the misidentification rate for either muons that may fail the tight identification criteria. To then extrapolate this contribution to the 4P region, the fake rate must once again be applied, this time to the respective complementary muon. This produces the final term

$$N_{\text{4P,exp.}}^{(\text{3P1F})} = \sum_{k}^{N_{\text{2P2F}}} \left(\frac{f_k^{(3)}}{(1-f_k^{(3)})}\frac{f_k^{(4)}}{(1-f_k^{(4)})} + \frac{f_k^{(4)}}{(1-f_k^{(4)})}\frac{f_k^{(3)}}{(1-f_k^{(3)})}\right) \tag{4.10}$$

$$= 2\sum_{k}^{N_{\text{2P2F}}} \left(\frac{f_k^{(3)}}{(1-f_k^{(3)})}\frac{f_k^{(4)}}{(1-f_k^{(4)})}\right) \tag{4.11}$$

An additional source of potential contamination is the contribution of four muon processes in which muons either fail the tight identification criteria or fall outside the detector acceptance. This is primarily relevant in the 3P1F region with contributions from ZZ production processes, which again lead to an overestimation of the 3P1F region. Contributions to the 4P region from this, denoted with $N_{\text{4P}}^{(\text{ZZ,3P1F})}$ are estimated via simulation with the same prescription as data events in the 3P1F region:

$$N_{\text{4P}}^{(\text{ZZ,3P1F})} = \sum_{k}^{N_{\text{3P1F}}^{ZZ}} \frac{f_k^{(4)}}{(1-f_k^{(4)})}. \tag{4.12}$$

The total contribution to the 4P region N_{4P} may finally be estimated by taking the sum of the 2P2F and 3P1F contributions and subtracting the discussed contamination terms. This leads to

$$N_{\text{4P}} = N_{\text{4P}}^{(\text{2P2F})} + N_{\text{4P}}^{(\text{3P1F})} - N_{\text{4P,exp.}}^{(\text{3P1F})} - N_{\text{4P}}^{(\text{ZZ,3P1F})} \tag{4.13}$$

$$= \sum_{k}^{N_{\text{3P1F}}} \frac{f_k^{(4)}}{(1-f_k^{(4)})} - \sum_{k}^{N_{\text{3P1F}}^{ZZ}} \frac{f_k^{(4)}}{(1-f_k^{(4)})} - \sum_{k}^{N_{\text{2P2F}}} \left(\frac{f_k^{(3)}}{(1-f_k^{(3)})}\frac{f_k^{(4)}}{(1-f_k^{(4)})}\right) \tag{4.14}$$

By applying this strategy, an inclusive yield of 6.59 events for the reducible background in the 4P region is observed. Directly around the Higgs mass peak of 120 GeV $< m(H) <$ 130 GeV, this reduces to a yield of 0.42 events. However, the derived observable distributions are significantly discontinuous. This arises from the strong influence of statistical fluctuations on the small yields of the individual control regions. The effects of these fluctuations are compounded by the subtractions performed in Eq. (4.13), which introduce a significant number of bins with negative, unphysical yields. Given the comparatively small contribution of this background, a simplification to the estimation is made to resolve this problem. Specifically, the shapes of the distributions for observables in the 2P2F region, which is the most populated control region in the reducible background estimation, are taken and scaled to the yield of the full estimation. This approach can be motivated by a comparison of the shapes of the 2P2F and 3P1F distributions, which show reasonable similarity in the regions relevant to this analysis, as seen in Appendix. Section C. The contribution of the reducible background estimation to distributions of observables that is derived in this way is shown in the following section (Section 4.3).

A number of systematic uncertainty sources may influence the reducible background estimation. This includes the effects of statistical uncertainties on each bin of the muon misidentification rate, effects related to differences in the background composition between different control regions as well as the discussed simplification introduced in the estimation strategy. However, due to the technical limitations resulting from the limited event yield, it is difficult to estimate the effect of these uncertainty sources. Considering this background estimation's comparatively small contribution, especially in the region enriched in signal, a conservative rate uncertainty of 100% is introduced instead. This allows the yield of this background to effectively be determined by a best fitting value in the statistical evaluation described in Section 4.4.

4.3 Validation of cH(ZZ$\rightarrow 4\mu$) event selection in sideband regions

Here, a validation of the described event selection is presented. Specifically, the 2018 dataset of events recorded by the CMS detector is compared to the sum of the previously presented estimation of background processes. The latter is also collectively referred to as the background estimation. This comparison is performed in sideband regions of the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ signal, which corresponds to the region in which no significant cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ contribution is expected to be found. To this end, all events outside the 100 GeV $< m$(H) $<$ 130 GeV region are chosen, a choice motivated by the spectrum of the mass of the Higgs boson candidate shown in Figure 4.14. Here, the entire mass spectrum is shown for the signal and background estimation. This illustrates the presence of the cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$ signal around the Higgs boson candidate mass peak at $m(H) = 125$ GeV, as well as the absence of a significant signal contribution in the previously defined sideband region.

In Figure 4.15a – Figure 4.15c, the spectrum of the Higgs boson candidate transverse momentum and the mass of the Z_1 and Z_2 candidates is shown for the sideband region, respectively. This is followed by the transverse momentum, the CvsL and CvsB discriminators, as well as the likelihood score of the selected jet in Figure 4.15d – Figure 4.15g. As can be observed from these distributions, a reasonable agreement between the data and background estimation is achieved, when accounting for the small event yields.

4.4 Statistical inference

For the statistical inference process, the Combine [85] tool is used. Specifically, a fit is performed to derive expected 95% CL upper limits on $\sigma\mathcal{B}$(cH(ZZ$\rightarrow 4\mu)_{\kappa_c}$) with the CL$_\mathrm{s}$ method [86]. This is done using distributions obtained from the signal and background estimations of cH(ZZ$\rightarrow$ 4μ) candidates for some chosen discriminating variable. To perform such a fit, a statistical model describing the discriminator distributions of the signal and background processes must be constructed. This is described in Subsection 4.4.1. Additionally, an uncertainty model describing

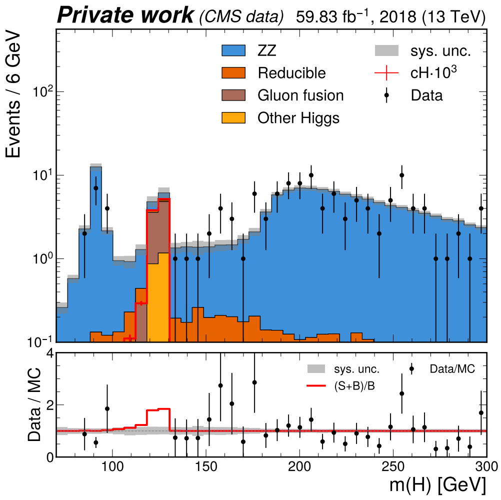


Figure 4.14: The mass spectrum of the Higgs boson candidate, showing the agreement between the 2018 dataset and simulation in the sideband regions of the spectrum. The contribution from a $\mathrm{cH}(\mathrm{ZZ} \rightarrow 4\mu)_{\kappa_c}$ signal, scaled by a factor 10^3 for better visibility, is also shown. The signal contribution is clearly concentrated around the Higgs boson mass of $m(H) = 125$ GeV, as expected.

the various systematic uncertainty sources that affect the analysis is required. This is described in Subsection 4.4.2.

4.4.1 Statistical model

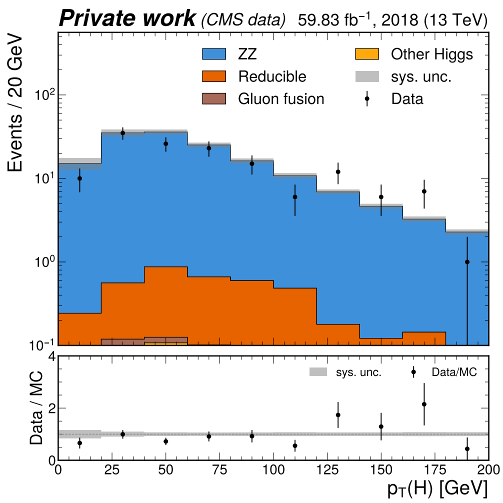
Assuming the absence of a measurable signal, given the very small signal cross section, an expected 95% CL upper limit on $\sigma\mathcal{B}(\mathrm{cH}(\mathrm{ZZ} \rightarrow 4\mu)_{\kappa_c})$ may can be derived. This in turn may be interpreted as an expected limit on κ_c using the prescription derived in Subsection 2.3.2. To do this, a statistical model is required that describes the predicted number entries in each bin of the discriminator histogram, according to a given hypothesis. Since each of the bins are filled in a discrete counting process, a Poissonian Ansatz with

$$\mathcal{P}(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{4.15}$$

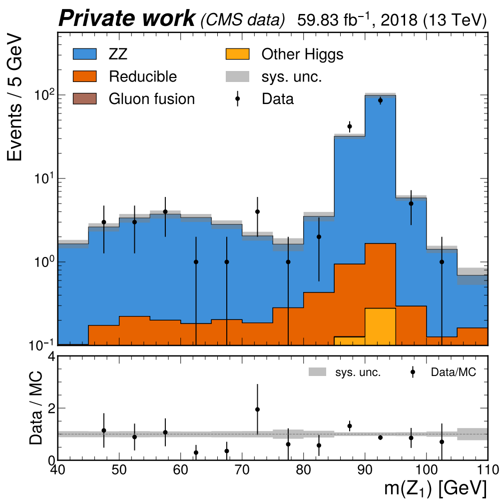
is a suitable choice. Here, λ denotes the expectation value with k observed events. In the i-th bin of the discriminator distribution, the expectation value λ_i is calculated using the models of the signal and background processes discussed in Section 4.2. This amounts to $\lambda_i = \mu s_i + b_i$ where s_i and b_i are the yields of the signal and background estimations respectively. The signal strength modifier μ allows for an arbitrary scaling of the signal contribution and is used as a floating parameter in the inference process. From this, a combined binned likelihood function

$$\mathcal{L}(d \,|\, \mu \cdot s(\boldsymbol{\theta}) + b(\boldsymbol{\theta})) = \prod_{i \in \mathrm{bins}} \mathcal{P}(d_i \,|\, \mu \cdot s(\boldsymbol{\theta}) + b(\boldsymbol{\theta})) \text{ x} \prod_{j \in \mathrm{nuis.}} \mathcal{C}(\hat{\theta}_j \,|\, \theta_j). \tag{4.16}$$

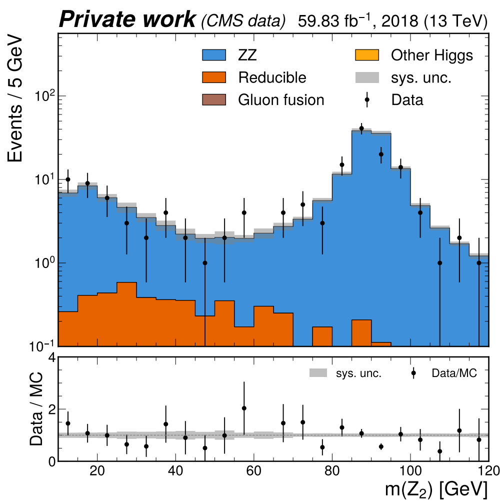
may be derived for some set of measured data d_i. Here, an uncertainty model is introduced via the nuisance parameters $\boldsymbol{\theta}$ that account for uncertainties related to the signal and background



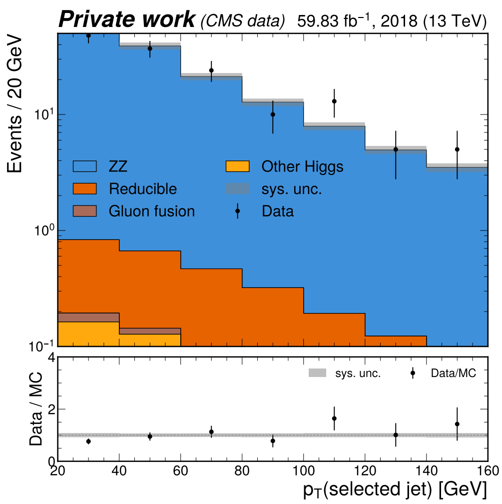
(a) Higgs boson candidate transverse momentum.



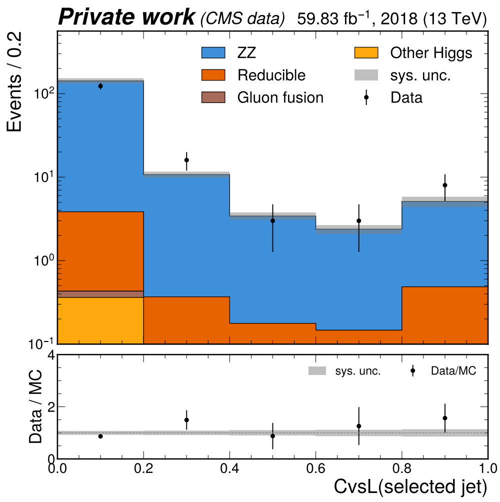
(b) Mass of the Z_1 candidate.



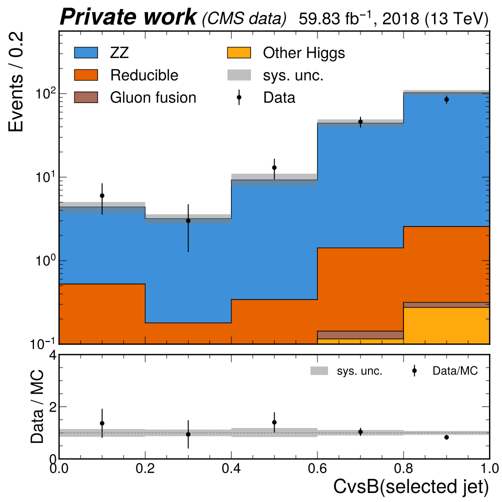
(c) Mass of the Z_2 candidate.



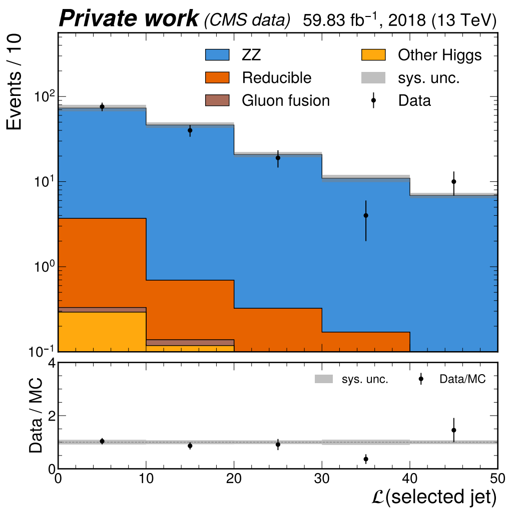
(d) Transverse momentum of the selected jet.



(e) CvsL discriminator of the selected jet.



(f) CvsB discriminator of the selected jet.



(g) Likelihood score of the selected jet.

Figure 4.15: Spectra of cH(ZZ $\rightarrow 4\mu$) candidate observables, showing the agreement between the 2018 dataset and the background estimation in the sideband region of the Higgs boson candidate mass. Variables related to the Higgs boson reconstruction and the selected jet in sideband regions of the Higgs boson candidate mass are shown.

estimation. These follow a distribution $\mathcal{C}$ and may alter the scaling of the signal and background contributions as well as their histogram shape with respect to the discriminator that is used. The estimate of $\boldsymbol{\theta}$ that is used to obtain estimations of s_i and b_i in the inference process is denoted by $\hat{\boldsymbol{\theta}}$ and is obtained by finding the global maximum of the likelihood. The specifics of the uncertainty model employed in this analysis, which is parametrised by the nuisance parameters, are discussed in Subsection 4.4.2. To characterise hypotheses, the test statistic that is used is the *profile likelihood ratio*:

$$q_\mu = -2\ln\left(\frac{\mathcal{L}(d\,|\,\mu\cdot s(\hat{\boldsymbol{\theta}}_{\boldsymbol{\mu}}) + b(\hat{\boldsymbol{\theta}}_{\boldsymbol{\mu}}))}{\mathcal{L}(d\,|\,\hat{\mu}\cdot s(\hat{\boldsymbol{\theta}}) + b(\hat{\boldsymbol{\theta}}))}\right), \tag{4.17}$$

where $(\hat{\mu}, \hat{\boldsymbol{\theta}})$ are the parameter values that globally maximise the likelihood while $\hat{\boldsymbol{\theta}}_{\boldsymbol{\mu}}$ maximises the likelihood for a given μ. A significant advantage of this test statistic is that in the large sample limit, the distribution $f(q\,|\mu)$ approaches the χ^2_k distribution with $k = 1$ degrees of freedom, according to Wilk's theorem [87]. This is very useful as it allows for a simple, analytic calculation of a p-value

$$p_\mu = \int_{q_{\mathrm{obs.}}}^{\infty} f(q\,|\,\mu)\mathrm{d}q. \tag{4.18}$$

With this p-value, the $\mathrm{CL_s}$ method may be applied [86]*. This involves calculating the value

$$\mathrm{CL_s} = \frac{p_\mu}{1-p_0} = \frac{\int_{q_{\mathrm{obs.}}}^{\infty} f(q\,|\,\mu)\,\mathrm{d}q}{\int_{q_{\mathrm{obs.}}}^{\infty} f(q\,|\,0)\,\mathrm{d}q}. \tag{4.19}$$

From this, a 95% CL upper limit on $\sigma\mathcal{B}(\mathrm{cH}(\mathrm{ZZ}\to 4\mu)_{\kappa_c})$ can be derived by performing a scan of μ, until a value $\mu^{(95\%)}$ is found that satisfies

$$\mathrm{CL_s} \leq \alpha = 0.05. \tag{4.20}$$

A feature of this method is that large overlaps of test statistic distributions between the H_μ and H_0 hypotheses, due to the very small signal cross section associated with H_μ, can be accounted for. This is achieved by weighting p_μ with $(1-p_0)^{-1}$, thus increasing the $\mathrm{CL_s}$ value for larger overlaps i.e. for similar test statistic distributions and decreasing the respective exclusion power. As a result, the $\mathrm{CL_s}$ method is considered to be well suited for setting upper exclusion limits on small, non-negative parameters, such as the signal strength of $\sigma\mathcal{B}(\mathrm{cH}(\mathrm{ZZ}\to 4\mu)_{\kappa_c})$ in this analysis.

4.4.2 Uncertainty model

The sources of uncertainty in the estimation of the signal and background processes are described in this section. These consist of two types. The first are referred to as shape uncertainties and may change the normalisation as well as the shape of the distributions in question. The effect of these uncertainties are captured by creating variations of these distributions, which are interpreted as $\pm 1\sigma$ variations of the underlying parameter describing the uncertainty source. During the statistical evaluation process, an interpolation is performed between the nominal distributions and the respective variations to characterise their effect. The second uncertainty type is

*Note, the specific convention with which the p-value associated with $f(q\,|\,0)$, p_0, is calculated in [86]. Specifically, the test statistic distribution below the observed test statistic value is integrated to give p_0. This is motivated by the fact that it is in this direction that the overlap with $f(q\,|\,\mu)$ lies.

referred to as normalisation uncertainties. These only affect the normalisation of distributions without affecting the shape and can thus be described with a single, real parameter. Both types of uncertainties are associated with a respective nuisance parameter in the statistical model introduced in Subsection 4.4.1. The nuisance parameters describing shape uncertainties are assumed to be normally distributed, while those that describe normalisation uncertainties are assumed to be log-normally distributed.

Theoretical uncertainties common to simulation

A number of theoretical uncertainties that are common to all simulated samples are included in the uncertainty model. This includes:

- Shape uncertainties related to the choice of the normalisation and factorisation scales μ_R and μ_F in the simulated Monte Carlo samples. These are varied independently by factors of 2 and 0.5 using a reweighting technique that is applied to the simulation, thus introducing two shape-changing nuisance parameters per process. Yield changes associated with the uncertainty on μ_R range from approximately 1% to 40%, with the largest change being observed for the ggH process. The yield change of the signal process that is associated with this uncertainty source is approximately 23%. Yield changes associated with the uncertainty on μ_F are comparatively smaller, ranging from 1% to 15%. The bH process accounts for the largest change while for the signal process a yield change of approximately 9% is observed.

- A shape uncertainty related to the modelling of the Parton Distribution Function (PDF). This uncertainty is obtained from the NNPDF3.1 PDF set that is used in simulation [88] and includes 100 PDF variations associated with variations of individual parameters. These are combined into a single nuisance parameter with the prescription

$$\sigma^+ = \sigma^- = \sqrt{\sum_{i=1}^{N_{\text{par}}} (F_i - F_0)^2}\,, \tag{4.21}$$

  where F_0 is the nominal value of the observable in question and F_i is the varied value associated with one of the N_{par} parameter variations [89]. A yield uncertainty of approximately 6.5% on the total yield of the background estimation is associated with this uncertainty. The yield uncertainty on the signal process is somewhat larger, at approximately 12%.

- An uncertainty related to the choice of the strong coupling constant α_s in the Monte Carlo simulation. This uncertainty is associated with a total yield change of approximately 2%.

- Shape uncertainties related to the tuning of the parton showering used in the Monte Carlo generator. These are implemented by varying the value of α_s used in the modelling of initial-state radiation and final-state radiation independently, introducing two nuisance parameters. Both uncertainties are associated with yield changes of less than 1% for the background processes. These uncertainties are associated with slightly larger yield changes for the signal process at 2–3%.

Experimental uncertainties common to simulation

The following experimental uncertainties are common to all simulated samples:

- Shape uncertainties related to the jet energy scale. For this, a simplified schema is used in which closely correlated uncertainty sources are grouped and described through 11 individual nuisance parameters. Sources of uncertainty include, for example, the limited set of data available for the calibration procedure or differences in calibration response to different jet flavours. The yield uncertainty introduced due to these uncertainties ranges between <1% to 7% for background and signal processes.

- A shape uncertainty related to the smearing method used to correct the jet energy resolution. This is described with a single nuisance parameter and is associated with small yield changes of approximately 1.5%.

- Shape uncertainties related to the calibration of the jet flavour-tagging algorithm that is used in the jet-driven approach described in Section 4.5. The effect of these are captured via 13 nuisance parameters and include, for example, uncertainty sources related to the individual phase spaces targeted in the calibration or the limited set of available data. No yield uncertainties are associated with these nuisance parameters. This is by design, as no flavour tagging discriminants are used for the event selection criteria. A number of jet flavour-tagging uncertainty sources such as the jet energy resolution, parton shower tuning and pile-up reweighting uncertainties also affect the calibration of the flavour-tagging algorithm. These are fully correlated to the corresponding, nominal uncertainties considered in the analysis.

- A shape uncertainty related to the pileup identification method that is used. This uncertainty is described with a single nuisance parameter and is associated with yield changes of less than 1%.

- Shape uncertainties related to the muon calibration used in the analysis. The response of the muon identification criteria, the isolation as well as the muon trigger path in simulation is calibrated to match data. Uncertainties associated with this calibration are combined into a single nuisance parameter. The yield uncertainty associated with this uncertainty source is approximately 4.5%. This is increased for the signal process at approximately 9%.

- A normalisation uncertainty related to the luminosity, which is known with a limited precision, and with which the simulation is scaled. This amounts to an uncertainty of 2.5% on the yields for the 2018 dataset [54] and is described with a single nuisance parameter.

- A shape uncertainty related to the application of the pileup reweighting procedure, with which the pileup profile of simulation is matched to that of data. This uncertainty is described with a single nuisance parameter. The uncertainty on process yields that is introduced by this uncertainty is almost negligible for both background and signal processes.

FS uncertainty in cH($\mathbf{ZZ} \to 4\mu$)$_{\kappa_c}$ and bH($\mathbf{ZZ} \to 4\mu$) modelling

As previously discussed Subsection 4.2.1, the FS uncertainty is a systematic uncertainty that is unique to cH($ZZ \to 4\mu$)$_{\kappa_c}$ and bH($ZZ \to 4\mu$) simulation. To parametrise this, an envelope representing a $\pm 1\sigma$ variation is extracted from the respective 3FS (4FS) and 4FS (5FS) samples and applied to the nominal cH($ZZ \to 4\mu$) 4FS FxFx (bH($ZZ \to 4\mu$) 5FS FxFx) sample. A shape changing nuisance parameter reflecting the FS uncertainty is thus introduced for both the cH($ZZ \to 4\mu$)$_{\kappa_c}$ and bH($ZZ \to 4\mu$) processes, respectively. For the former, this corresponds to an uncertainty on the yield of approximately 59%, and 42% for the latter.

Uncertainty related to reducible background modelling

Due to the minor nature of the reducible background contribution, a simple normalisation uncertainty is introduced to account for uncertainties associated with the reducible background estimation procedure. The uncertainty is set as 100% of the yield, giving significant freedom in the fitting procedure to find a best-fit value of the corresponding nuisance parameter.

Uncertainties related to statistical precision of background estimation

The employed simulation estimation methods themselves include a statistical uncertainty due to the limited number of events that are generated to estimate each process. This results in the introduction of a nuisance parameter per process and per bin of the final discriminator. This can be simplified to introducing only a single, per-bin nuisance parameter by using the Barlow-Beeston approach [90][91].

Additional uncertainties

The following additional uncertainties are also included:

- Two normalisation uncertainties are introduced for the theoretical uncertainty associated with the application of the K-factors for the qq→ZZ and gg→ZZ processes. These amount to an uncertainty of 4% and 1.5% on the yields, respectively.
- A normalisation uncertainty originating from the theoretical uncertainty on the branching ratio $\mathcal{B}(H \to ZZ \to 4\ell)$ [20]. This is applied to simulated samples in which a Higgs boson is produced and corresponds to an uncertainty of 2% on the yields.

A technical list of all nuisance parameters that parametrise the above discussed uncertainty sources is given in Appendix Section D.

4.5 Results

In this section, the results obtained by performing the statistical evaluation described in Section 4.4 on a simulated Asimov dataset [87] are discussed. The Asimov dataset consists of the nominal estimation of the signal and background processes, which is taken to be representative of the dataset that would be obtained from applying the analysis to data. It is used instead of data since the region sensitive to the $\mathrm{cH}(\mathrm{ZZ} \to 4\mu)_{\kappa_c}$ signal remains blinded at the time of publishing of this thesis. The reasons for this is that a final, unblinded analysis is intended to encompass both the Run-2 and Run-3 datasets recorded by the CMS detector. Given that the recording of the latter dataset is still ongoing at the time of writing, this work can thus be understood as a development of the methods required for such an analysis.

Here, two different discriminators are considered and expected 95% CL upper limits on $\sigma\mathcal{B}(\mathrm{cH}(\mathrm{ZZ} \to 4\mu)_{\kappa_c})$ and κ_c are presented for each. Additionally, the effects of the previously discussed individual uncertainty sources on the derivation of the nominal result are estimated. Subsequently, a fit of the background estimation to the 2018 dataset recorded by the CMS experiment is presented in the sideband regions of the Higgs boson candidate mass. This is followed by extrapolations of the obtained results to full Run-2 and High Luminosity LHC scenarios. Finally, a number of potential improvements to the analysis are discussed.

4.5.1 Expected upper limits on $\sigma\mathcal{B}(\mathbf{cH(ZZ} \rightarrow 4\mu)_{\kappa_c})$

As a baseline scenario, the mass of the Higgs boson candidate is used as a discriminator for the statistical evaluation. This is performed with the aforementioned simulated Asimov dataset, which is scaled to an integrated luminosity of 59.83 fb^{-1} to correspond to the dataset recorded by the CMS detector in the 2018 era of Run-2. The distribution used for this corresponds to the spectrum previously shown in Figure 4.14. For this scenario, an expected 95% CL upper limit of $\sigma\mathcal{B}(\mathrm{cH(ZZ} \rightarrow 4\mu)_{\kappa_c}) = 1088$ is derived with respect to the SM expectation. This corresponds to an expected 95% CL upper limit of $|\kappa_c| = 191$.Given the resonant nature of the peak around the Higgs boson candidate mass, which is shared by all Higgs production processes, this approach to setting upper limits on κ_c can be interpreted as a measurement of when an excess in jet-associated Higgs boson production becomes statistically significant at the 95% CL.

To improve upon this baseline scenario, it is investigated whether using the information of the selected jet can improve upon this result. This is henceforth referred to as the *jet-driven approach* To this end, the DeepJet CvsB and CvsL discriminators are used in a pseudo two-dimensional fit, again using the same Asimov dataset. Specifically, an *unrolling* procedure is applied to project the two-dimensional information provided by the CvsB and CvsL discriminators into a single, final discriminator†. This involves constructing a histogram of the CvsB discriminator in bins of the corresponding CvsL value. The chosen bins for the CvsL discriminator are [0, 0.4, 0.7, 1.0]. The CvsB discriminator is split into five equidistant bins. This choice is made to find a good compromise between the separation of signal enhanced and background dominated regions, and the smoothness of the distribution. Additionally, it is required that the Higgs boson candidate mass be in the [100, 130] GeV window to reduce the presence of non-Higgs boson backgrounds. The resulting histogram can be seen in Figure 4.16a. In this scenario, an expected 95% CL upper limit of $\sigma\mathcal{B}(\mathrm{cH(ZZ} \rightarrow 4\mu)_{\kappa_c}) = 999$ times the SM expectation is derived. This corresponds to an expected upper limit of $|\kappa_c| = 176$. While this is an improvement with respect to the baseline scenario, the gain in analysis sensitivity that is achieved is comparatively small. This behaviour can be understood by the fact that a large portion of the jets of the $\mathrm{cH(ZZ} \rightarrow 4\mu)_{\kappa_c}$ signal are associated with a light jet-like discriminator (bins $0-4$). As a result, a large component of the signal process exhibits similarity to the discriminator distributions associated with background processes such as the ggH process. These are also typically associated with a light jet-like discriminator and thus impact the sensitivity to the signal. Despite the signal-like region enriched with the $\mathrm{cH(ZZ} \rightarrow 4\mu)_{\kappa_c}$ process (for example bins $11-14$), the impact of this signal enriched region appears to be subdominant since its relative proportion is smaller compared to the other signal component. Motivated by this finding, it is investigated whether an additional cut on the CvL discriminator yields an improvement in the expected upper limit that is obtained. However, it is found that this is not the case for a number of potential CvL discriminator cut values. Instead, the expected 95% CL upper limit that is derived in this way consistently increases, likely due to the fact that a significant portion of the signal is also removed with this requirement.

Overall, a significant drawback to the described jet-driven approach is that the predicted number of events is around one tenth of an event in a majority of the discriminator bins for the 2018 dataset, as indicated by the fashed line in Figure 4.16a. In total, only two bins have a yield predicted to be greater than one. This impairs the interpretability of this approach when applying it to 2018 data, as the shape of the discriminator distributions predicted by simulation cannot be reflected in the data due to the many fractional predicted bin yields. This problem is expected to remain even for an expansion of the analysis to the full Run-2 dataset recorded

†This is done since the Combine tool only accepts 1D input distributions

by the CMS detector, corresponding to an integrated luminosity 138 fb^{-1}. Only with significantly larger datasets such as the expected dataset of the HL-LHC, assuming a total integrated luminosity of 3000 fb^{-1} [12], would this issue be resolved. A possible solution to reduce the impact of this statistical limitation would be to reduce the number of bins used in constructing the discriminator. However, given the significant reduction in bin numbers that would be required to achieve this, the information gained from the shape of the discriminator distributions is effectively lost. Accordingly, such a version of the measurement would become very similar to baseline scenario, albeit with reduced control over the non-Higgs backgrounds that dominates in the sideband regions of the Higgs boson candidate mass distribution.

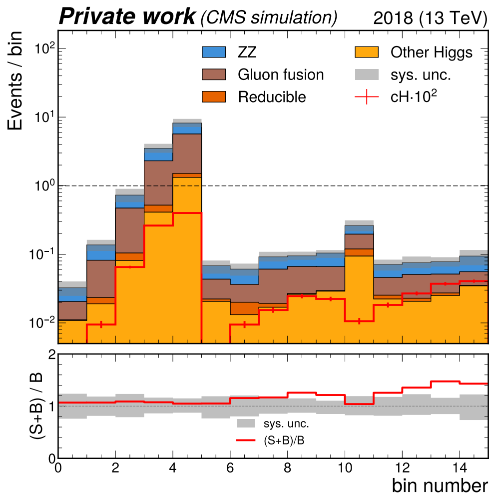
Since the described statistical limitation originates from the criteria imposed on the Higgs boson candidate mass, an additional variation on the jet-based approach is trialed in which the criteria are not applied. The resulting discriminator distribution is shown in Figure 4.16b. However, the expected 95% CL upper limit of $\sigma\mathcal{B}(\mathrm{cH}(\mathrm{ZZ}\rightarrow 4\mu)_{\kappa_c}) = 3244$ that is derived is significantly worse than in the baseline scenario. This can again be explained by the large overlap in the distributions of the signal and background processes in the light jet-like region of the discriminator. However, here the impact of the background processes is greater than previously, due to the large contributions of ZZ production processes that is reintroduced. As a result, there is a significant loss of sensitivity to the signal process.

Given the described challenges associated with the jet-driven approach, the result of the baseline approach is taken as the nominal result.

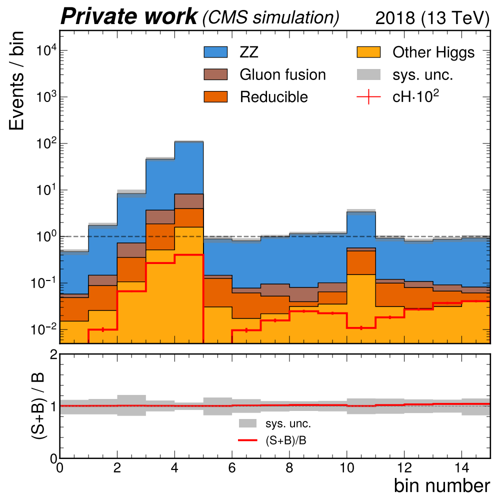
4.5.2 Impact of nuisance parameters

For the nominal result, the impacts of the nuisance parameters on the derived expected 95% CL upper limit are investigated. This can be achieved by performing a series of fits on the previously introduced Asimov dataset, which is now injected with a $\mathrm{cH}(\mathrm{ZZ}\rightarrow 4\mu)$ signal at a signal strength r corresponding to the nominal result. Generally speaking, the goal of these fits is to maximise the profile likelihood ratio and thus find a best fit value of the $\mathrm{cH}(\mathrm{ZZ}\rightarrow 4\mu)_{\kappa_c}$ signal strength $\hat{r}$. Accordingly, in a nominal fit on the Asimov dataset, $\hat{r}$ corresponds to the signal strength with which the $\mathrm{cH}(\mathrm{ZZ}\rightarrow 4\mu)_{\kappa_c}$ signal is injected into the dataset. The impact of the nuisance parameters on this nominal fit can now be gauged by performing a series of additional fits. In these fits, the individual nuisance parameters θ_i are varied by $\pm\sigma$ to find the best fit values $\hat{r}_{\pm\theta_i}$. The size of the differences $\Delta\hat{r}_i = \hat{r} - \hat{r}_{\pm\theta_i}$ thus give an indication of the uncertainty on $\hat{r}$ that each nuisance parameter introduces. With this technique, the leading uncertainty sources that limit the sensitivity of the nominal result can be identified. These are shown in Figure 4.17, which shows the $\Delta\hat{r}_i$ values of the leading 30 (of a total of 70) uncertainty sources.

As shown, the flavour scheme uncertainty of the $\mathrm{cH}(\mathrm{ZZ}\rightarrow 4\mu)_{\kappa_c}$ simulation is clearly the leading uncertainty source. Given the relatively large yield changing effect associated with this uncertainty and the very small expected signal strength of the $\mathrm{cH}(\mathrm{ZZ}\rightarrow 4\mu)_{\kappa_c}$ process, this is expected. Other leading uncertainty sources are also of theoretical origin and similarly introduce significant yield changes to signal and background processes. These include, for example, the uncertainty on the renormalisation scale (especially for the ggH process), the PDF modelling, and the uncertainty associated with the factorisation scale in the signal estimation. The leading experimental uncertainty is the uncertainty associated with the muon calibration. Uncertainties associated with the jet energy scale and jet energy resolution produce comparatively subdominant effects. This is also expected, given that in the nominal result, the discriminator is not directly related



(a) Higgs boson candidate mass ∈ [100,130] GeV.



(b) Inclusive Higgs boson candidate selection.

Figure 4.16: The unrolled discriminator distributions for the jet-driven approach. These represent the CvsB discriminator, in three bins of the CvsL discriminator, for the selected jet of the reconstructed cH(ZZ → 4μ) candidates. It is shown once inclusively in the Higgs boson candidate mass spectrum and once with Higgs boson candidate mass criteria applied. As seen in the lower panel of both figures, the relative contribution of a cH(ZZ → 4μ)$_{\kappa_c}$ signal with respect to the background estimation generally increases with increasing bin number in both distributions. The region most enriched with signal are the right-most bins of both distributions. This is as expected, since these bins correspond to selected jets with both a high CvsL and CvsB value. Here, the cH(ZZ → 4μ)$_{\kappa_c}$ signal contribution is scaled by a factor of 10^2 to improve its visibility. Note the dashed line that represents the threshold where at least one, or more, events are expected per bin.

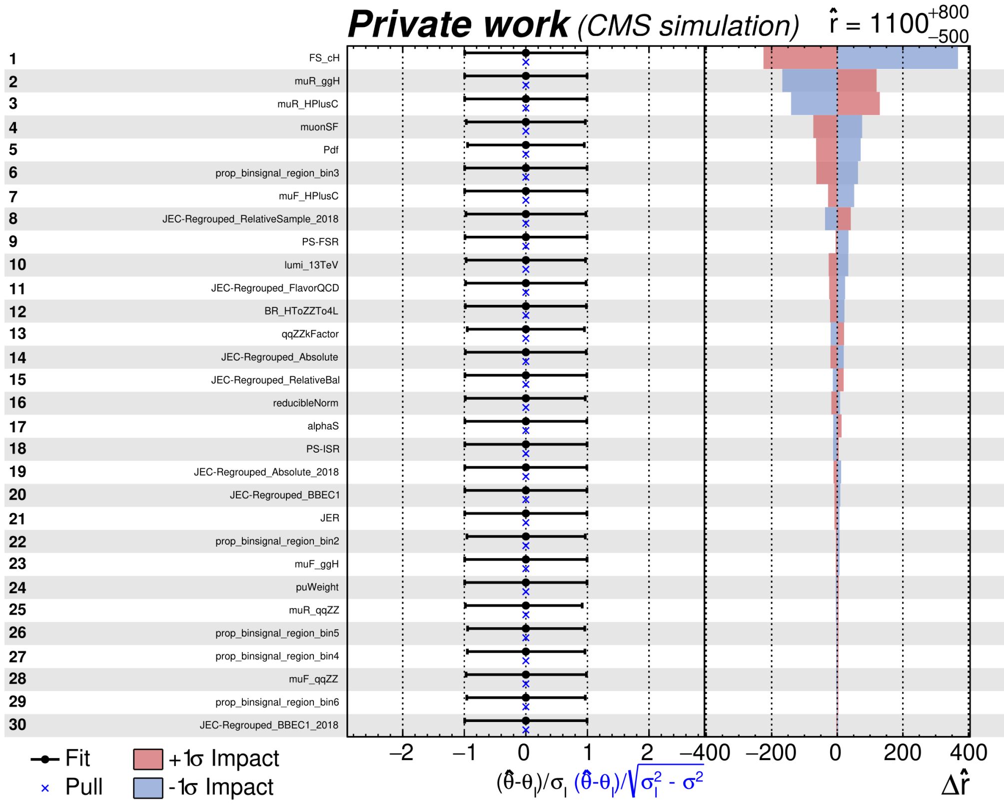


Figure 4.17: Figure showing the 30 leading uncertainty sources in a fit of the $\sigma\mathcal{B}(\text{cH}(\text{ZZ} \rightarrow 4\mu)_{\kappa_c})$ signal strength, ordered by their relative impact on the fit. In the left panel the nuisance parameter names are shown. In the central panel, the fitted nuisance parameter values as well as their so-called pulls are shown. On the right, the $\Delta\hat{r}_i$ values are shown for the $\pm\sigma$ variations associated with each nuisance parameter.

to the jet. The discriminator distributions associated with the 30 leading uncertainty sources are shown in Appendix Section D. Only the shape changing uncertainties are shown since normalisation uncertainties only modify the scaling of distributions.

Lastly, to ensure that the statistical model does not behave significantly differently during the fitting procedure in the absence of signal, the impacts of the nuisance parameters in a fit scenario where no signal is injected into the Asimov dataset is also investigated. The leading 30 uncertainty sources in this scenario can be seen in Appendix Section E. No unexpected changes are observed, demonstrating the robustness of the model. Here, the uncertainties associated with the signal model no longer play a noticeable role, as expected in the absence of a signal in the Asimov dataset.

4.5.3 Fitting nuisance parameters in sideband regions

By performing a fit of the estimated background contributions to data in sideband regions, which are defined as the region outside of the [100,130] GeV window in the Higgs boson candidate mass spectrum, it can be demonstrated that a number of nuisance parameters can already be fixed to best-fit values in this way. This fit produces some improvement in agreement between the background estimation and data in these sideband regions, as shown in Figure 4.18. Here, *pre-fit* refers to the unmodified, nominal background estimation, while *post-fit* refers to the background estimation that has been modified by fixing the associated nuisance parameters to their best-fit values. The observed difference between pre-fit and post-fit distributions is small, indicating that already the pre-fit background estimation model describes the data reasonably well. This also indicates that the effect of the considered systematic uncertainty sources is subdominant to the significant statistical fluctuations observed in data. It should be noted that, looking at the lower panel of Figure 4.18, a slight trend may be interpreted in the ratio of the post-fit distribution to the data. However, it is unclear whether this is an effect of the background process estimation in the sideband region or also a result of statistical fluctuations. This behaviour could be re-evaluated in future analyses of this phase space with higher yields.

The best-fit values of the individual nuisance parameters can be seen in Figure 4.19. These fall into three categories. The first contains nuisance parameters associated with processes where a Higgs boson is produced. Since contributions from these are primarily in the [100,130] GeV Higgs boson candidate mass region, no information can be gained about the respective nuisance parameters from a fit in the sideband region. The second category contains nuisance parameters associated with non-Higgs processes, such as ZZ production. These are indeed fixed to best-fit values that best reflect the data and are distributed within the $\pm 1\sigma$ band around the respective nominal values. This reflects the indication that no unexpectedly large deviations between the nominal background process estimation and data are observed. Additionally, for some nuisance parameters, the relative uncertainty on the nuisance parameter value can be constrained. This is, for example, the case for the nuisance parameter describing the renormalisation scale of the qq$\rightarrow$ZZ process. However, the uncertainties for the majority of nuisance parameters remain unconstrained. The last category contains the nuisance parameters assigned to the per-bin statistical uncertainty associated with the estimation of background processes. These best-fit values of these are again reasonably distributed around their respective nominal values.

4.5.4 Extrapolation to larger datasets & comparison to other cH channels

To make a comparison to published cH(WW) and cH($\gamma\gamma$) results of the CMS collaboration, a statistical extrapolation of the baseline scenario to the full Run-2 dataset of the CMS detector is performed. This consists of scaling the prediction of the Asimov dataset derived for the 2018 era to an integrated luminosity of 138 fb^{-1}. However, it should be noted that this extrapolation neglects any changes in detector operating conditions and assumes no changes in any of the uncertainty sources. This means this result should only be taken as an indication of the analysis sensitivity that may be expected. In this scenario, an expected 95% CL upper limit of $\sigma\mathcal{B}(\mathrm{cH}(\mathrm{ZZ}\rightarrow 4\mu)_{\kappa_c}) = 725$ is derived with respected to the SM expectation ($|\kappa_c| = 128$). With this result, the presented cH(ZZ$\rightarrow 4\mu$) analysis is expected to be comparatively less sensitive to κ_c than the aforementioned analyses, which report an expected 95% CL upper limit of $|\kappa_c|$ = 95 in the cH(WW) channel and $|\kappa_c|$ = 72.5 in the cH($\gamma\gamma$) channel, respectively. Since the sensitivity of the analysis presented in this work appears to be strongly constrained in part by

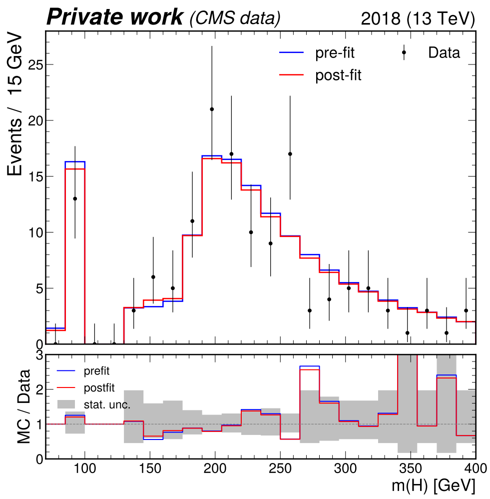


Figure 4.18: The pre- and post-fit distributions of the background estimation model in the sideband region of the Higgs boson candidate mass spectrum for the 2018 data taking era of the CMS detector. The distribution observed for data is also shown and, in the lower panel, the ratio of the pre-fit and post-fit distributions with respect to data can be seen. A reasonable agreement to data is observed already for the pre-fit distribution, indicating that the model is able to describe the data well within the statistical fluctuations observed in data.

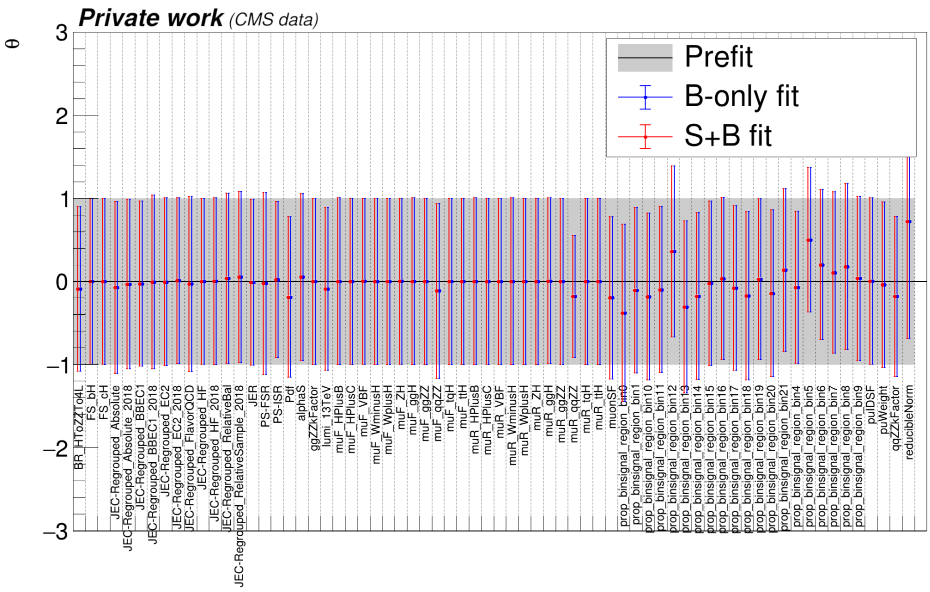


Figure 4.19: The best-fit values of the nuisance parameters θ that describe the uncertainty model of the background estimation in the baseline scenario. These values are determined in a fit to data in the sideband region of the Higgs boson candidate mass.

signal selection efficiency, possible explanations for this might be the larger Higgs boson decay branching ratios in the H(WW) and H($\gamma\gamma$) channels compared to H(ZZ$\rightarrow$ 4μ). However, a direct comparison is difficult due to significantly different background process compositions and signal reconstruction efficiencies in all three analyses.

It is, however, worth noting that performing any cuts on the CvL jet flavour tagging discriminator is found to decrease the analysis sensitivity in this work. This can be explained by the fact that a significant part of the κ_c-sensitive cH signal is associated with light-jet like discriminator values, either due to signal contributions that indeed produce associated light flavour jets, or due to jet flavour misidentification. The previous cH(WW) and cH($\gamma\gamma$) analyses however, perform their initial c-jet selection by indeed making such a cut. Since the properties of the associated jet that are observed in this work are not expected to differ significantly across cH channels, it may be that the kinematics-based jet association developed here would thus positively impact the sensitivity of those analyses.

Similarly to the extrapolation to the full Run-2 dataset, the baseline scenario may also be extrapolated to the expected 3000 fb^{-1} dataset of the High-Luminosity LHC (HL-LHC). Again, it should be noted that this neglects changes to the CMS detector and operating conditions, which are expected to differ significantly during HL-LHC operation [92]. In this case, an expected 95% CL upper limit of $\sigma\mathcal{B}$(cH(ZZ$\rightarrow$ 4μ)$_{\kappa_c}$)$=$ 318 times the SM expectation ($|\kappa_c| = 58$) is derived. Additionally, an extrapolation to the expected HL-LHC for the jet-driven approach is made, since the previously described statistical limitations become less significant in this scenario. Here, an expected 95% CL upper limit of $\sigma\mathcal{B}$(cH(ZZ$\rightarrow$ 4μ)$_{\kappa_c}$)$=$ 154 times the SM expectation ($|\kappa_c| = 29$) is derived. Clearly, this represents a significant improvement with respect to the baseline approach in a HL-LHC scenario, thus motivating a jet-driven approach in analysis scenarios with significantly larger datasets. The reasons for this is that, in such a scenario, information from differential distributions can be leveraged more effectively as the limited signal efficiency becomes less influential.

4.5.5 Potential analysis improvements

A number of potential analysis improvements may be considered for future implementation, besides using a larger dataset. One such possibility would be to loosen the muon selection criteria that are applied. This would allow for a more reliable estimation of the reducible background component as well as increase the signal selection efficiency, potentially enhancing the analysis sensitivity as a result. Additionally, the nuisance parameters of the uncertainty model could also potentially be more effectively constrained, which may also contribute to such an enhancement. The motivation to pursue this modification is especially clear in the jet-driven analysis approach, where it could alleviate the discussed statistical constraints. Another method through which an increase in event yields can also be achieved, would be the inclusion of final states with electrons in the analysis. This would introduce $H(ZZ \rightarrow 4e)$ as well as $H(ZZ \rightarrow 2\mu 2e)$ final states, clearly broadening the targeted phase space and thus increasing the expected signal yield. However, the efficiency with which electrons can be identified differs to the identification efficiency of muons and would also introduce additional background processes to the analysis. As a result, the effect of such an inclusion would need to be studied in greater detail.

An additional factor that clearly affects the signal acceptance significantly is the minimum jet transverse momentum criterion of 25 GeV that is applied. Given that the majority of the cH signal is associated with jets at lower momenta, lowering this value could accordingly increase the signal yield. However, this would also require a dedicated calibration of the jet-related properties that are used in the analysis, such as the flavour tagging discriminants. Accordingly, the development of dedicated low transverse momentum jet flavour-tagging algorithms could be very interesting to pursue, given such an algorithm would naturally be calibrated for the low transverse momentum region. Note this not only benefits the presented analysis but can clearly also be applied to other cH analysis channels, since the analyses of the cH($\gamma\gamma$) and cH(WW) channels are similarly limited by such a minimum jet transverse momentum criterion.

Also of potential interest are, for example, multivariate event classification methods. These have already been implemented in the cH($\gamma\gamma$) and cH(WW) channels, where they are used to differentiate κ_c-sensitive cH process contributions from the continuous background as well as the Higgs boson backgrounds. As such, the impact of such methods should also be explored in future analyses of the cH(ZZ $\rightarrow 4\mu$) channel. Note however, that the expected event yields in the cH(ZZ $\rightarrow 4\mu$) channel are significantly lower than in the other two channels, which means the methods developed in those channels may have to be adapted accordingly. For example, in the cH($\gamma\gamma$) analysis a events are categorised into 27 separate regions and each region is used in a simultaneous fit of the associated diphoton mass distributions to extract the signal. Clearly, such an approach further dilutes the event yields and is thus of limited interested for the analysis presented here, where low event yields already present a problem. In the cH(WW) analysis on the other hand, the output of the multivariate classifiers is used to construct the final discriminator for the statistical evaluation. Such an approach may be of greater interest for future iterations of the analysis presented in this work. This is especially true since such an approach facilitates the combination of flavour-tagging discriminators with other, kinematic event properties for more optimal signal discrimination, as is done in the cH(WW) analysis. Furthermore the information of the differential flavour-tagging discriminator distributions may still be incorporated in this way, while the issue of the previously discussed low event yields associated with unrolling the full flavour-tagging discriminator distributions could potentially be avoided.

Lastly, it is evident that theoretical uncertainties associated with the modelling of the processes relevant to the analysis presented here, especially the signal process, have a significant impact on the derived expected upper limits. As a result, improvements in this domain could greatly improve the analysis sensitivity. This is especially true to the FS uncertainty associated with modelling of the cH process, which affects not only this analysis, but also the existing cH(WW) and cH($\gamma\gamma$) analyses where similar sensitivity constraints are observed.

Two approaches to reducing the impact of the FS may be currently envisioned for future cH analysis efforts. The first is an experimental approach, where the nuisance parameter associated with the FS uncertainty could be constrained using control regions. An example of this would be a control region targeting charm quark-associated Z boson production (henceforth referred to as the cZ process), a process which is analogous to the cH process and is thus expected to suffer from the same modelling uncertainties. Example Feynman diagrams of the cZ process are shown in Figure 4.20. A key difference with respect to the cH process is however, that cZ production is expected to be a far more common process at the LHC, since its cross section is not suppressed by the presence of the charm-Yukawa coupling. As a result, the inclusion of such a control region could plausibly allow for a relatively precise measurement of the cZ process. This measurement could in turn be used to determine which values of the nuisance parameter associated with the FS uncertainty are preferred in the final statistical evaluation, thus potentially reducing the impact of the FS uncertainty on the analysis sensitivity towards κ_c. The second approach is a theoretically driven approach. Specifically, a modification of the PDF description

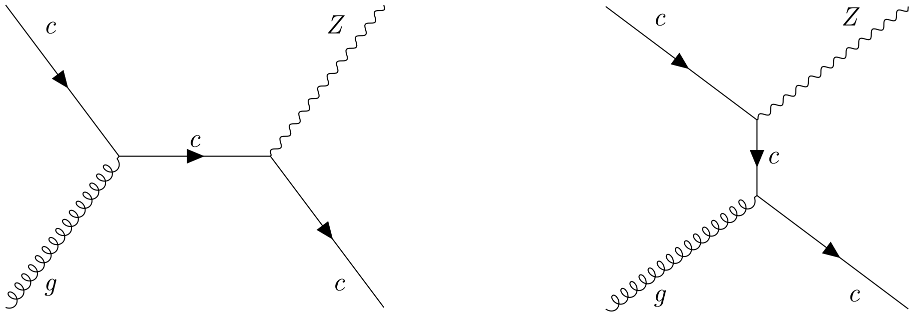


Figure 4.20: The leading-order cZ processes. Note that these look just like the κ_c-sensitive cH process, though with a Z boson instead of a Higgs boson. The corresponding diagrams with an anti-charm quark $\overline{c}$ are implied.

of the proton to also allow for the simulation of massive initial state quarks would naturally remove the need for the inclusion of a FS uncertainty in the signal modelling. First steps to such an approach have been demonstrated in [93], where a proof of concept using the SHERPA[94] event generator is presented. The adoption of such a description in the modelling of the cH process could thus potentially be very promising. Regardless of the approach taken, a reduction of the impact of the FS uncertainty should be a central element to future cH analysis efforts.

Also worth mentioning is the strong scale dependence associated generally with the emission of light quarks alongside the production of a Higgs boson. The resulting uncertainty affects particularly the gluon fusion process, and since this background common to all channels of the cH process, the cH($\gamma\gamma$) and cH(WW) channels are similarly affected by this uncertainty. Improvements in this aspect of the background modelling through, for example, higher order descriptions in perturbative QCD, could thus also have a significant impact on the sensitivity of analyses targeting the cH process.

Lastly, a large scale combination of cH analysis results in different Higgs boson decay channels is clearly also of future interest. Such an initial combination has already been performed for the cH($\gamma\gamma$) and cH(WW) channels and a 95% CL upper limit of $\kappa_c = 47$ is derived. However, the inclusion of additional, unexplored cH channels is clearly relevant here for a more fitting comparison with the other strategies to measure κ_c, such as via the direct decay of the Higgs boson to charm quark pairs or the indirect methods discussed in Section 2.5. Furthermore, since the cH process has only very recently come into focus as an analysis target, there are many potential analysis improvements yet to be made, including those discussed here. As a result, significant increases in the sensitivity of cH analyses towards κ_c may still be to come.

Chapter 5

An EFT interpretation of the cH(ZZ$\to$4μ) process

The methods developed in Chapter 4 may also be used to investigate EFT interpretations of the cH signal. In this chapter, a demonstration of this idea is presented by targeting a modified cH process to which the previously discussed CMD SMEFT operator is introduced. This follows the ideas presented in [1]. Since the vast majority of analysis components remain unchanged from the SM interpretation of the cH(ZZ$\to 4\mu$) analysis, only the changes relevant to this variant of the analysis are explicitly discussed. This includes a brief discussion in Section 5.1 on the simulation of the signal associated with the CMD operator, here referred to as cH(ZZ$\to 4\mu)_{cG}$, and how the properties of this signal motivate a change in the jet selection strategy. Subsequently, an expected 95% CL upper limit on the Wilson coefficient C_{cG} is derived.

5.1 The cH(ZZ$\to 4\mu)_{cG}$ process

5.1.1 Simulation of the cH(ZZ$\to 4\mu)_{cG}$ process

Here, the signal associated with the $\hat{O}_{cG}$ operator is again simulated using MadGraph5_aMC@NLO. To specifically simulate SMEFT, the SMEFTSim model [95][96] is used, which provides an implementation of dimension-six SMEFT operators in the Warsaw basis. For this analysis, the contributions of all SMEFT operators except for the $\hat{O}_{cG}$ operator are turned off. While the MadGraph5_aMC@NLO syntax used for the event generation is very similar to that presented in Subsection 4.2.1, a key difference is that unlike in the SM, SMEFT models such as this one are currently limited to providing simulations at LO in perturbative QCD. A consequence of this is that in the simulation of the cH(ZZ$\to 4\mu)_{cG}$ process, the ambiguity associated with a choice of flavour scheme in the simulation falls away. The reason is that a 3FS version of the cH(ZZ$\to 4\mu)_{cG}$ process cannot be generated, since the process is only defined at NLO in perturbative QCD in the 3FS. Accordingly, the 4FS becomes the default choice.

The simulation of the cH(ZZ$\to 4\mu)_{cG}$ process consists of three components, the first of which is the purely SM component. This corresponds to the SM cH(ZZ$\to 4\mu$) process, simulated with a cross section that is fixed to the SM expectation. Second is the so-called quadratic component, which consists exclusively of SMEFT contributions introduced by the $\hat{O}_{cG}$ operator. The last component is the so-called *interference term*, consisting of a mix of SM and SMEFT contributions. The effect of this behaviour on the cH(ZZ $\rightarrow 4\mu)_{cG}$ cross section σ_{total} can be parametrised as

$$\sigma_{\text{total}} = A\sigma_{\text{SM}} + B\sigma_{\text{int.}} + C\sigma_{\text{EFT}}, \tag{5.1}$$

where A, B, and C are coefficients describing the relative strength of each contribution. This expression can be normalised to the SM cH cross section and expressed in terms of C_{cG} and the new physics scale Λ as

$$\sigma_{\text{EFT}} = \sigma_{\text{SM}}(1 + \frac{C_{cG}}{\Lambda^2}\mu_{\text{int.}} + \frac{C_{cG}^2}{\Lambda^4}\mu_{\text{EFT}}). \tag{5.2}$$

Here, the signal strength modifiers $\mu_{\text{int.}}$ and μ_{EFT} are introduced for the interference and quadratic terms respectively. By simulating the cH(ZZ $\rightarrow 4\mu)_{cG}$ for a range of Wilson coefficient C_{cG} values, the scaling behaviour of Eq. (5.2) with C_{cG} can be obtained through a polynomial fit. Using the LO in perturbative QCD CM cross section of $\sigma_{\text{SM}} = 0.058$ pb and $\Lambda = 1$ TeV, it is found that $\mu_{\text{int.}} = 2.2$ and $\mu_{\text{EFT}} = 55.8$ [1].

A technical complication in setting upper limits on C_{cG} arises from the fact that the interference and quadratic terms are, for some chosen discriminator, associated with different distribution shapes. The relative contributions of these distributions scale non-linearly with C_{cG} and so, to obtain separate distributions, separate simulations must be performed for these components. However, given the relatively small size of the interference contribution in comparison to the quadratic contribution, the interference term is neglected and the total contribution sensitive to C_{cG} is approximated using only the quadratic term. This allows for a simple, linear scaling of the signal cross section during the limit setting procedure, to which the SM component now becomes a background process. It should be noted that this only holds approximately true for $C_{cG} > 1$, since the relative strength of the interference term starts to become increasingly larger for $C_{cG} < 1$. As such, the parameter space of C_{cG} that is probed in the limit setting procedure should not be significantly smaller than $C_{cG} = 1$ for the modelling of the signal observables to remain accurate. To put into perspective the transition from the region where the quadratic term is dominant to one where the linear term is dominant, the relative cross section contributions of the linear term can be calculated with Eq. (5.2) for fixed values of C_{cG}. For example, for $C_{cG} = 1$, the relative contribution of the linear term to the full cH(ZZ $\rightarrow 4\mu)_{cG}$ signal is approximately 4%. For $C_{cG} = 0.1$ this increases to approximately 30% and for $C_{cG} = 0.01$ to approximately 80%.

5.1.2 Modifications to the analysis strategy

The event selection used in this analysis variant is kept almost identical to the selection presented in Section 4.1. The only change is a simplification with respect to the jet selection. Instead of applying the previously introduced jet selection algorithm, the jet with the highest transverse momentum in a candidate event is selected. This choice is motivated by the $\hat{O}_{cG}$ operator introducing a significant kinematic modification to the cH process, in which the associated charm quarks can be produced at much higher transverse momenta than in the SM case. This behaviour is shown in Figure 5.1 for a simulation of the cH(ZZ $\rightarrow 4\mu)_{cG}$ process. The transverse momentum of the jet that is selected in this way for cH(ZZ $\rightarrow 4\mu)_{cG}$ candidate events is shown in Figure 5.2, along with the full background estimation. Clearly, the cH(ZZ $\rightarrow 4\mu)_{cG}$ signal produces a strong enhancement in the high transverse momentum regime of the selected jet, as expected. This quality makes the transverse momentum of the selected jet an excellent candidate as the final discriminator in the statistical evaluation.

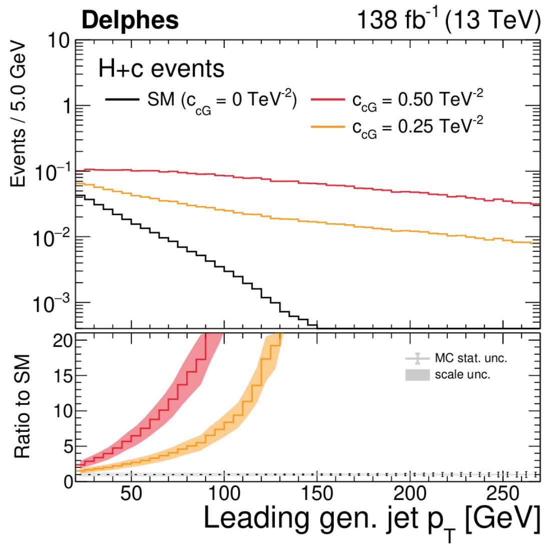


Figure 5.1: The transverse momentum of the leading generated jet for a cH process modified by the CMD operator $\hat{O}_{cG}$. The enhancement in the high p_T region associated with $\hat{O}_{cG}$ can be seen for a number of different C_{cG} values. Figure taken from [1].

The same statistical evaluation strategy that is presented in Chapter 4 is used. The background process estimation remains unchanged with the exception being that the SM cH(ZZ$\rightarrow 4\mu$) process is now also considered a background process. Again, a range of systematic uncertainty sources are considered in the uncertainty model. This includes all previously introduced uncertainty sources, except those associated with jet flavour tagging, since no jet flavour tagging is performed. Since only the jet selection method has changed, the relative magnitude of the uncertainties remains unchanged with respect to the numbers quoted in Chapter 4.

For the cH(ZZ$\rightarrow$ $4\mu)_{cG}$ signal simulation, the same uncertainty sources as for the cH(ZZ$\rightarrow 4\mu$) process are considered. The only exception to this is the flavour scheme uncertainty, which is not defined for the cH(ZZ$\rightarrow$ $4\mu)_{cG}$ signal simulation. The magnitude of the theoretical uncertainties on the cH(ZZ$\rightarrow$ $4\mu)_{cG}$ signal are, like for the cH(ZZ$\rightarrow 4\mu$) process, significant. This includes especially the uncertainty associated with the PDF set, which produces a yield change of approximately 20%. Also, the renormalisation and factorisation scales uncertainties remains significant and are associated with yield changes of approximately 18% and 10%, respectively.

5.2 Results

The transverse momentum of the selected jet is used as a final discriminator, which corresponds to the distribution shown in Figure 5.2. An additional cut of p_T(jet)$>$ 50 GeV is implemented, since this removes a significant portion of background process contributions. Note that, as discussed in Section 2.4, the redefinition of the Wilson coefficient as $\tilde{C}_{cG} = \frac{C_{cG}}{\Lambda^2}$ is used in the following. Additionally, the perturbativity requirements discussed in Subsection 2.4.2 are ensured by introducing a $M_{cut} = 1$ TeV criterion that is applied to candidate events. Here, M_{cut} is estimated via the invariant mass of the Higgs boson candidate and jet system. At an integrated

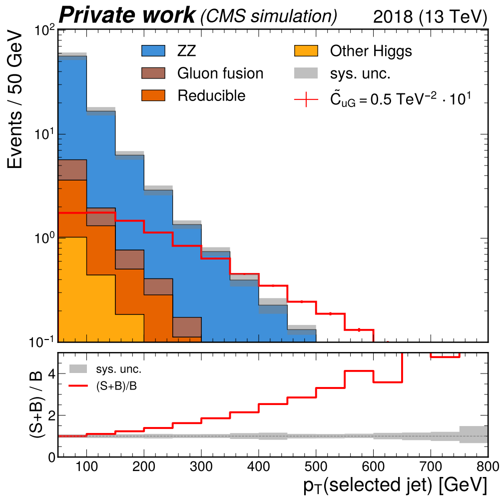


Figure 5.2: The spectrum of the transverse momentum of the jet selected in a SMEFT variant of the cH(ZZ → 4μ) analysis. Contributions from the $\hat{O}_{cG}$ operator, simulated for a nominal value of $\tilde{C}_{cG} = 0.5$ TeV^{-2}, are scaled by a factor 10 to improve visibility and can be clearly see to produce jets at high transverse momenta.

luminosity equivalent to the 2018 dataset of the CMS detector, an expected 95% CL upper limit of $\tilde{C}_{cG} = 1.99$ is set.

Like for the SM oriented analysis, a decomposition of the effects of the individual uncertainty sources on this result can be performed. Again, the 30 leading uncertainty sources are considered for a signal strength corresponding to the upper limit on $\tilde{C}_{cG}$ that is set, which are listed in Figure 5.3. Leading theoretical uncertainties include the uncertainty associated with the PDF used in simulation, the renormalisation and factorisation scale of the cH(ZZ → 4μ)$_{cG}$ signal, as well as the uncertainties associated with $\mathcal{B}(H \to ZZ \to 4\ell)$ and the qq→ZZ K-factor. Leading experimental uncertainties include the normalisation of the reducible background estimation, the uncertainty on the luminosity as well as the calibration of the muons. The contributions from the reducible background here appear to become relevant due to the presence of a handful of events scattered throughout the very high transverse momentum region of the spectrum. Since the expected contributions of other backgrounds in this high-p_T region become largely negligible, and it is the region most enriched in signal, the normalisation of these events stemming from the reducible background estimation thus plays a more significant role. A common factor to all these uncertainty sources is that they are associated with larger relative yield changes in the background and signal processes, which can potentially mask the presence of the signal. Also noticeable is that, as a result of the sparse population of bins in the discriminator bins at very high transverse momenta, the statistical modelling uncertainties associated with these bins is also significant.

A statistical extrapolation to a full Run-2 and HL-LHC scenario is again performed. In a Run-2 scenario, an expected upper limit of $\tilde{C}_{cG} < 1.61$ is derived. For a HL-LHC scenario, this improves to $\tilde{C}_{cG} < 0.76$. It should be noted that in the regime of the latter result, the contribution

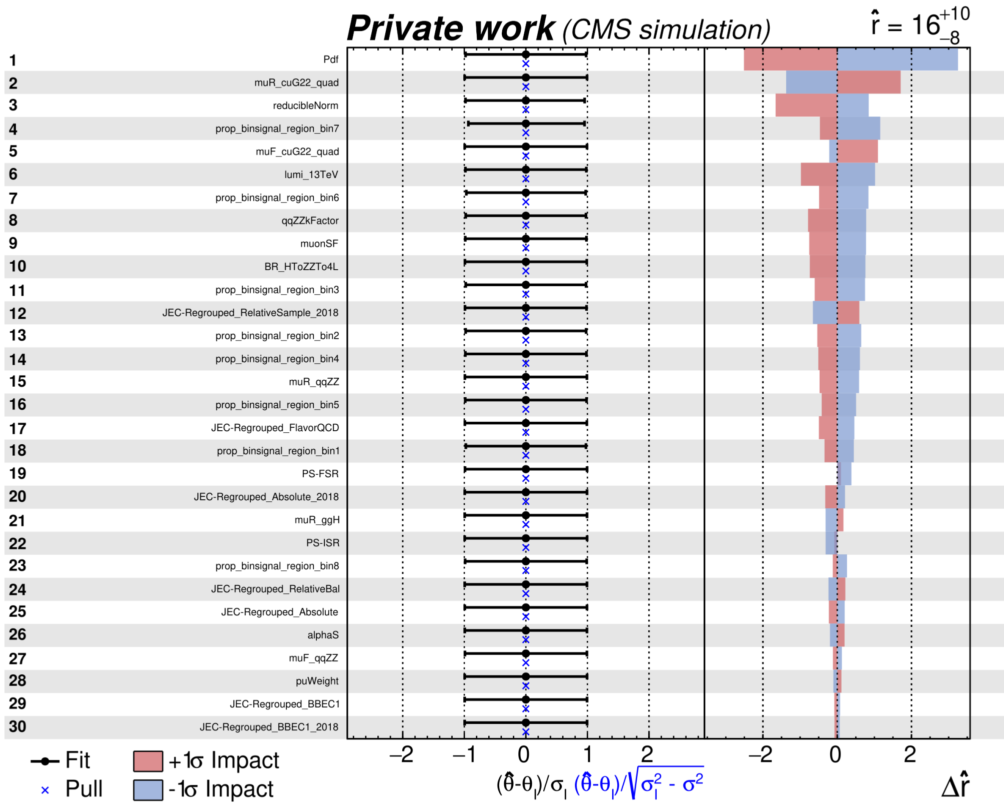


Figure 5.3: Figure showing the 30 leading uncertainty sources in a fit of $\tilde{C}_{cG}$, orderd by their relative impact on the fit. In the left panel the nuisance parameter names can be seen. In the central panel, the fitted nuisance parameter values as well as their so-called pulls are shown. On the right, the $\Delta\hat{r}_i$ values can be seen for the $\pm\sigma$ variations associated with each nuisance parameter. Here $\hat{r}$ represents the signal strength associated with the expected upper limit that is set for $\tilde{C}_{cG}$.

from the linear (interference) term associated with $\hat{O}_{cG}$ starts to becomes larger and should be accounted for. However, for a value of $\tilde{C}_{cG} = 0.76$, the linear term still only accounts for approximately 5% of the total cross section associated with the cH(ZZ $\rightarrow 4\mu)_{cG}$ signal. The constraints derived here are approximately in line with those reported in [1], indicating that the expected results presented there are expected to translate well into an experimental setting.

A number of potential analysis improvements can be made in future iterations of this analysis. Like in the SM case, methods with which to increase the overall signal selection efficiency would be interesting to explore. While the low transverse momentum region of the jet is not relevant in the way it was previously, the loosening of the muon selection criteria used in the Higgs boson candidate reconstruction is again of interest. Additionally, other optimisations could be explored, such as multivariate methods. These could be promising due to the significant differences in event kinematics for the signal and background processes. While these differences are clearly already captured by the transverse momentum of the associated jet, correlation with other kinematic features could be explored and leveraged to enhance this separation between signal and background processes. Since theoretical uncertainties associated with the modelling of the background and signal processes again constrain the analysis sensitivity significantly, improvements in this domain are also of significant interest here. Lastly, contributions of other dimension-six SMEFT operators such as those discussed in [1] may also be considered in future analysis iterations.

Conclusion

In this thesis, a first-time analysis of the cH process in the H→ZZ→4μ channel is presented. To this end, the 2018 data taking era of the CMS detector is considered, with the ultimate goal of placing constraints on the charm quark Yukawa interaction modifier κ_c. Initially, the SM of particle physics is discussed with a particular focus on the Higgs boson, as it is the Higgs boson which is responsible for the generation of particle masses in the SM. This is also true for the charm quark, which gains its mass through a Yukawa interaction with the Higgs boson. Unlike the Yukawa interactions of other heavier fermions in the SM, the charm-Yukawa interaction is yet to be measured experimentally. Motivated by this, the cH process is identified as a process that is of experimental interest for such a measurement and is discussed in the context of the SM. A method by which a signal strength measurement of the cH process can be translated to a constraint on κ_c is also introduced. Subsequently, the interpretation of the cH process in a beyond-the-SM physics scenario, parametrised through SMEFT, is discussed. Here, specifically the chromomagnetic dipole operator is identified as a SMEFT operator that may be uniquely constrained with the cH process. This is followed by a discussion of the latest results in analyses targeting κ_c.

Following this, the CMS detector is introduced. This is a multipurpose particle physics detector that can reconstruct pp collisions at the LHC to identify cH candidate events. The various sub-detector systems of the CMS detector and the reconstruction methods and algorithms relevant to the analysis are discussed. This includes jet flavour tagging algorithm DeepJet. Additionally, the Monte Carlo simulation methods that can be used to estimate the physics processes that occur in the pp collisions recorded by the CMS detector at the LHC are presented.

Subsequently, the analysis strategy employed in this work to target the cH(ZZ $\rightarrow 4\mu)_{\kappa_c}$ process is detailed. This includes the reconstruction of Higgs boson candidates and the selection of an associated jet candidate. From these individual components, candidate cH(ZZ $\rightarrow 4\mu$) events are obtained. The simulation of the cH(ZZ $\rightarrow 4\mu)_{\kappa_c}$ signal process is discussed, alongside the methods that are used to estimate a variety of background processes. This is followed by a presentation of the statistical methods that are used to derive expected 95% CL upper limit on the signal strength of the cH(ZZ $\rightarrow 4\mu)_{\kappa_c}$ process, as well as the model of systematic uncertainty sources that is implemented. With this method, expected 95% CL upper limits on $\sigma\mathcal{B}$(cH(ZZ $\rightarrow 4\mu)_{\kappa_c}$) as well as κ_c are set for a number of scenarios. In the nominal scenario where the Higgs boson candidate mass is used as a discriminator, this corresponds to an expected 95% CL upper limit of $\sigma\mathcal{B}$(cH(ZZ $\rightarrow 4\mu)_{\kappa_c}$) = 1088 times the SM expectation ($|\kappa_c| = 191$) for the 2018 dataset recorded by the CMS detector. Additional scenarios using different discriminators are considered and, for the nominal scenario, a statistical extrapolation to a full Run-2 dataset and the anticipated High-Luminosity LHC dataset is performed. The corresponding expected 95% CL upper limits that are derived are $\sigma\mathcal{B}$(cH(ZZ $\rightarrow 4\mu)_{\kappa_c}$) = 725 times the SM expectation ($|\kappa_c| = 128$) and

$\sigma\mathcal{B}(\mathrm{cH(ZZ \to 4\mu)}_{\kappa_c}) = 318$ times the SM expectation ($|\kappa_c| = 58$), respectively. Additionally, the impact of leading systematic uncertainty sources in the evaluation procedure are discussed, along with a fit to data in sidebands region of the analysis. Potential analysis improvements are also discussed.

Finally, the potential of the presented analysis to constrain the Wilson coefficient of the chromomagnetic dipole operator in SMEFT is demonstrated and discussed. An expected constraint of $\tilde{C}_{cG} < 1.94$ is set at 95% CL. Again, a statistical extrapolation to full Run-2 and High Luminosity-LHC scenarios is performed, with the expected constraints being set at $\tilde{C}_{cG} < 1.58$ and $\tilde{C}_{cG} < 0.75$ at 95% CL, respectively. Potential improvements to this interpretation of the analysis are also discussed, including the modelling uncertainties that affect analyses of the cH process.

The expected sensitivity to κ_c that is derived in this work is lower than in other, similar analyses. This includes analyses of the cH(WW) and cH($\gamma\gamma$) processes, as well analyses targeting H$\to$ c$\overline{\mathrm{c}}$ decays. However, the presented analysis remains to be an important, complementary contribution in the quest to determine the nature of the charm-Yukawa coupling, especially given the significant challenges associated with such a measurement. Additionally, the novel results of the presented SMEFT interpretation of the cH process represent a unique method with which to probe certain beyond-the-SM scenarios that may provide exciting hints at deviations from expected SM behaviour.

Outlook

As data taking at the LHC draws to a close and particle physicists look towards the HL-LHC, the measurement of the charm-Yukawa coupling remains to be a highly relevant, yet challenging facet of our characterisation of the Higgs boson. A key component of this challenge arises from the very small cross sections associated with processes sensitive to this coupling. Accordingly, the continued development of a variety of analysis strategies that target such processes remains to be a key strategic component in ultimately making such a measurement. As a complement to analysis channels in which the Higgs boson decays directly to a pair of charm quarks, strategies targeting charm quark-associated Higgs production have gained increased attention. Given their novel status, the continued development of such analysis strategies remains to be of significant interest and is a multifaceted endeavour, considering the variety of Higgs boson decay channels that may be targeted. These include, so far, the H$\to 4\mu$ channel presented in this work, as well as the H$\to WW$ and H$\to \gamma\gamma$ channels. However, channels such as H$\to 4e$, H$\to 2e2\mu$, and H$\to \tau\tau$ remain largely unexplored, much like SMEFT interpretations of the cH process. With a variety of improvements, such as developing experimental methods to specifically target low p_T jets or improvements in the simulation of the cH process, significant gains in the sensitivity of these analyses to the charm-Yukawa coupling or a SMEFT interpretation thereof may be achieved. These improvements are especially relevant given that, at first order, they are independent of the Higgs boson decay channel of the cH process. With the many different analysis strategies and improvements to be explored with the full Run-2, Run-3 and HL-LHC datasets, the future of a charm-Yukawa coupling measurement surely continues to look promising.

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Appendix

A Columnar analysis framework

For this work, an analysis framework built upon the Coffea [97] software package and based on columnar analysis methods was conceived and written. To better understand the idea of columnar analysis, it helps to consider the computations are required of a typical physics analysis. On a per-event basis, an analyser must repeatedly perform a set of operations on some input data (event data delivered for example via the CMS NanoAOD [98] format) to produce a desired set of outputs. This may consist of applying object selection criteria or producing new, higher level variables from the input. In the context of an entire dataset, the subset of data associated with a single event can thus be thought of as a single row in a larger matrix. With this matrix picture in mind, two approaches to producing the desired outputs may be implemented. The first is a row based approach where the necessary computations are performed serially, once per event, with one event following another. This represents a 'traditional' event loop approach to analysis. The second is the column based approach where operations are performed on entire columns of event data, thus lending the *columnar* analysis name. This means a single computational step is effectively performed simultaneously across a set of events. A result of this is that it allows for an improved optimisation of computations, as relevant data may be stored in contiguous memory and is thus accessed with more ease. Both of these approaches are visualised in Figure 4. The implementation of the columnar approach is here facilitated by the use of the Awkward array [99] python package, which generalises numpy [100] arrays to data with non-regular (or *jagged*) shapes.

To perform the analysis presented in Chapter 4 it is necessary to process approximately $5.6 \cdot 10^{10}$ events, equating to approximately 8.4 TB worth of data. Given the scope of these processing requirements, the motivation to optimise the associated computing effort is clear. Using the described columnar approach, the analysis framework is able to achieve a turn around time of approximately one working day for the complete analysis (including the handling of systematic uncertainties) using the distributed grid computing resources available at the IIHE. This represents an almost tenfold speed increase to a previous framework iteration using a traditional CMS software C++ event loop structure. Along with the implementation of the algorithms described in Chapter 4 and Chapter 5, the framework design includes an automatic, central handling of samples and relevant metadata, job submission to distributed computing resources, as well as plotting tasks and production of histogram templates for the CMS Combine tool. The development of the analysis framework thus represents a central element of the presented analysis. Additionally, a variant of this framework was also used for the analysis presented in [1].

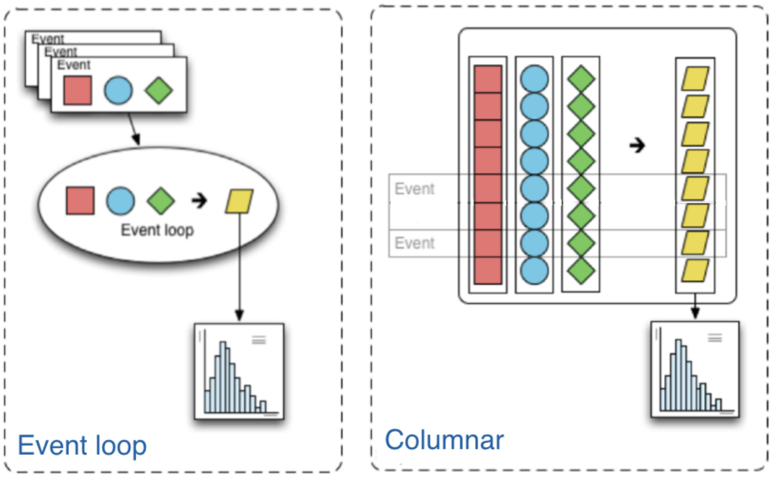


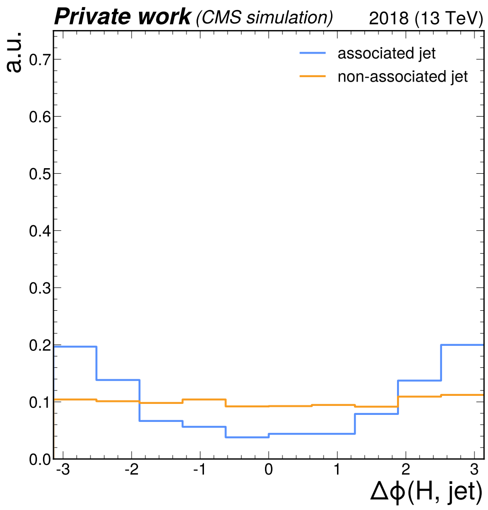
Figure 4: A representation of event loop based and columnar based analysis approaches can be seen on the left and right respectively. The squares, circles and diamonds represent event data that is transformed via an algorithm into output that is histogrammed, represented by the trapezoids. Figure taken from [97].

B Likelihood templates derived for jet selection algorithm

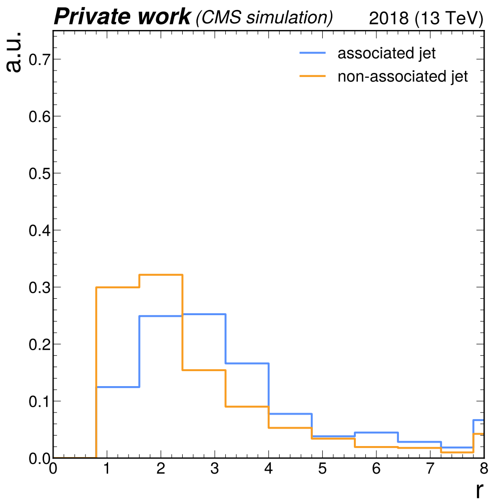
A number of likelihood templates are derived for the jet selection algorithm using the $\Delta\phi(\mathrm{H}, \mathrm{jet})$ and $r = p_T(\mathrm{jet})/p_T(\mathrm{H})$ variables, as described in Subsection 4.1.2. These are shown in Figure 5 to Figure 10.

C Comparison of discriminators in 3P1F and 2P2F regions

A comparison between the 2P2F and 3P1F regions of the reducible background estimation for the DeepJet c-tagging discriminators, as well as the Higgs mass and selected jet transverse momentum is shown in Figure 11 and Figure 12. Here, the 3P1F contribution observed in data is corrected by subtracting the expected contribution from ZZ production, which is estimated from simulation. Overall, there exists a reasonable agreement between the c jet identification discriminator shapes, when accounting for the small overall contribution of the reducible background estimation to the full background estimation. Though there are significant differences when considering, for example, the Higgs boson candidate mass spectrum, there is a reasonable agreement in the signal enriched region around $m(H) = 125$ GeV. There is also an acceptable agreement observed for the transverse momentum of the selected jet, which is only relevant for the EFT analysis that is presented. Given this reasonable agreement of the shapes of these discriminator distributions, and considering the small relative total contribution of the reducible background, the 2P2F discriminator shapes are used to estimate the distribution shape of the full reducible background estimation presented in Subsection 4.2.3. The reason that this simplification is made is that a continuous distribution shape cannot be obtained for the full reducible background estimation due to statistical limitations.

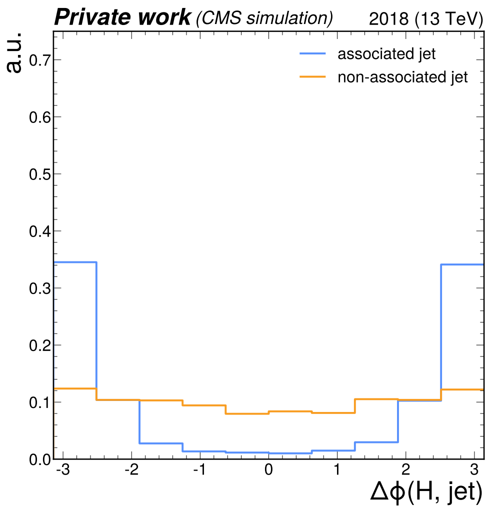


(a) The $\Delta\phi(H, jet)$ template.

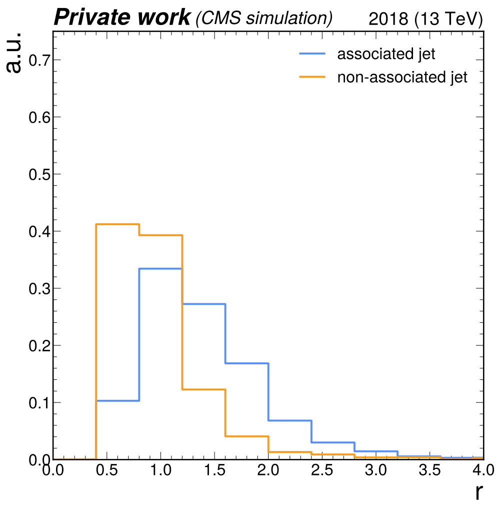


(b) The r template.

Figure 5: The $\Delta\phi(H, jet)$ and r templates in the 0-15 GeV bin of the Higgs boson candidate p_T.

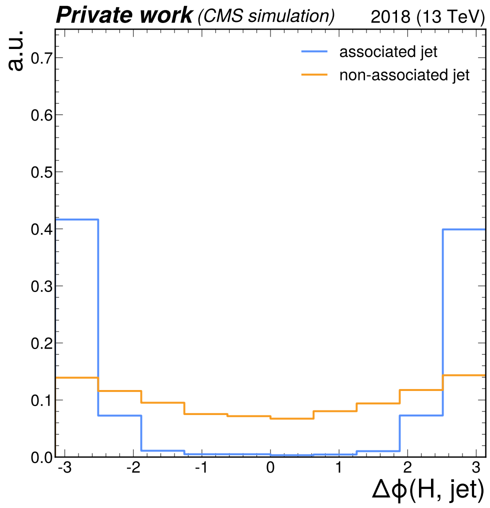


(a) The $\Delta\phi(H, jet)$ template.

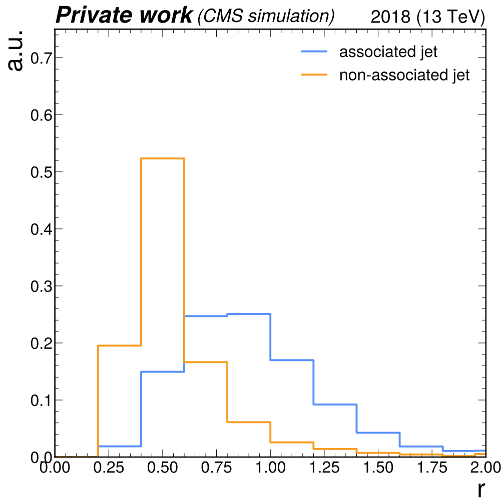


(b) The r template.

Figure 6: The $\Delta\phi(H, jet)$ and r templates in the 15-30 GeV bin of the Higgs boson candidate p_T.

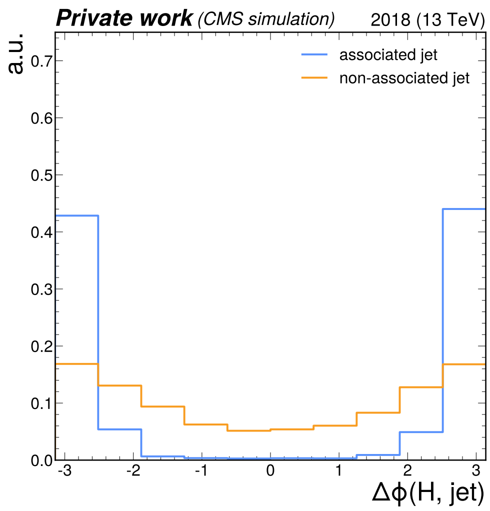


(a) The $\Delta\phi(H, jet)$ template.

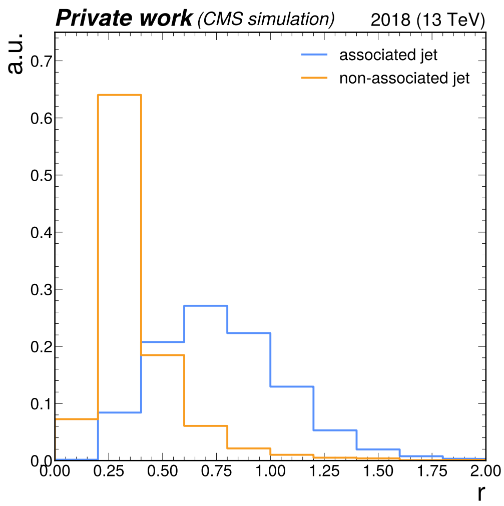


(b) The r template.

Figure 7: The $\Delta\phi(H, jet)$ and r templates in the 30-50 GeV bin of the Higgs boson candidate p_T.

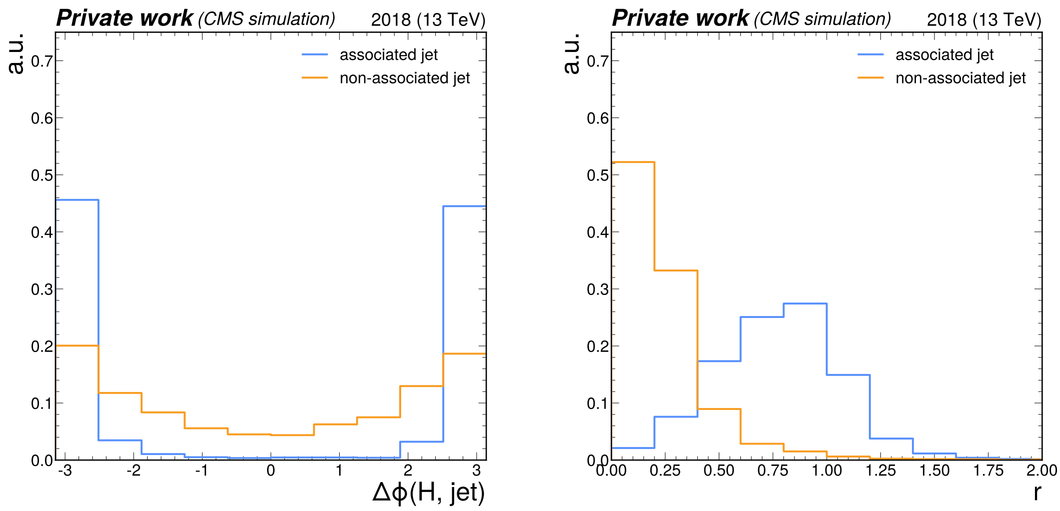


(a) The $\Delta\phi(H, jet)$ template.



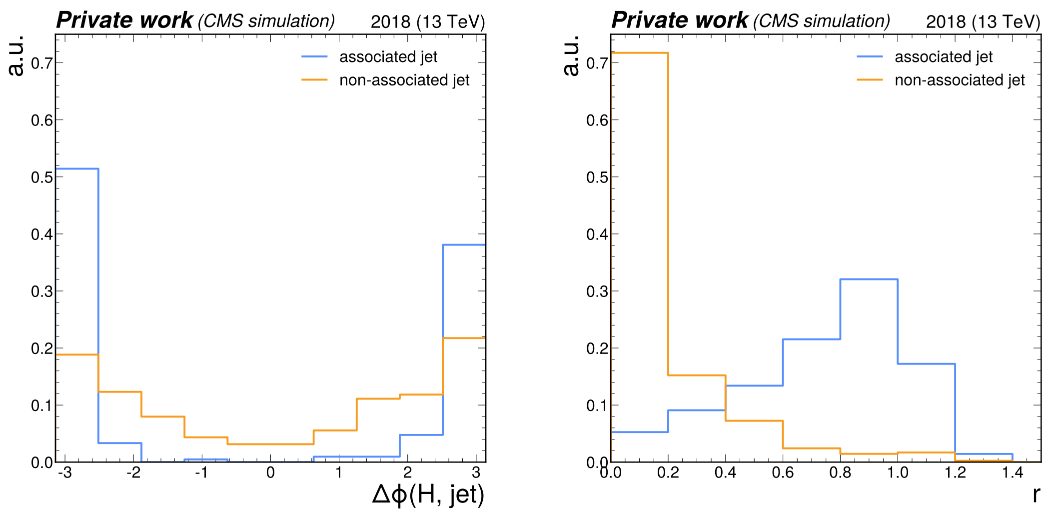
(b) The r template.

Figure 8: The $\Delta\phi(H, jet)$ and r templates in the 50-100 GeV bin of the Higgs boson candidate p_T.



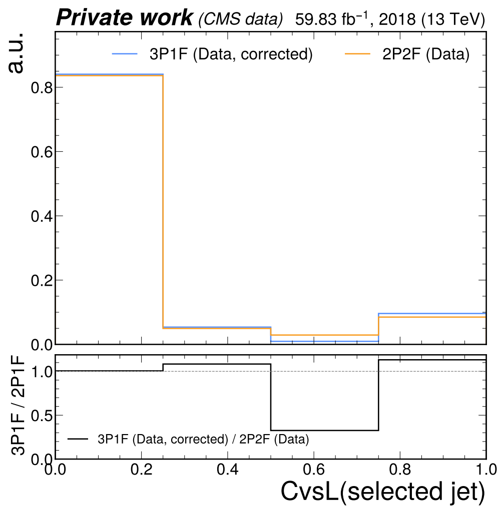
(a) The $\Delta\phi(H, jet)$ template. (b) The r template.

Figure 9: The $\Delta\phi(H, jet)$ and r templates in the 100-200 GeV bin of the Higgs boson candidate p_T.

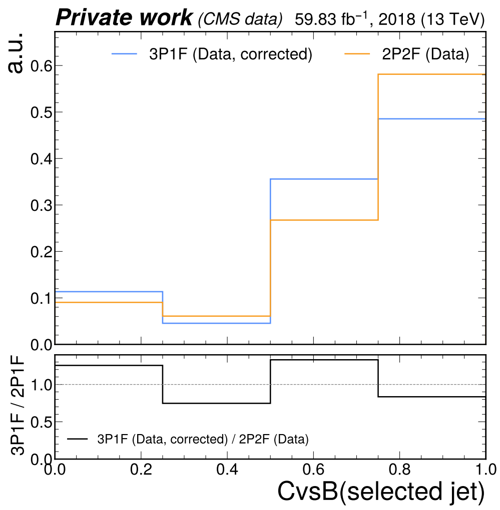


(a) The $\Delta\phi(H, jet)$ template. (b) The r template.

Figure 10: The $\Delta\phi(H, jet)$ and r templates in the >200 GeV bin of the Higgs boson candidate p_T.



(a) The CvsL discriminator.



(b) The CvsB discriminator.

Figure 11: A comparison between the 2P2F and 3P1F regions of the reducible background estimation, looking at the shape of the DeepJet c-tagging discriminator distributions. A reasonable agreement is observed, when accounting for the small overall contribution of the reducible background to the full background estimation.

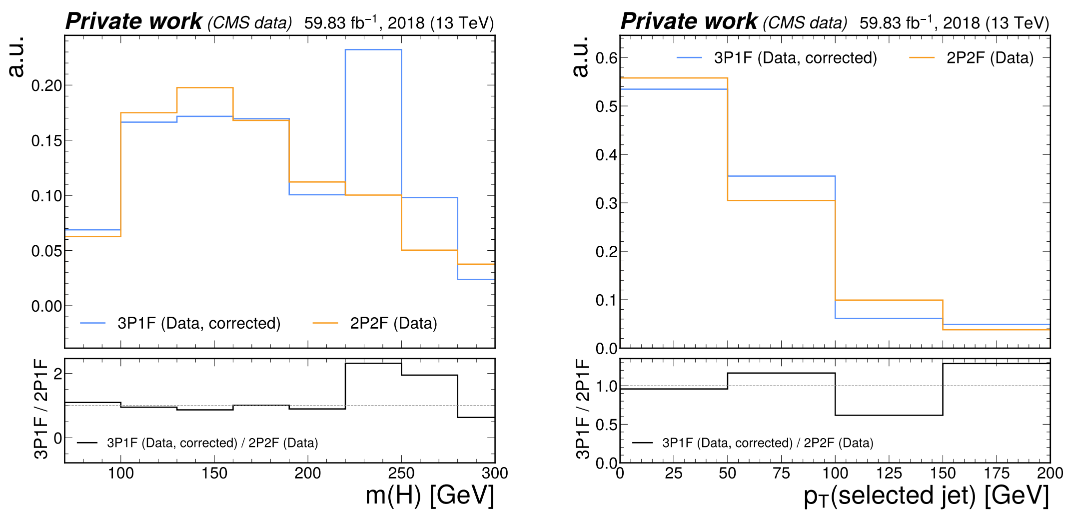


Figure 12: A comparison between the 2P2F and corrected 3P1F regions of the reducible background estimation, looking at the Higgs boson candidate mass spectrum and transverse momentum of the selected jet. For Higgs boson candidate mass, a reasonable agreement is observed in the region around $m(H) = 125$ GeV, with the most significant differences being observed regions not sensitive to the signal. An acceptable agreement for the transverse momentum of the selected jet (here for the EFT analysis interpretation) is also observed.

D List of nuisance parameters and corresponding distributions

Here, the full list of nuisance parameters that are considered in the statistical evaluation that is performed in Section 4.5 is given with a brief, corresponding description:

- **reducibleNorm**: the normalisation uncertainty of the reducible background estimation contribution.
- **lumi_13TeV**: the uncertainty associated with the integrated luminosity of the 2018 dataset.
- **BR_HToZZTo4L**: the uncertainty associated with the branching ratio of the Higgs boson to four leptons.
- **Pdf**: the uncertainty associated with the choice of PDF.
- **JEC-Regrouped_FlavourQCD**: an uncertainty associated with the responses of the energy scale calibration to different jet flavours.
- **JEC-Regrouped_RelativeBal**: an uncertainty originating from differences that are observed between different methods with which the energy scale calibration can be performed.
- **JEC-Regrouped_HF**: a grouping of uncertainties from a variety of sources for the $|\eta| > 3$ detector region.
- **JEC-Regrouped_HF_2018**: a grouping of uncertainties from a variety of sources for the $|\eta| > 3$ detector region.
- **JEC-Regrouped_BBEC1**: a grouping of uncertainties from a variety of sources for the $|\eta| < 2.5$ detector region.
- **JEC-Regrouped_BBEC1_2018**: a grouping of uncertainties from a variety of sources for the $|\eta| < 2.5$ detector region.
- **JEC-Regrouped_EC2**: a grouping of uncertainties from a variety of sources for the $2.5 <|\eta| < 3$ detector region.
- **JEC-Regrouped_EC2_2018**: a grouping of uncertainties from a variety of sources for the $2.5 <|\eta| < 3$ detector region.
- **JEC-Regrouped_Absolute**: a grouping of uncertainties on the flat, absolute energy scale.
- **JEC-Regrouped_Absolute_2018**: a grouping of uncertainties on the flat, absolute energy scale.
- **JEC-Regrouped_RelativeSample_2018**: an uncertainty arising from residual differences observed in the energy scale calibration, when looking at Z+jet, γ+jet and dijet events.
- **JER**: the uncertainty associated with the jet energy resolution calibration.
- **cTagSF_Extrap**: an uncertainty originating from an extrapolation that is applied in the calibration of DeepJet.

- **cTagSF_Interp**: an uncertainty originating from an interpolation that is applied in the calibration of DeepJet.
- **cTagSF_LHEScaleWeight_muF**: an uncertainty originating from factorisation scale uncertainties of the simulated samples used in the calibration of DeepJet.
- **cTagSF_LHEScaleWeight_muR**: an uncertainty originating from renormalisation scale uncertainties of the simulated samples used in the calibration of DeepJet.
- **cTagSF_PSWeightFSR**: an uncertainty originating from modelling of final state radiation in simulated samples used in the calibration of DeepJet.
- **cTagSF_PSWeightISR**:an uncertainty originating from modelling of initial state radiation in simulated samples used in the calibration of DeepJet.
- **cTagSF_PUWeight**: an uncertainty originating from pileup reweighting performed for simulated samples used in the calibration of DeepJet.
- **cTagSF_Stat**: an uncertainty originating from the limited size of simulated samples used in the calibration of DeepJet.
- **cTagSF_XSec_BRUnc_DYJets_b**: an uncertainty originating from an uncertainty of the cross section in simulated samples used in the calibration of DeepJet.
- **cTagSF_XSec_BRUnc_DYJets_c**: an uncertainty originating from an uncertainty of the cross section in simulated samples used in the calibration of DeepJet.
- **cTagSF_XSec_BRUnc_WJets_b**: an uncertainty originating from an uncertainty of the cross section in simulated samples used in the calibration of DeepJet.
- **cTagSF_XSec_jer**: an uncertainty originating from uncertainty associated with the jet energy resolution calibration, which affects the calibration of DeepJet.
- **cTagSF_XSec_jesTotal**: an uncertainty originating from uncertainty associated with the jet energy scale calibration, which affects the calibration of DeepJet.
- **muonSF**: the uncertainty associated with the calibration of muons.
- **puIDSF**: the uncertainty associated with the calibration of the pileup identification method.
- **puWeight**: the uncertainty associated with the pileup reweighting procedure that is applied.
- **alphaS**: the uncertainty associated with the choice of $\alpha_s(Q^2)$ in the Monte Carlo simulation.
- **muR_HPlusC**: the uncertainty associated with the choice of the renormalisation scale of the cH process.
- **muR_ggH**: the uncertainty associated with the choice of the renormalisation scale of the ggH process.
- **muR_ggZZ**: the uncertainty associated with the choice of the renormalisation scale of the gg$\rightarrow$ZZ process.

- **muR_tqH**: the uncertainty associated with the choice of the renormalisation scale of the tqH process.
- **muR_ttH**: the uncertainty associated with the choice of the renormalisation scale of the ttH process.
- **muR_VBF**: the uncertainty associated with the choice of the renormalisation scale of the VBF process.
- **muR_WminusH**: the uncertainty associated with the choice of the renormalisation scale of the $W^{-}H$ process.
- **muR_WPlusH**: the uncertainty associated with the choice of the renormalisation scale of the $gW^{+}H$ process.
- **muR_ZH**: the uncertainty associated with the choice of the renormalisation scale of the ZH process.
- **muR_qqZZ**: the uncertainty associated with the choice of the renormalisation scale of the $qq \to ZZ$ process.
- **muR_HPlusB**: the uncertainty associated with the choice of the renormalisation scale of the bH process.
- **muR_HPlusC**: the uncertainty associated with the choice of the factorisation scale of the cH process.
- **muR_ggH**: the uncertainty associated with the choice of the factorisation scale of the ggH process.
- **muR_ggZZ**: the uncertainty associated with the choice of the factorisation scale of the $gg \to ZZ$ process.
- **muR_tqH**: the uncertainty associated with the choice of the factorisation scale of the tqH process.
- **muR_ttH**: the uncertainty associated with the choice of the factorisation scale of the ttH process.
- **muR_VBF**: the uncertainty associated with the choice of the factorisation scale of the VBF process.
- **muR_WminusH**: the uncertainty associated with the choice of the factorisation scale of the $W^{-}H$ process.
- **muR_WPlusH**: the uncertainty associated with the choice of the factorisation scale of the $gW^{+}H$ process.
- **muR_ZH**: the uncertainty associated with the choice of the factorisation scale of the ZH process.
- **muR_qqZZ**: the uncertainty associated with the choice of the factorisation scale of the $qq \to ZZ$ process.
- **muR_HPlusB**: the uncertainty associated with the choice of the factorisation scale of the bH process.

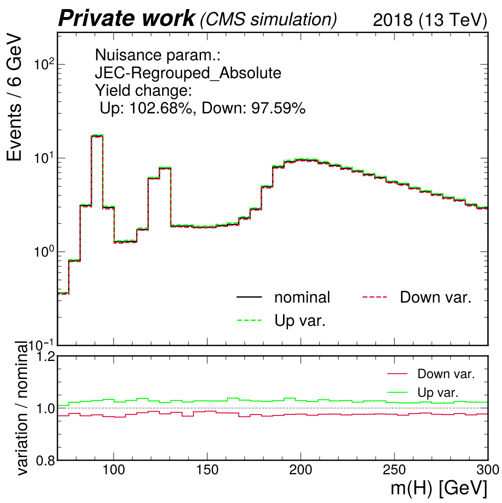
- **PS-FSR**: the uncertainty associated with the modelling of final state radiation.
- **PS-ISR**: the uncertainty associated with the modelling of initial state radiation.
- **FS_bH**: the uncertainty associated with the modelling of the cH process, resulting from the choice of FS.
- **FS_cH**: the uncertainty associated with the modelling of the bH process, resulting from the choice of FS.
- **qqZZkFactor**: the uncertainty associated with the K-factor that is applied to the qq→ZZ process.
- **ggZZkFactor**: the uncertainty associated with the K-factor that is applied to the gg→ZZ process.

A number of additional nuisance parameters are additionally generated during the fitting procedure that represent the statistical modelling uncertainty associated with each bin. Also note that a number of the jet energy scale nuisance parameter names appear twice, with an additional suffix of '2018' in their second appearance. The reasons for this is that the associated uncertainties are split into a part that is not correlated across Run-2 data taking eras (i.e. 2016, 2017, 2018) and one that is. The uncorrelated part is denoted with the additional suffix. Such correlations are not relevant to this analysis, given that only the 2018 era is analysed. However, the distributions associated with both the respective uncorrelated and correlated parts of these uncertainty sources are still obtained separately and thus implemented as individual nuisance parameters.

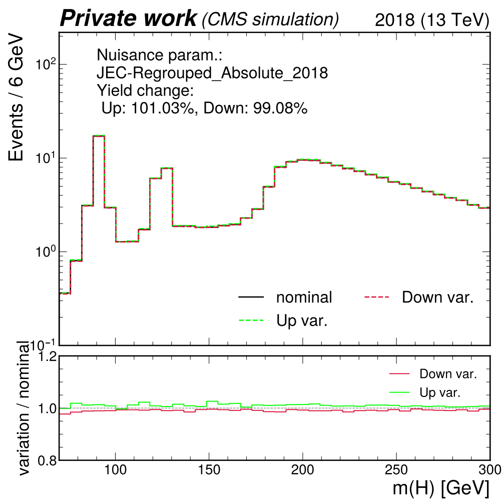
The distributions associated with dominant uncertainty sources used in the statistical evaluation (in the nominal scenario) that is presented in Section 4.5, are shown in Figure 13, Figure 14 and Figure 15. Unless explicitly stated otherwise, the distributions shown are the sum of the background processes. Distributions for individual processes (such as the cH($ZZ \to 4\mu$) signal) are shown separately, per uncertainty, in case significant differences in the distribution shapes associated with the uncertainty are observed for that process, with respect to the sum of background processes.

E Nuisance parameter impacts for $r = 0$ fit of cH($ZZ \to 4\mu$)

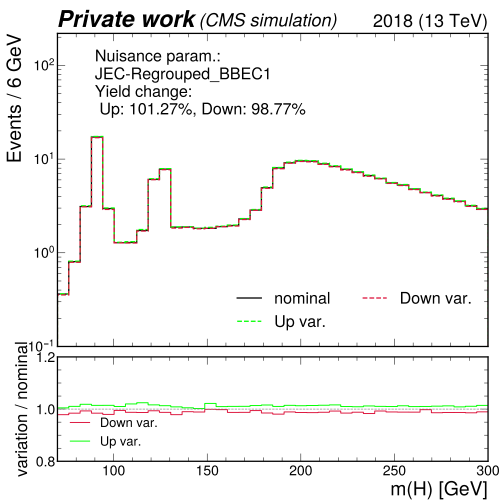
In Figure 16, the impacts of the nuisance parameters on a fit with the baseline scenario can be seen for an expected signal strength of zero. This can be understood as a test of the robustness of the uncertainty model. As expected, the uncertainties associated with the cH($ZZ \to 4\mu$) signal modelling no longer have a significant impact on the fit, while the remaining nuisance parameters otherwise behaves similarly to when a signal is injected into the Asimov dataset. This indicates that the uncertainty model is working as expected.



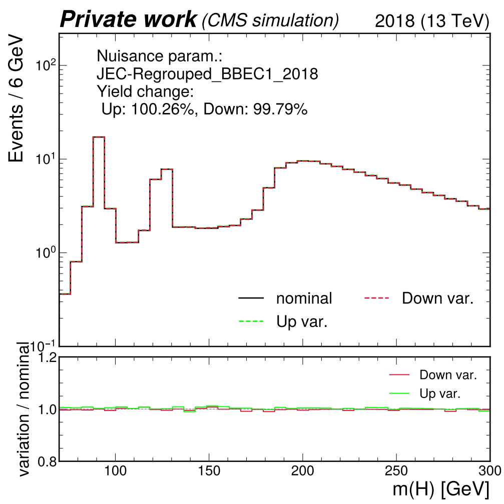
(a) JES: Regrouped_Absolute



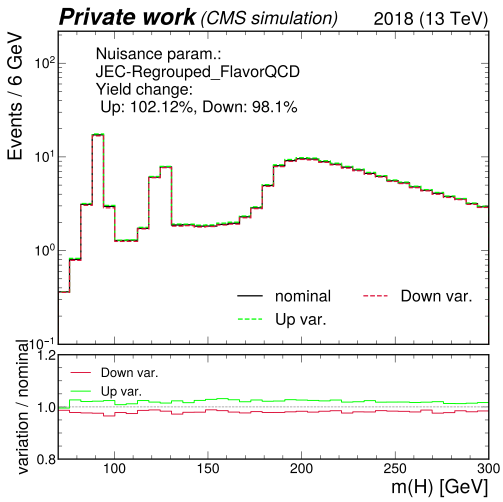
(b) JES: Regrouped_Absolute_2018



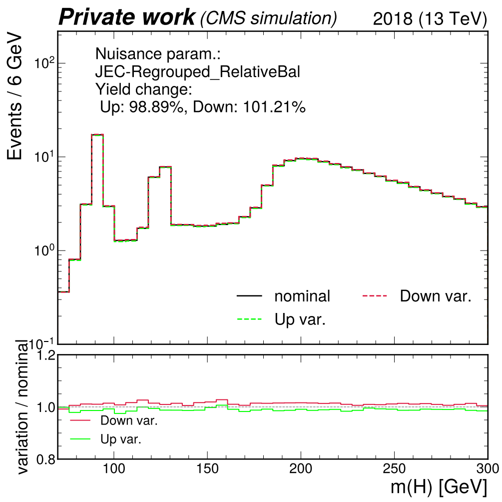
(c) JES energy scale: Regrouped_BBEC1



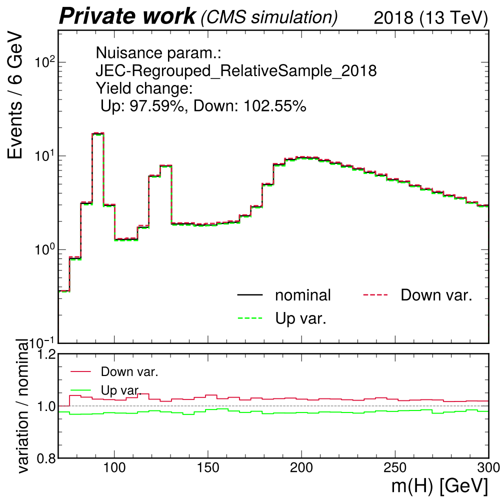
(d) JES: Regrouped_BBEC1_2018



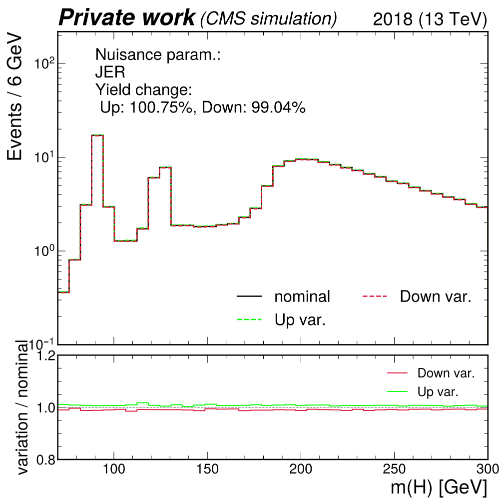
(e) JES: Regrouped_FlavorQCD



(f) JES: Regrouped_RelativeBal

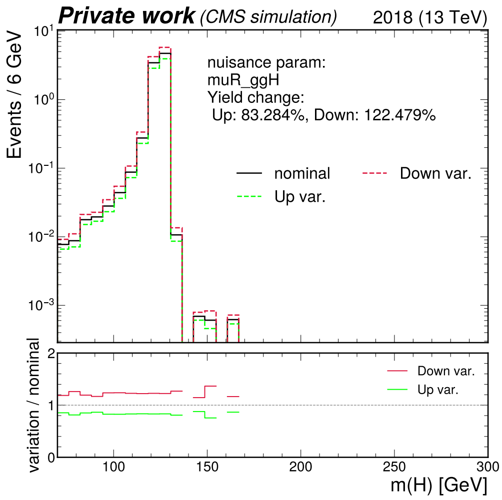


(g) JES: Regrouped_RelativeSample_2018

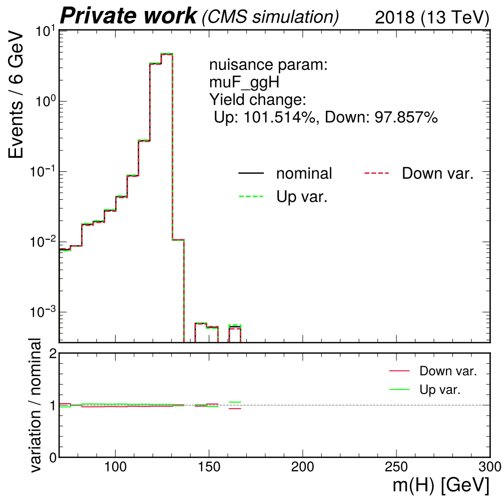


(h) JER

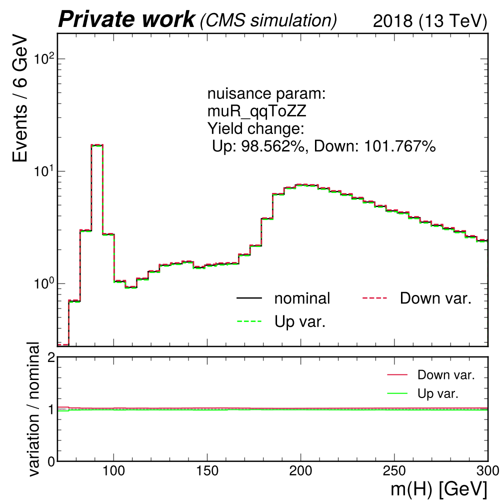
Figure 13: Leading uncertainty sources associated with the jet energy scale (JES) and jet energy resolution (JER) calibration.



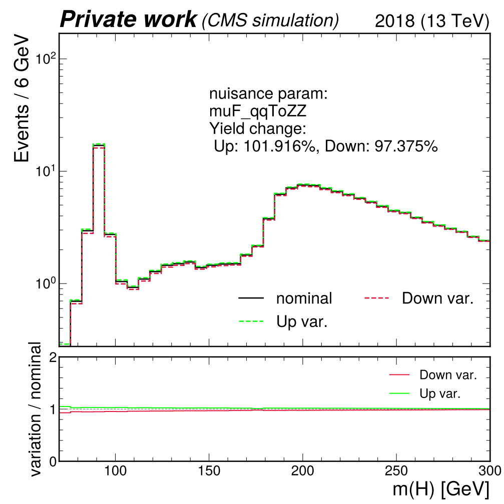
(a) Renormalisation scale: ggH



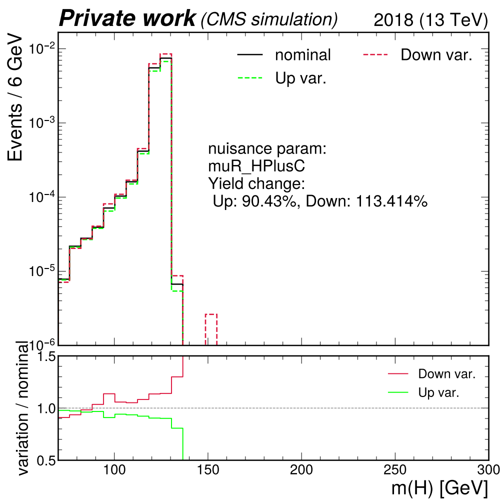
(b) Factorisation scale: ggH



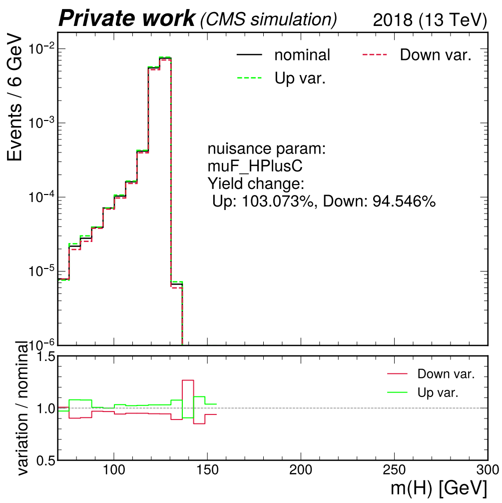
(c) Renormalisation scale: qq→ZZ



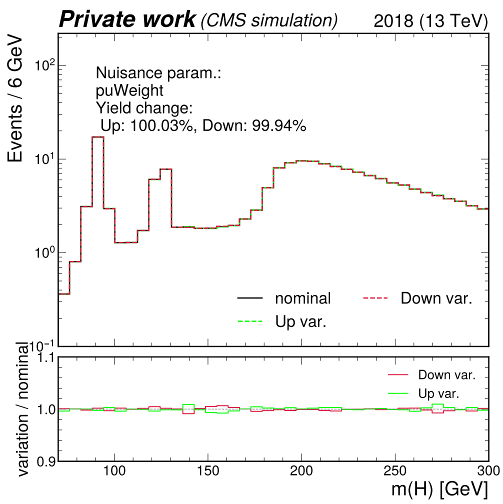
(d) Factorisation scale: qq→ZZ



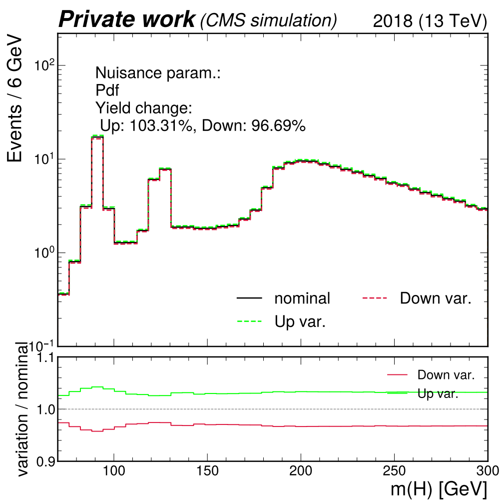
(e) Renormalisation scale: cH($ZZ \to 4\mu$)



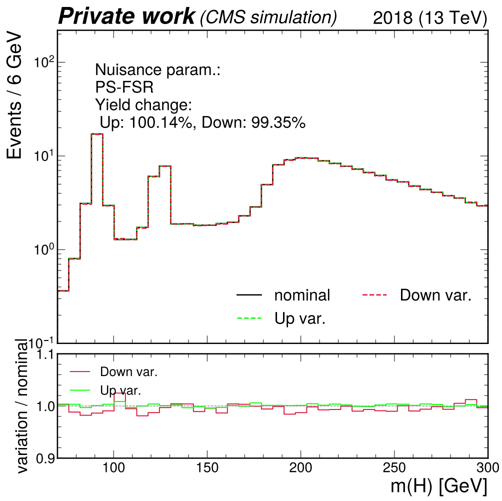
(f) Factorisation scale: cH($ZZ \to 4\mu$)



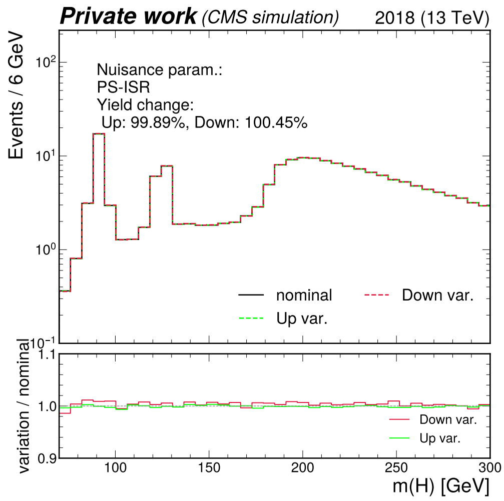
(g) Pileup reweighting



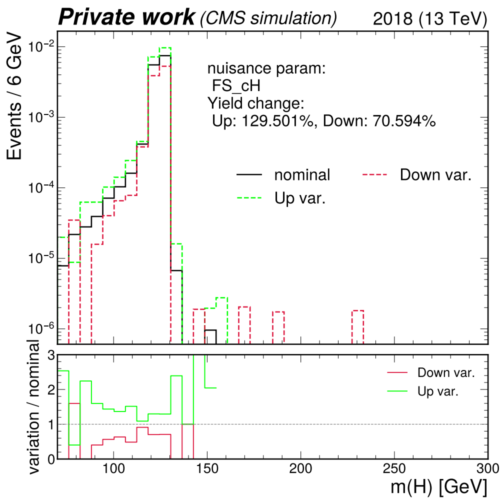
(h) Parton distribution function



(i) Parton showering: FSR



(j) Parton showering: ISR



(k) Flavour scheme: $cH(ZZ \rightarrow 4\mu)$

Figure 14: Leading theoretical uncertainty sources related to the modelling of the signal and background processes.

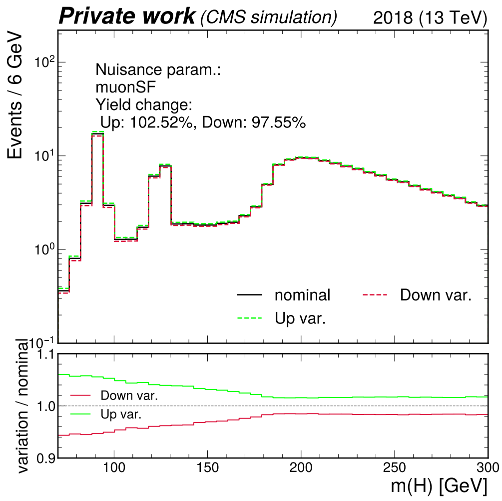


Figure 15: The uncertainty associated with the calibration of muons.

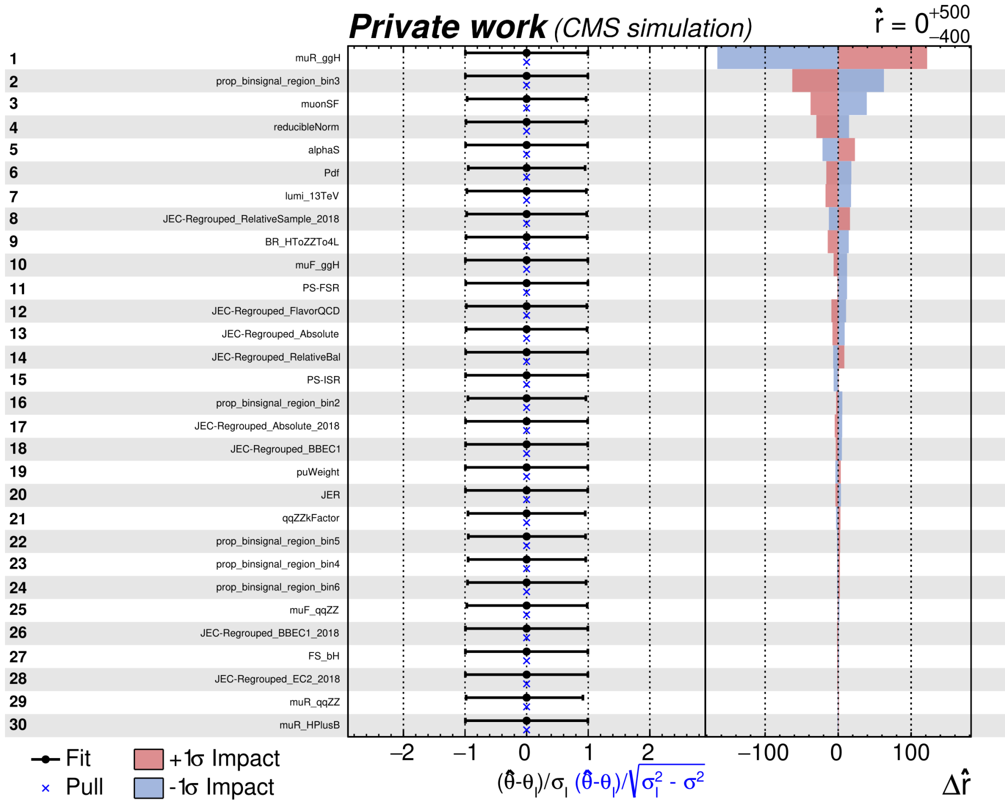


Figure 16: The 30 leading uncertainty sources in a fit of $\sigma\mathcal{B}(\mathrm{cH}(\mathrm{ZZ}\to 4\mu)_{\kappa_c})$ with $r = 0$, ordered by their relative impact on the fitted signal strength.